THE CLASSIFICATION OF SNC (p,k)-PLANES WITH HODGE STRUCTURES OF K3 TYPE

FEDERICO FALLUCCA, MATTEO PENEGINI AND ANNA ULIVI

ABSTRACT. A SNC (p,k)-plane is a $(\mathbb{Z}/p)^k$-Galois cover $\pi\colon X \longrightarrow \mathbb{P}^2$ with a simple normal crossing branch locus. In this setting, there exist several intermediate quotients $Y_1, Y_2, \ldots, Y_{\phi_{p,k}}$ corresponding to subgroups of $(\mathbb{Z}/p)^k$ of index p. We focus on the case where each Y_i is either a surface with geometric genus 0 or 1, with at least one with geometric genus 1. We classify all SNC (p,k)-planes satisfying these conditions obtaining that an extensive list of cases are minimal regular surfaces of general type with positive geometric genus. Moreover, these assumptions allow us to have strong control over the weight-2 Hodge structure of the minimal model of the covering surface X. Therefore, we also prove that Tate and Mumford–Tate conjectures hold for such surfaces.
As an independent result of our methods, we obtain a classification of SNC $(p,1)$-planes with geometric genus 1, which turn out to be precisely all K3 surfaces, and a classification of SNC $(p,1)$-planes with geometric genus 0, which turn out to be instead rational surfaces.

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0. Introduction

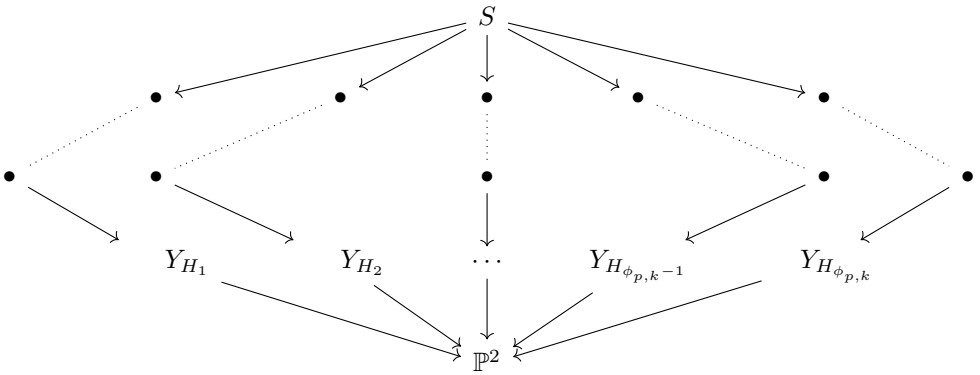
The theory of abelian covers was developed over the last decades of the twentieth century by various authors and was ultimately codified in the seminal paper [**39**]. This theory has proven to be a powerful tool for constructing varieties with rich and intriguing geometric properties. In this work we shall exploit the theory of abelian covers to study the weight–2 Hodge structure of surfaces of general type.

Our approach is inspired by ideas of Morrison [**34**], who showed that to any surface S with $p_g(S) = 1$ one can associate a K3 surface whose transcendental second cohomology is isomorphic to that of S. In general, however, this isomorphism does not arise from an actual morphism between the surface and its associated K3 surface.

In this paper, we provide explicit examples of surfaces with $p_g \geq 1$ whose weight–2 Hodge structures are completely encoded by the Hodge structures of certain K3 surfaces. Moreover, in all our examples these K3 surfaces arise as geometric quotients of the original surfaces.

To be more precise, we consider the following situation, which combines previous work of the first author [**16**] and of the second author [**23**]. A (p, k)-plane is a finite morphism $\pi\colon X \longrightarrow \mathbb{P}^2$ where X is a normal surface and with the property that there is a subgroup $(\mathbb{Z}/p)^k$ of $\mathrm{Aut}(X)$ such that π factors as the composition of the quotient map $X \longrightarrow X/(\mathbb{Z}/p)^k$ and an isomorphism $X/(\mathbb{Z}/p)^k \cong \mathbb{P}^2$. Assume that the branch divisor of π is simple normal crossing (SNC for short). This cover has several intermediate quotients $Y_H \coloneqq X/H$ corresponding to the subgroups H of $(\mathbb{Z}/p)^k$. In particular, we focus on the $\phi_{p,k} \coloneqq \frac{p^k-1}{p-1}$ subgroups $H_i \leq (\mathbb{Z}/p)^k$ of index p, and on the associated intermediate covers $Y_{H_i} \longrightarrow \mathbb{P}^2$ with Galois group $G/H_i \cong \mathbb{Z}/p$. Hence, from any (p, k)-plane we obtain a diagram of the following type.

(1)



We focus our attention on SNC (p, k)-planes with the following

Property ★: Each Y_{H_i} is a surface with geometric genus either 0 or 1, and at least one Y_{H_i} has geometric genus 1.

We shall prove in Theorems 4.1 and 3.7 that the intermediate quotients Y_{H_i} with geometric genus zero are rational surfaces, while those with geometric genus one are $K3$ surfaces. In general, the surface Y_{H_i} need not be smooth; nevertheless, with a slight abuse of terminology, we shall refer to Y_{H_i} as a *K3 surface* whenever its minimal model is a K3 surface.

This situation has already been considered in the case $(p, k) = (2, 2)$ by several authors, e.g., [**7**, **8**, **20**, **23**, **27**–**29**, **41**], who investigated various specific properties under the assumption that the cover is smooth. Indeed, in this case, the transcendental cohomology of the cover splits into transcendental parts of K3 type, making the study of some very deep conjectures accessible, such as the Tate and Mumford-Tate conjectures, and the Voisin conjecture together with the

more general Bloch-Beilinson conjecture. Furthermore, Laterveer asked in [**29**] for the existence of other examples with the same property and high geometric genus ≥ 4:

Question. *[**29**, Question 7.7] Do there exist surfaces S of general type with $p_g(S) = 4$ such that the transcendental cohomology of S splits into 4 pieces of K3 type? And what about $p_g > 4$?*

In this paper, we provide a positive and exhaustive answer to this question. Moreover, we present a systematic study of the above-mentioned construction for primes $p \geq 2$ and integers $k \geq 1$, allowing also for (p,k)-planes with singularities. More precisely, we give a complete description of SNC (p,k)-planes satisfying Property ★: we not only classify all such coverings in families for all possible primes p and all positive integers k, but also determine their invariants and explicit defining equations for every resulting surface X in suitable weighted projective spaces. Furthermore, we study the weight-2 Hodge structure of the minimal model of X. This cohomology group decomposes into Hodge substructures, and we show that the rational transcendental part is completely determined by the cohomology of those intermediate quotients Y_{H_i} which are K3 surfaces. Thus, the extensive list of surfaces of general type with $p_g \geq 4$ (see Tables 7 and 8) provide new relevant examples for the question posed by Laterveer.

This remarkable property allows us also to draw deeper conclusions. In particular, we prove the Mumford–Tate conjecture, and consequently the Tate conjecture, for all the surfaces constructed, following an argument similar to that in [**11**, **12**, **23**].

We can summarize our main results in the following theorems.

Theorem 0.1. *Let $p, k \in \mathbb{N}$ with p a prime number and $k \geq 2$. Moreover, let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane. Let $\rho\colon \widetilde{X} \longrightarrow X$ be the minimal resolution of singularities of X, and $\widetilde{X}_{min}$ its minimal model.*

Let us assume that each intermediate $(p,1)$-plane Y_{H_i} of π is either a surface with geometric genus 0 or 1, and at least one of them has geometric genus 1.

Then π falls in one of the cases listed in tables 17 from 29. More precisely:

(1) *The only admissible pairs (p,k) are the following:*

p \ k	2	3	4	5	6
2	48	63	42	14	3
3	13	5	1	–	–
5	3	1	–	–	–
7	2	–	–	–	–

TABLE 1. Number of families of SNC (p,k)-planes.

(2) *The surface $\widetilde{X}$ is regular. Moreover, $\widetilde{X}$ is a minimal surface except for the cases $C-(3,2)$, $A-(5,2)$, and $B-(7,2)$, whose minimal models are obtained by contracting further a single (-1)-curve. The surface $\widetilde{X}$ is a surface of general type, except in the following cases:*
   - *$\widetilde{X}_{min}$ is a K3 surface in cases: $A-(2,2)$, $A-(2,3)$, $B-(2,3)$, $A-(2,4)$, $A-(2,5)$, $C-(3,2)$, $A-(5,2)$;*
   - *$\widetilde{X}$ is a minimal properly elliptic surface (with Kodaira dimension one) in cases $B-(3,2)$ and $C-(3,3)$.*

   In tables 7 and 8, we list for each case the self-intersection of a canonical divisor of $\widetilde{X}_{min}$, along with its geometric genus, topological Euler characteristic, and surface type.

We also list in tables 17 to 29 the Picard number together with the Hodge numbers of the transcendental part of the Hodge structure of a very general member of the family.

(3) *$\widetilde{X}_{min}$ has the following decomposition of its transcendental lattice:*

$$T_{\widetilde{X}_{min}} \otimes \mathbb{Q} \simeq (T_{Y_{H_i}} \oplus \ldots \oplus T_{Y_{H_{\phi_{p,k}}}}) \otimes \mathbb{Q},$$

where all the transcendental Hodge structures of the surfaces Y_{H_i} are either trivial or the ones of a K3 surface.

(4) *Finally, $\widetilde{X}_{min}$ satisfies the Mumford–Tate and Tate conjectures.*

As a byproduct of our work, we also obtain a classification of all SNC $(p,1)$-planes $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$. Note that these (p,k)-planes are exactly those satisfying Property ★ for $k = 1$; hence it provides together with Theorem 0.1 an exhaustive conclusion of the classification.
We remark once more that X is not smooth in general; nevertheless, we analyze its minimal resolution of singularities and minimal model, and in doing so we establish the following theorem. This result is of independent interest and may well be known, although we have been unable to locate it in the existing literature.

Theorem 0.2. *Let $p \geq 2$ be a prime number and let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC $(p,1)$-plane. Let $\widetilde{X}_{min}$ denote the minimal model of X, and assume that $p_g(\widetilde{X}_{min}) = 1$. Then:*

(1) *$p \in \{2,3,5,7,11,13,17,19\}$;*
(2) *$\widetilde{X}_{min}$ is a K3 surface;*
(3) *for a very general member of the family holds $\rho(X) = 1$;*

Furthermore, π falls in one of the cases listed in tables 9 to 16, where we also provide the Picard number $\rho(\widetilde{X}_{min})$ together with the Hodge numbers of the transcendental part of the Hodge structure of a very general member of the family.

A second byproduct of our work consists in the classification of SNC $(p,1)$-planes with geometric genus 0, which turns out to be rational surfaces.

Theorem 0.3. *Let $p \geq 2$ be a prime number and let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC $(p,1)$-plane with $p_g(X) = 0$. Then X is a rational surface and it falls in one of the cases listed in tables 2 to 5.*

As the reader will see, the amount of computation required in the classification process is far too large to be carried out by hand. For this reason, we rely on several computer algebra scripts, which can be found at the webpage

`MAGMA` scripts and results repository

We emphasize that the group-theoretic computations were performed using `MAGMA` [**5**].

In the same webpage one can find the outputs of the classification algorithm together with a commented version which studies the intermediate quotients of each SNC (p,k)-plane turned out. As a consequence of this study, we also highlight the following remark.

Remark 1. There are two significant intermediate (p,k)-planes:

- a $(2,3)$-plane branched over seven lines in general position, which is a Campdelli surface with maximal Picard rank 8, see [**1**, Sec. 3]. This appears as one of the intermediate quotients of the surfaces belonging to the families $I-(2,4), O-(2,4), C-(2,5), F-(2,5), I-(2,5), J-(2,5), A-(2,6), B-(2,6)$. It has been already met in [**16**, Subsec. 5.8] and was first constructed in this way by Persson in [**43**, Ex. 5.8]. We also mention [**13**, Thm. 1.1(1)] where the authors give it through equations of a different (not normal) birational model.
- a $(5,2)$-plane branched over four lines in general position, which is the original Godeaux surface with maximal Picard rank 9 constructed by Godeaux. Indeed, it can be also

viewed as the étale cyclic quotient of $A-(5,3)$, which is the Fermat quintic $x_0^5+x_1^5+x_2^5+x_3^5=0$ in $\mathbb{P}^3$, where $(\mathbb{Z}/5)^3$ acts on each coordinate by a different 5-root of unity. See also [**9**, Theorem 65] and [**40**].

We conclude the introduction by outlining the structure of the paper.

In Section 1 we briefly recall the theory of abelian covers of algebraic varieties, with particular emphasis on the case of Galois covers with group $(\mathbb{Z}/p)^k$, following [**16**, **39**].

In Section 2 we develop the necessary tools to bound the admissible values of p and k, thereby providing all the ingredients required to construct an explicit algorithm for the classification of SNC (p,k)-planes satisfying Property ★. Consequently, we present the algorithm implementing the classification and we prove Theorem 0.1 point (1).

Section 3 is devoted to the proof of Theorem 0.2: any $(p,1)$-plane with geometric genus 1 is a K3 surface. We provide a detailed analysis of the $(p,1)$-planes appearing in the theorem; specifically we study their minimal resolution of singularities to determine the minimal model via the contraction of (-1)-curves, see Figures 1, 2, 3, 4, 5.

In Section 4, we prove Theorem 0.3, see Theorem 4.1, where we classify all $(p,1)$-planes with geometric genus 0 in (infinite) families, and then proving that each of those surfaces are rational.

In Section 5, we discuss some general results on the Hodge structures of SNC (p,k)-planes. More precisely, we investigate the Tate and Mumford-Tate conjectures of these surfaces following mainly the ideas contained in [**23**]. Thus, we prove Theorem 0.1 points (3) and (4).

In Section 6 we prove Theorem 0.1 point (2), see Theorem 6.1. More precisely, we analyze each SNC (p,k)-plane satisfying Property ★ in the classification list 17 to 29, explicitly computing its numerical invariants and determining its Kodaira dimension. For the reader's convenience, the tables produced by the algorithm are collected in Appendix 8, in order to keep the exposition streamlined.

Finally, in Section 7, we present detailed and instructive examples of SNC (p,k)-planes that we used as a guideline of the entire work. In the same examples, we also show how to determine the Picard number and the rank of the transcendental part of the Hodge structure of a very general member of their family.

Notation and conventions

We work over the field $\mathbb{C}$ of complex numbers.

With $\lfloor x \rfloor$ we denote the greatest integer less or equal than the real number x.

We denote for short by $(\mathbb{Z}/p)^k$ the elementary abelian p-group $(\mathbb{Z}/p\mathbb{Z})^k$.

Let G be an abelian group. The set of characters, that is, group homomorphisms $\chi\colon G\longrightarrow\mathbb{C}^\times$, forms an abelian group under pointwise multiplication. We denote the character group by G^*.

Given a character $\chi\colon G\longrightarrow\mathbb{C}^\times$ of $G=(\mathbb{Z}/p)^k$ and $g\in G$, define $\langle\chi,g\rangle\in\{0,\dots,p-1\}$ by

$$\chi(g)=\exp\left(\frac{2\pi i}{p}\langle\chi,g\rangle\right).$$

By *surface* we mean a projective, connected normal two dimensional complex variety S, and for such a surface $\omega_S=\mathcal{O}_S(K_S)$ denotes the canonical class, $p_g(S)=h^0(S,\omega_S)$ is the *geometric genus*, $q(S)=h^1(S,\omega_S)$ is the *irregularity*, $\chi(\mathcal{O}_S)=1-q(S)+p_g(S)$ is the *Euler-Poincaré characteristic*, $e(S)$ is the *topological Euler characteristic*, and $\rho(S)$ is the *rank of the Picard group* of S. If $q(S)=0$, we call S a *regular surface*.
$T(S)_{\mathbb{Q}}$ denotes the *transcendental lattice* $T(S)=\mathrm{NS}(S)^\perp$ with rational coefficients.
$H^{1,1}_{tra}(S)$ denotes the transcendental part of the complex vector space $H^{1,1}(S)$, and by $h^{1,1}(S)$ its dimension.

By an abuse of notation we will call *K3 surface* also singular surfaces, whose minimal model is a K3 surface.

Throughout the paper, we denote Cartier (or Weil) divisors on a variety by capital letters and the corresponding line bundles by italic letters, so we write for instance $\mathcal{L} = \mathcal{O}_S(L)$.

We denote by $\equiv$ the numerical equivalence relation among Cartier divisors, while $\equiv_{lin}$ denotes the linear equivalence relation.

A cyclic quotient singularity x of a normal surface X is called of type $\frac{1}{n}(1,a)$, with $\gcd(a,n) = 1$, if there exists an open neighborhood of x isomorphic to the quotient $\mathbb{C}^2/(\mathbb{Z}/n)$, where $\mathbb{Z}/n$ acts diagonally on $\mathbb{C}^2$ by sending $(z_1, z_2) \mapsto \big(\exp(\frac{2\pi i}{n})z_1, \exp(\frac{2\pi i}{n}a)z_2\big)$.

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1. General results of abelian covers with Galois group $(\mathbb{Z}/p)^k$

We briefly recall the theory of abelian covers following the notation developed in [**16**].

A *Galois cover* with a finite group G is a finite morphism $\pi\colon X \longrightarrow Y$ onto a smooth complete algebraic variety Y, with X normal admitting a subgroup of Aut(X) isomorphic to G and such that π factors through the composition of the quotient map $X \longrightarrow X/G$ and an isomorphism $X/G \cong Y$. The group G is the Galois group of π.

Definition 1.1. *An* abelian cover *is a Galois cover with Y smooth, X normal, and G a finite abelian group. A* (p,k)-cover *is an abelian cover with Galois group $G = (\mathbb{Z}/p)^k$. A* (p,k)-plane *is a (p,k)-cover with $Y \cong \mathbb{P}^2$.*

Let $\pi\colon X \longrightarrow Y$ be an abelian cover with Galois gruop G. Following [**16**] and [**39**], the direct image of the structure sheaf $\mathcal{O}_X$ on Y splits as a direct sum of line bundles $\mathcal{L}_\chi^{-1}$ indexed by characters $\chi \in G^*$

$$\pi_*\mathcal{O}_X = \bigoplus_{\chi \in G^*} \mathcal{L}_\chi^{-1}. \tag{2}$$

By Zariski-Nagata purity theorem, the branch locus of π is a divisor. The reduced structure of this divisor will be denoted by D. As is customary for Galois covers, we call the divisor $\pi^{-1}(D)$, endowed with the reduced structure, the *ramification divisor* of π, and we denote it by R. Let T be an irreducible component of R. The elements of G fixing all the points of T form a cyclic subgroup H of G, the so called *inertia group* of T. Furthermore, there is a unique character ψ of H and a uniformizing parameter t for $\mathcal{O}_{X,T}$ such that any $h \in H$ acts on t as $t \mapsto \psi(h)t$. In addition, as observed in [**17**], we can choose a *unique* element $g \in G$ such that $H = \langle g \rangle$ with the property that $\chi(g) = \exp\left(\frac{2\pi i}{|H|}\right)$, i.e., h acts on t by the multiplication of the first $|H|$-root of the unity. In this case, g is called the *local monodromy* of the component T. This discussion leads to the following natural decomposition of the ramification divisor

$$R = \sum_{g \in G} R_g,$$

where R_g is the (possibly zero) reduced divisor given by the sum of those irreducible components $T \subset R$ whose inertia group is generated by g and whose local monodromy character corresponds to g. In particular, two irreducible components T_1 and T_2 appear in the same summand R_g if and only if they have the same local monodromy. Furthermore, since G is a abelian group, then

R_g is a union of G-orbits. For this reason, there exists a unique reduced divisor $D_g \subset Y$ such that

$$R_g = \pi^{-1}(D_g).$$

Therefore, we have also a decomposition of the reduced branch locus

$$D = \sum_{g \in G} D_g.$$

Definition 1.2. *The* building data *of the abelian cover $\pi\colon X \longrightarrow Y$ consist of the collection of line bundles $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ together with the collection of reduced divisors $\{D_g\}_{g \in G}$ constructed above.*

Remark 2. By construction, D_0 is the zero divisor and $\mathcal{L}_1 \cong \mathcal{O}_Y$. Furthermore, X is connected if and only if $h^0(Y, \mathcal{L}_\chi^{-1}) = 0$ for all $\chi \neq 1_G$. Indeed, from the decomposition (2) one has $h^0(X, \mathcal{O}_X) = \sum_{\chi \in G^*} h^0(Y, \mathcal{L}_\chi^{-1})$.

The most powerful result regarding abelian coverings is Pardini's Existence Theorem [**39**, Theorem 2.1], which states that the building data completely determine an abelian cover of a smooth variety Y. This allows one to work with abelian coverings simply by managing the building data, in the sense that all the geometric properties of X, such as its smoothness, canonical class, and numerical invariants, can be expressed in terms of the building data, and vice versa. For the sake of simplicity and for our purposes, we only present the theorem in the case where $G \cong (\mathbb{Z}/p)^k$ is an elementary abelian p-group and the base of the cover is smooth, i.e., for (p,k)-covers with p prime number and $k \geq 1$.

Theorem 1.3. *[**39**, Thm. 2.1] Let $\pi\colon X \longrightarrow Y$ be a (p,k)-cover. Then for any pairs of characters $\chi, \eta \in G^*$ it holds*

$$L_\chi + L_\eta \equiv_{lin} L_{\chi\eta} + \sum_{g \in (Z/p)^k} \left\lfloor \frac{\langle \chi, g\rangle + \langle \eta, g\rangle}{p} \right\rfloor D_g. \tag{3}$$

Conversely, given a smooth variety Y and

- *a collection of line bundles $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ on Y indexed by the characters of G,*
- *a collection of* divisors *$\{D_g\}_{g \in G}$ on Y indexed by the elements of G,*

with the property that $D \coloneqq \sum_{g \in G} D_g$ is reduced and the linear equations (3) *hold for any pair $\chi, \eta \in G^*$, then there exists a unique (p,k)-cover, up to isomorphisms of G-covers, $\pi\colon X \longrightarrow Y$ whose building data are $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ and $\{D_g\}_{g \in G}$.*

As evident, the number of linear equivalence equations required to guarantee the existence of a (p,k)-cover grows quadratically with the cardinality of the group. As a consequence, the explicit determination of a (p,k)-cover can be computationally demanding. The following Theorem 1.4 reduces the number of equations that need to be checked to k, the rank of $(\mathbb{Z}/p)^k$.

Theorem 1.4. *[**39**, Prop. 2.1] Let $\{e_1, \dots, e_k\}$ be a set of generators of $(\mathbb{Z}/p)^k =: G$ and let $\{\epsilon_1, \dots, \epsilon_k\} \subset G^*$ be the corresponding dual set of generators of G^*.*

Let $\pi\colon X \longrightarrow Y$ be a (p,k)-cover. Then

$$p \cdot L_{\epsilon_j} \equiv_{lin} \sum_{g \in G} \langle \epsilon_j, g\rangle D_g, \qquad \text{for any} \qquad j = 1, \dots, k. \tag{4}$$

Conversely, given a smooth variety Y and

- *a collection of divisors $\{D_g\}_{g \in G}$ on Y and*
- *line bundles $\{\mathcal{L}_{\epsilon_1}, \dots, \mathcal{L}_{\epsilon_k}\}$ on Y*

satisfying equations (4) *for any* $j = 1, \dots, k$. *Then for any* $\chi \in G^*$ *the divisor* $\sum_g \chi(g) D_g$ *is* p*-divisible, in the sense that there exists a divisor* L_χ *such that* $p \cdot L_\chi \equiv \sum_g \chi(g) D_g$. *Setting* $\mathcal{L}_\chi \coloneqq \mathcal{O}_Y(L_\chi)$, *then* $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ *and* $\{D_g\}_{g \in G}$ *satisfy equations* (3).

Consequently, there exists a unique (p, k)*-cover* $\pi \colon X \longrightarrow Y$ *(up to isomorphisms of* G*-covers) whose building data are* $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ *and* $\{D_g\}_{g \in G}$.

It is not so difficult to see that divisors D_g determine L_χ (namely the line bundle $\mathcal{L}_\chi$) up to torsion of $\mathrm{Pic}(Y)$.

Lemma 1.5. *[**16**, Cor. 1.6] Any line bundle* $\mathcal{L}_\chi$ *in the building data of a* (p, k)*-cover satisfies the linear equivalence equation*

$$p \cdot L_\chi \equiv_{lin} \sum_{g \in G} \langle \chi, g \rangle D_g. \tag{5}$$

In particular, if $\mathrm{Pic}(Y)$ is torsion-free (as in the case when Y is a rational or K3 surface) then the divisors D_g uniquely determine the line bundles.

Remark 3. By Lemma 1.5, if the base space Y has torsion-free Picard group, the building data of a (p, k)-cover are uniquely determined by the collection of divisors $\{D_g\}_{g \in G}$. Indeed, in this case there is a unique collection of line bundles $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ for which the relations (5) are satisfied, and these are implicitly fixed throughout.

Moreover, in order to test whether a given collection of divisors $\{D_g\}_{g \in G}$ defines a (p, k)-cover, it suffices to verify that the divisors

$$\sum_{g \in G} \langle \epsilon_i, g \rangle \, D_g$$

are p-divisible for a suitable dual set of generators $\{\epsilon_1, \dots, \epsilon_k\}$ of G^*, as follows from Theorem 1.4.

Remark 4. If Y has a torsion-free Picard group and X is connected, then the set of local monodromies $\{g \in G | D_g \neq 0\}$ generates the entire group G. Otherwise, if the subgroup H generated by local monodromies were proper in G, one could define a non trivial character χ vanishing on H. However, it would follow from Equation (5) that $\mathcal{L}_\chi$ has p-torsion, and so thus it must be the trivial bundle. This would imply $h^0(Y, \mathcal{L}_\chi^{-1}) = 1$, which contradicts the connectedness of X, according to Remark 2.

As will become apparent below, most geometric and numerical invariants of a (p, k)-cover are naturally expressed in terms of the line bundles $\mathcal{L}_\chi$, rather than directly in terms of the divisors D_g. Consequently, although the cover can be defined by specifying the divisors D_g, constructing a (p, k)-cover with prescribed numerical invariants requires passing between these two types of building data, a process which can be computationally demanding.

For this reason, the authors in [**16**] investigated whether the line bundles $\mathcal{L}_\chi$ determine the divisors D_g for a general abelian cover. As observed in [**16**, Example 1.7], it is not so difficult to construct (singular) abelian covers having the same $\mathcal{L}_\chi$ but different D_g, so it is not always possible to determine D_g in terms of $\mathcal{L}_\chi$. However, a positive answer is given in [**16**, Thm. 1.11] for groups $G = (\mathbb{Z}/2)^k$. We generalize this crucial result for elementary abelian p-groups. Before doing that let us recall the following definition.

Definition 1.6. *The natural isomorphism* $G \xrightarrow{\sim} G^{**}$ *allows us to regard each element* $g \in G$ *as a character of* G^*, *which we denote again by* g, *by setting*

$$g(\chi) \coloneqq \chi(g).$$

The kernel of g *is then the subgroup of* G^* *consisting of the characters* χ *such that* $\chi(g) = 1$. *In other words,*

$$\chi \in \ker g \iff g \in \ker \chi.$$

Let $\mathcal{H} \subset G^*$ be a subgroup, possibly of the form $\ker g$. For any $g \in G$, we denote by $g_{|\mathcal{H}}$ the element of $\mathcal{H}^*$ obtained by restricting g to $\mathcal{H}$. Since G is an elementary abelian p-group, the order $o(g_{|\mathcal{H}})$ of $g_{|\mathcal{H}}$ is either 1 or p. We define the orthogonal subgroup of $\mathcal{H}$ as

$$\mathcal{H}^{\perp} \coloneqq \{g \in G \mid g_{|\mathcal{H}} = 1\}.$$

Theorem 1.7. *Let $\pi\colon X \to Y$ be a (p,k)-cover with building data $\{\mathcal{L}_\chi\}_{\chi \in G^*}$, $\{D_g\}_{g\in G}$. Then, given a subgroup $\mathcal{H} \leq G^*$ of index p^t in G^*, we have*

$$\frac{p^k(p-1)}{2} \sum_{h \in \mathcal{H}^{\perp}} D_h \equiv_{lin} p \left(\sum_{\chi \notin \mathcal{H}} L_\chi - (p^t - 1) \sum_{\chi \in \mathcal{H}} L_\chi \right). \tag{6}$$

In particular, for $\mathcal{H} = G^*$, we have

$$\frac{p^k(p-1)}{2} \sum_{g \in G} D_g \equiv_{lin} p \sum_{\chi \in G^*} L_\chi. \tag{7}$$

Proof. Consider a subgroup $\mathcal{H}$ of G^*. Then applying equation (5) to each $\chi \in \mathcal{H}$, one has

$$\begin{aligned} p \sum_{\chi \in \mathcal{H}} L_\chi &\equiv_{\text{lin}} \sum_{\chi \in \mathcal{H}} \left(\sum_{g \in G} \langle \chi, g \rangle D_g \right) \\ &\equiv_{\text{lin}} \sum_{g \in G} \left(\sum_{\chi \in \mathcal{H}} \langle \chi, g \rangle \right) D_g \\ &\equiv_{\text{lin}} \sum_{g \in G} \frac{|\mathcal{H}|}{2} p \left(1 - \frac{1}{o(g_{|\mathcal{H}})} \right) D_g, \end{aligned} \tag{8}$$

where the last equality follows from [**16**, Lemma 1.9], which is true for an arbitrary abelian group. Furthermore, when $\mathcal{H} = G^*$, since $o(g_{|\mathcal{H}}) = p$, the last equation (8) reduces to (7).

Now, using both (8) and (7) we obtain

$$\begin{aligned} \sum_{g \in G} \frac{p^{k+1}}{2} \left(\frac{1}{o(g_{|\mathcal{H}})} - \frac{1}{p} \right) D_g &\equiv_{\text{lin}} \sum_{g \in G} \frac{p^{k+1}}{2} \left(1 - \frac{1}{p} \right) D_g - \sum_{g \in G} \frac{p^{k+1}}{2} \left(1 - \frac{1}{o(g_{|\mathcal{H}})} \right) D_g \\ &\equiv_{\text{lin}} p \sum_{\chi \in G^*} L_\chi - p[G^* : \mathcal{H}] \sum_{\chi \in \mathcal{H}} L_\chi \\ &\equiv_{\text{lin}} p \left(\sum_{\chi \notin \mathcal{H}} L_\chi - \left(p^t - 1\right) \sum_{\chi \in \mathcal{H}} L_\chi \right). \end{aligned}$$

However, the coefficient on the left member of the equation is either $\frac{p^k}{2}(p-1)$ or 0 according if $g \in \mathcal{H}^{\perp}$ or not. Thus the left summand is over $\mathcal{H}^{\perp}$ and (6) follows. □

Corollary 1.8. *Let $g \in G$ with $g \neq 0$. Then*

$$\frac{p^k(p-1)}{2} \sum_{h \in \langle g \rangle} D_h \equiv_{lin} p \left(\sum_{\chi \notin \ker g} L_\chi - (p-1) \sum_{\chi \in \ker g} L_\chi \right). \tag{9}$$

Proof. It follows directly from Theorem 1.7 applied to $\mathcal{H} \coloneqq \ker g$. Indeed, in this case $\mathcal{H}^{\perp} = \langle g \rangle$: the inclusion $\langle g \rangle \subseteq \mathcal{H}^{\perp}$ is straightforward, whilst the other direction holds directly by the fact that each character defined over a proper subgroup can be extended to a non-trivial character of the entire group. □

Remark 5. Formula (9) shows that the numerical class of $\sum_{h\in\langle g\rangle} D_h$ can be expressed in terms of the line bundles $\mathcal{L}_\chi$. Note that when working with an elementary abelian 2-group, each subgroup $\langle g\rangle$, $g\neq 0$, consists of exactly two elements, namely the identity and g itself, and the above sum simply reduces to D_g. Thus, in the case $G\cong(\mathbb{Z}/2)^k$, the formula in [**16**, Thm. 1.11] becomes a particular instance of Formula (9).

1.1. **Smooth and singular (p,k)-covers and their invariants.**

Definition 1.9. *A* smooth (p,k)-cover *is a (p,k)-cover $\pi\colon X\longrightarrow Y$ such that the reduced branch divisor D has normal crossings, and for each $g\in G$, the divisor D_g is either empty or a disjoint union of smooth irreducible components. Furthermore, we require that*

$$\langle g\rangle\cap\langle h\rangle=\{0\}, \tag{10}$$

for any two distinct irreducible components $\Delta_g\leq D_g$ and $\Delta_h\leq D_h$ such that $\Delta_g\cap\Delta_h\neq 0$.

Proposition 1.10. *Let $\pi\colon X\longrightarrow Y$ be a smooth (p,k)-cover. Then X is smooth.*

Proof. This is a special case of [**39**, Prop. 3.1]. First, by [**39**, Cor. 3.1] the variety X is normal. Moreover, the only points of X that may be singular are those lying over intersection points of two irreducible components of the reduced branch divisor D. However, by [**39**, Prop. 3.1], condition (10) ensures that all points lying above such intersections are smooth. Hence the variety X is smooth. □

To consider more general (p,k)-covers, we must allow certain singularities. Thus, we will introduce a slightly weaker condition on the reduced branch divisor D than the one required for a smooth (p,k)-cover.

Definition 1.11. *A* Simple Normal Crossing (p,k)-cover *(SNC for short) is a (p,k)-cover $\pi\colon X\longrightarrow Y$ of a smooth surface Y such that the reduced branch divisor D has normal crossings, i.e. all irreducible components of D are smooth, each two of them either do not intersect or intersect transversally, and no point of Y belongs to more than two of them.*

For the rest of this section, we shall consider SNC (p,k)-covers of a surface Y. By Proposition 1.10 (see also [**39**, Prop. 3.1]), the singular points of X arise from the intersection points of pairs of distinct irreducible components $\Delta_g\subseteq D_g$ and $\Delta_h\subseteq D_h$ such that $\langle g\rangle\cap\langle h\rangle\neq 0$. In other words, the image in Y of the singular locus of X is

$$\pi(\mathrm{Sing}(X))=\bigcup_{\substack{\langle g\rangle\cap\langle h\rangle\neq\{0\}\\ \Delta_g\leq D_g,\,\Delta_h\leq D_h}}(\Delta_g\cap\Delta_h).$$

Such a surface X has only rational singularities, that are determined precisely as follows.

Proposition 1.12. *[**39**, Prop. 3.3] Let $\pi\colon X\longrightarrow Y$ be an SNC (p,k)-cover of a smooth surface Y, and let $\Delta_g\subset D_g$ and $\Delta_h\subset D_h$ be smooth irreducible components such that $\langle g\rangle\cap\langle h\rangle\neq 0$. Let $1\leq a\leq p-1$ be the integer such that $h=(p-a)g$. If Δ_g and Δ_h intersect transversely at a point $y\in Y$, then the fiber $\pi^{-1}(y)$ consists of p^{k-1} points, each of which is a cyclic quotient singularity of X of type $\frac{1}{p}(1,a)$.*

Proof. The statement is a special case of [**39**, Proposition 3.3]. □

Let x be a singularity of type $\frac{1}{n}(1,a)$ with $\gcd(n,a)=1$. The minimal resolution of singularities of x consists of replacing x by a Hirzebruch-Jung string $E_1\cup\cdots\cup E_l$, where the E_i are smooth rational curves with $E_iE_{i+1}=1$, for $i=1,\ldots,\ell-1$, and $E_iE_j=0$ otherwise. Their self-intersection is $E_i^2=-b_i$, where b_i are the coefficients of the continued fraction

$$\frac{n}{a}=b_1-\cfrac{1}{b_2-\cfrac{1}{b_3-\ldots}}=[b_1,\ldots,b_\ell].$$

Definition 1.13. *Let x be a singularity of type $\frac{1}{n}(1,a)$ with $\gcd(n,a)=1$ and let $1 \leq a' \leq n-1$, such that $a' = a^{-1}$ in $(\mathbb{Z}/n)^*$. We define the following*

- $k_x := k(\frac{1}{n}(1,a)) := -2 + \frac{2+a+a'}{n} + \sum_{i=1}^{\ell}(b_i - 2) \geq 0$;
- $e_x := e(\frac{1}{n}(1,a)) := \ell + 1 - \frac{1}{n}$;
- $E_x := E(\frac{1}{n}(1,a)) := \sum_{i=1}^{\ell}\left(1 - \frac{D_{i,\ell}+D_{i,1}}{n}\right)E_i$, *where $D_{i,\ell}$ and $D_{i,1}$ are the denominators of the truncated continued fractions*

$$\frac{N_{i,\ell}}{D_{i,\ell}} := [b_i, b_{i+1}, \ldots, b_\ell], \qquad \text{and} \qquad \frac{N_{i,1}}{D_{i,1}} := [b_i, b_{i-1}, \ldots, b_1].$$

Clearly, $N_{1,\ell} = n$, $D_{1,\ell} = a$, and $N_{\ell,\ell} = b_\ell$, $D_{\ell,\ell} = 1$. It is well known that $N_{\ell,1} = n$, $D_{\ell,1} = a'$, and $N_{1,1} = b_1$, $D_{1,1} = 1$.

Theorem 1.14. *Let $\pi\colon X \longrightarrow Y$ be an SNC (p,k)-cover of a smooth surface Y. Let $\rho\colon \widetilde{X} \longrightarrow X$ be the minimal resolution of singularities of X, and consider the generically finite morphism $\widetilde{\pi} := \pi \circ \rho\colon \widetilde{X} \longrightarrow Y$. Then:*

$$p_g(\widetilde{X}) = \sum_{\chi \in G^*} h^0(Y, K_Y + \mathcal{L}_\chi), \tag{11}$$

$$q(\widetilde{X}) = q(Y) + \sum_{\chi \in G^* \setminus \{1\}} h^1(Y, -\mathcal{L}_\chi), \tag{12}$$

$$\chi(\mathcal{O}_{\widetilde{X}}) = p^k \chi(\mathcal{O}_Y) + \frac{1}{2}\sum_{\chi \in G^*} L_\chi(L_\chi + K_Y), \tag{13}$$

$$K^2_{\widetilde{X}} = p^{k-2}\left(pK_Y + (p-1)\sum_{g \in G} D_g\right)^2 - \sum_{x \in Sing(X)} k_x, \tag{14}$$

(15)

$$e(\widetilde{X}) = p^k\left(e(Y) - \left(1 - \frac{1}{p}\right)\cdot \sum_{\Delta_g}(2 - 2g(\Delta_g)) + \frac{1}{2}\left(1 - \frac{1}{p}\right)^2 \sum_{\Delta_g \neq \Delta_h} \Delta_g \Delta_h\right) + \sum_{x \in Sing(X)} e_x.$$

Furthermore, we have

$$pK_{\widetilde{X}} \equiv_{lin} \widetilde{\pi}^*\left(pK_Y + (p-1)\sum_{g \in G} D_g\right) - \sum_{x \in Sing(X)} pE_x. \tag{16}$$

Remark 6. Theorem 1.14 generalizes [**39**, Proposition 4.2] to the case where cyclic quotient singularities are allowed. Indeed, if the covering surface is smooth, then $\mathrm{Sing}(X)$ is empty and the formulae reduce to those in [**39**, Proposition 4.2].

Proof. The formulae for the geometric genus (11) and the irregularity (12) follow directly from (2), [**18**, Satz 1] and [**39**, Proposition 4.1], since the singular locus of X consists only of rational singularities from Proposition 1.12. As for (13), the holomorphic Euler characteristic satisfies

$$\chi(\mathcal{O}_{\widetilde{X}}) = 1 - q(\widetilde{X}) + p_g(\widetilde{X}) = \sum_{\chi \in G^*} \chi\big(K_Y + \mathcal{L}_\chi\big),$$

and the claim follows from the Riemann–Roch theorem.

Formula (16) is obtained by first applying the canonical bundle formula for the $(\mathbb{Z}/p)^k$-cover to express pK_X as the pullback of $pK_Y + (p-1)\sum_{g\in G} D_g$, and then passing to the minimal

resolution $\rho\colon \widetilde{X} \longrightarrow X$, where the discrepancy contributions of the cyclic quotient singularities are computed explicitly using the Hirzebruch-Jung resolution data (see e.g. [**14**] and [**25**, page 286]).

The self-intersection of the canonical class (14) is a standard computation obtained from (16) (see [**21**, Remark 3.3] and [**15**, Thm. 2.4]).

Formula (15) for $e(\widetilde{X})$ is obtained by applying the additivity of the Euler characteristic with respect to stratifications. More precisely, one first computes $e(X)$ by decomposing Y into the complement of the branch locus and its natural strata, using that π is a $(\mathbb{Z}/p)^k$-cover with simple normal crossings branch divisor, and then corrects for the singular points of X by passing to the minimal resolution of singularities $\rho\colon \widetilde{X} \longrightarrow X$. □

Remark 7. Notice that, by Noether's formula, the Euler characteristic could also be recovered from the previously computed invariants. Nevertheless, in the sequel we shall compute the Euler characteristic using Formula 15.

We automatically obtain the following useful result for SNC (p,k)-planes.

Lemma 1.15. *The minimal resolution of singularities $\widetilde{X}$ of a SNC (p,k)-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ is a regular surface, namely $q(\widetilde{X}) = 0$.*

Proof. Indeed, every line bundle of $\mathbb{P}^2$ has trivial first cohomology group; thus, from Theorem 1.14, we have

$$q(\widetilde{X}) = \sum_{\chi \in G^*} h^1(\mathbb{P}^2, -L_\chi) = 0.$$

□

1.2. **Decomposition of the branch curves under resolution of singularities.** We conclude this section studying the decomposition of the pullback $\widetilde{\pi}^*\Delta_g$ of a smooth irreducible component $\Delta_g \subset D_g$, keeping the terminology and notation of Theorem 1.14. We denote by $\widetilde{\Delta}_g$ the strict transform of $\pi^*\Delta_g$ under the resolution $\rho\colon \widetilde{X} \longrightarrow X$.

Proposition 1.16. *Let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane, and let $\widetilde{\pi} := \pi \circ \rho\colon \widetilde{X} \longrightarrow Y$. The reduced pullback $\widetilde{\pi}^*\Delta_g$ decomposes as the sum of a smooth irreducible curve $\widetilde{\Delta}_g$ (the strict transform of Δ_g under the resolution ρ) and the Hirzebruch-Jung strings arising from the singular points of X lying over Δ_g.*

More precisely, let $x \in Sing(X)$ be such that $\pi(x) \in \Delta_g$. In this case, there exists a unique integer $1 \leq a_x \leq p-1$ such that $\pi(x) \in \Delta_g \cap \Delta_h$, where $h = (p-a_x)g$, and x is a cyclic quotient singularity of type $\frac{1}{n}(1, a_x)$ (cf. Proposition 1.12).

Then, $\widetilde{\Delta}_g$ meets the exceptional chain $E_{1,x} \cup \cdots \cup E_{\ell_x,x}$ transversally in a single point, and this intersection point lies on the first component $E_{1,x}$. Moreover, we have

$$\widetilde{\Delta}_g^2 = p^{k-2}\Delta_g^2 - \sum_{x \in Sing(X),\, \pi(x) \in \Delta_g} \frac{a_x}{p} \tag{17}$$

$$2g(\widetilde{\Delta}_g) - 2 = p^{k-1}\left(2g(\Delta_g) - 2 + \sum_{\Delta_t,\, t \notin \langle g \rangle} \left(1 - \frac{1}{p}\right) \Delta_g \cdot \Delta_t\right). \tag{18}$$

Furthermore,

$$\widetilde{\pi}^*\Delta_g = p\widetilde{\Delta}_g + \sum_{x \in Sing(X),\, \pi(x) \in \Delta_g} \left(\sum_{i=1}^{\ell_x} D_{i,\ell_x} E_{i,x}\right), \tag{19}$$

where $\frac{N_{i,\ell_x}}{D_{i,\ell_x}} = [b_i, b_{i+1}, \ldots, b_{\ell_x}]$ *is the* i*-th truncated continued fraction of* $\frac{p}{a_x} = [b_1, \ldots, b_{\ell_x}]$.

Proof. The description of the intersection between the strict transform $\widetilde{\Delta}_g$ and the Hirzebruch–Jung string $E_{1,x} \cup \cdots \cup E_{\ell_x,x}$ follows from a local analysis in a neighborhood of the point $x \in X$. Indeed, since the divisors Δ_g and Δ_h intersect transversely at $\pi(x)$, the local equation of the cover near x shows that $\widetilde{\Delta}_g$ meets the exceptional chain transversely in a single point on the first component.

We now determine the coefficients $\alpha_{1,x}, \ldots, \alpha_{\ell_x,x}$ of the decomposition

$$\widetilde{\pi}^*\Delta_g = p\widetilde{\Delta}_g + \sum_{x \in \mathrm{Sing}(X),\, \pi(x) \in \Delta_g} \left(\sum_{i=1}^{\ell_x} \alpha_{i,x} E_{i,x} \right),$$

Since $\widetilde{\pi}^*\Delta_g$ is the pullback of a divisor on Y, it has zero intersection with each exceptional curve $E_{j,x}$. Using the fact that the strict transform $\widetilde{\Delta}_g$ meets the Hirzebruch–Jung chain only on the first component $E_{1,x}$, this yields a tridiagonal linear system for the coefficients $\alpha_i := \alpha_{i,x}$, whose unique solution is expressed explicitly in terms of the continued fraction data of the corresponding cyclic quotient singularity. More precisely, these coefficients are the solution to the linear system $\widetilde{\pi}^*\Delta_g \cdot E_{i,x} = 0$, $i = 1, \ldots, \ell_x$, which takes the form

$$\begin{cases} 0 &= p + (-b_1)\alpha_1 + \alpha_2 \\ 0 &= 0 + \alpha_1 - b_2\alpha_2 + \alpha_3 \\ \vdots & \\ 0 &= 0 + \alpha_{\ell_x - 1} - b_{\ell_x}\alpha_{\ell_x} \end{cases}.$$

The coefficient matrix of the linear system is the intersection pair matrix of the exceptional chain, and the constant vector is given by $(-p, 0, \ldots, 0)^t$. Thus, the explicit solution is obtained by multiplying the constant vector by the inverse of the intersection pair matrix, which can be found in [**14**]. Specifically, α_i is the denominator D_{i,ℓ_x} of the i-th partial continued fraction $\frac{N_{i,\ell_x}}{D_{i,\ell_x}} = [b_i, b_{i+1}, \ldots, b_{\ell_x}]$ of $\frac{p}{a_x} = [b_1, \ldots, b_{\ell_x}]$.

We can now compute the self-intersection of $\widetilde{\Delta}_g$. Intersecting the decomposition of $\widetilde{\pi}^*\Delta_g$ with $\widetilde{\Delta}_g$ and using the projection formula, we obtain

$$(\widetilde{\pi}^*\Delta_g) \cdot \widetilde{\Delta}_g = \Delta_g \cdot \widetilde{\pi}_* \widetilde{\Delta}_g = p^{k-1}\Delta_g^2.$$

On the other hand, since $\widetilde{\Delta}_g$ meets each exceptional chain only on the first component E_1^x, we have

$$(\widetilde{\pi}^*\Delta_g) \cdot \widetilde{\Delta}_g = p\,\widetilde{\Delta}_g^2 + \sum_{x \in \mathrm{Sing}(X),\, \pi(x) \in \Delta_g} \alpha_{1,x},$$

where $\alpha_{1,x} = D_{1,x} = a_x$. Therefore,

$$\widetilde{\Delta}_g^2 = p^{k-2}\Delta_g^2 - \sum_{x \in \mathrm{Sing}(X),\, \pi(x) \in \Delta_g} \frac{a_x}{p}.$$

The last formula determines the genus $g(\widetilde{\Delta}_g)$ of the curve $\widetilde{\Delta}_g$. We compute it by applying Riemann-Hurwitz to the restriction $\widetilde{\pi}_{|\widetilde{\Delta}_g} : \widetilde{\Delta}_g \longrightarrow \Delta_g$, which is a $G/\langle g \rangle \cong (\mathbb{Z}/p)^{k-1}$ Galois cover. Geometrically, restricting the cover to Δ_g amounts to quotienting the monodromy by $\langle g \rangle$, so that $\widetilde{\pi}_{|\widetilde{\Delta}_g}$ is branched precisely at the intersection points $\Delta_g \cap \Delta_t$ for which the monodromy class of t is nontrivial in $G/\langle g \rangle$. This means, more or less, that in a schematic picture of the branch locus, one can recognize the components Δ_t contributing to the branching of $\widetilde{\pi}_{|\widetilde{\Delta}_g}$ simply as those curves intersecting Δ_g whose associated group element t does not lie in $\langle g \rangle$; intersections with

components indexed by $t \in \langle g \rangle$ produce no branching and are therefore invisible in the induced cover. In virtue of this, we claim:

Claim: The branch locus of the restriction

$$\widetilde{\pi}_{|\widetilde{\Delta}_g} : \widetilde{\Delta}_g \longrightarrow \Delta_g$$

consists of the points $y \in \Delta_g$ such that $y \in \Delta_t$ for some component $\Delta_t \neq \Delta_g$ with $t \notin \langle g \rangle$. Moreover, the local monodromy at such a point y is given by the class $[t] \in G/\langle g \rangle$.

This claim implies that $\widetilde{\Delta}_g$ is a disjoint union of smooth connected components. In fact, $\widetilde{\Delta}_g$ is connected. Indeed, the number of connected components of $\widetilde{\Delta}_g$ equals the index of the subgroup of $G/\langle g \rangle$ generated by the local monodromies of $\widetilde{\pi}_{|\widetilde{\Delta}_g}$. We show that this subgroup coincides with the whole group $G/\langle g \rangle$.

Since X is connected and, by Remark 4 (and denoting by $d_g \coloneqq \deg D_g$),

$$G = \langle\, h \in G \mid d_h \neq 0 \,\rangle,$$

Let $h \in G$ with $d_h \neq 0$, then the curve Δ_h intersects Δ_g in at least one point, since any two curves in $\mathbb{P}^2$ intersect. If $h \in \langle g \rangle$, then $[h] = 0$ in $G/\langle g \rangle$. Otherwise, each point of $\Delta_g \cap \Delta_h$ has local monodromy $[h]$ for the map $\widetilde{\pi}_{|\widetilde{\Delta}_g}$. Hence we obtain

$$G/\langle g \rangle = \langle\, [h] \mid d_h \neq 0 \,\rangle \subseteq \langle\, [t] \mid \Delta_g \cap \Delta_t \neq 0,\ t \notin \langle g \rangle \,\rangle,$$

which shows that $\widetilde{\Delta}_g$ is connected.

Therefore $\widetilde{\Delta}_g$ is a smooth irreducible curve in $\widetilde{X}$. Its genus is then computed by applying the Riemann–Hurwitz formula to the morphism $\widetilde{\pi}_{|\widetilde{\Delta}_g}$.

Proof of the Claim: Let us consider a point $y \in \Delta_g$. We distinguish three cases:

- The point y does not belong to any other component of the branch locus of π. In this case, the fibre $\pi^{-1}(y)$ consists of p^{k-1} points of X, which are smooth by [**39**, Prop. 3.1]. Since the resolution is an isomorphism over smooth points, the fibre $\widetilde{\pi}^{-1}(y)$ also consists of p^{k-1} points, hence y is not a branch point;
- The point y belongs to another component $\Delta_t \neq \Delta_g$ of the branch locus of π, and $t \in \langle g \rangle$. Here, the fibre $\pi^{-1}(y)$ still consists of p^{k-1} points on X, which are cyclic quotient singularities by Proposition 1.12. Since $\widetilde{\Delta_g}$ intersects the Hirzebruch-Jung string $E_1^x \cup \cdots \cup E_{l_y}^x$ at a single point on E_1^x, there is a one-to-one correspondence between the points of $\widetilde{\pi}^{-1}_{|\widetilde{\Delta_g}}(y)$ and $\pi^{-1}(y)$. Thus, y is not a branch point.
- The point $y \in \Delta_t$, for some $\Delta_t \neq \Delta_g$, with $t \notin \langle g \rangle$. In this case, the fibre $\pi^{-1}(y)$ consists of $|G/\langle g, t \rangle| = p^{k-2}$ of X, which are smooth for X by [**39**, Prop. 3.1]. Consequently, the fibre $\widetilde{\pi}^{-1}(y)$ consists of p^{k-2} points, making y a branch point for $\widetilde{\pi}_{|\widetilde{\Delta_g}}$.

  To determine the local monodromy of y, let $x \in \widetilde{X}$ lie over y. Since $\rho(x)$ is a smooth point for X and the intersection of Δ_g and Δ_t is transversal, there exists an open neighborhood of x with coordinates (z_0, z_1) where $\widetilde{\Delta_g} = (z_0 = 0)$, $\widetilde{\Delta_t} = (z_1 = 0)$. The action is $g\colon (z_0, z_1) \mapsto (\exp\left(\frac{2\pi i}{p}\right) z_0, z_1)$ and $t\colon (z_0, z_1) \mapsto (z_0, \exp\left(\frac{2\pi i}{p}\right) z_1)$. The restriction to $\widetilde{\Delta_g}$ shows that $[t]$ acts locally as $(0, z_1) \mapsto (0, \exp\left(\frac{2\pi i}{p}\right) z_1)$, identifying $[t]$ as the local monodromy.

□

Remark 8. Notice that in case $k = 1$, hence the group $G = \mathbb{Z}/p$, then Δ_g is isomorphic to $\widetilde{\Delta}_g$.

1.3. **Equations of (p,k)-planes in weighted projective spaces.** Let us consider a collection of building data $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ and $\{D_g\}_{g\in G}$ on $\mathbb{P}^2$, with $G=(\mathbb{Z}/p)^k$. We conclude this section by constructing the associated (p,k)-plane $\pi\colon X\longrightarrow\mathbb{P}^2$, realizing X as a subvariety of a suitable weighted projective space, following the approach described in [**16**].

Let $f_g(x_0,x_1,x_2)$ be an homogeneous polynomial of degree d_g in the variables x_0,x_1,x_2, whose zero locus is the support of the divisor D_g on $\mathbb{P}^2$.

We consider the weighted projective space $\mathbb{P}\left(1,1,1,(l_\chi)_{\chi\in G^*\setminus 1}\right)$ whose the first three variables are x_0,x_1,x_2, and the remaining ones are $(y_\chi)_{\chi\in G^*\setminus 1}$ (e.g., for $(p,k)=(2,2)$, we consider the weighted projective space $\mathbb{P}(1,1,1,l_{\epsilon_1},l_{\epsilon_2},l_{\epsilon_1+\epsilon_2})$ with coordinates $(x_0,x_1,x_2,\,y_{\epsilon_1},y_{\epsilon_2},y_{\epsilon_1+\epsilon_2})$).

Notice that the projection on the first three variables $\mathrm{pr}\colon\mathbb{P}\left(1,1,1,(l_\chi)_{\chi\in G^*\setminus 1}\right)\setminus\{x_0=x_1=x_2=0\}\longrightarrow\mathbb{P}^2$ is the vector bundle of $\mathbb{P}^2$ of rank p^k-1 corresponding to $\bigoplus_{\chi\in G^*\setminus 1}\mathcal{O}_{\mathbb{P}^2}(l_\chi)$.

We have a natural action of the group $G=(\mathbb{Z}/p)^k$:

$$g\cdot(x_0,x_1,x_2,(y_\chi)_{\chi\in G^*\setminus 1})=(x_0,x_1,x_2,(\chi(g)\cdot y_\chi)_{\chi\in G^*\setminus 1}),\qquad g\in G.$$

We define $X\subseteq\mathbb{P}\left(1,1,1,(l_\chi)_{\chi\in G^*\setminus 1}\right)\setminus\{x_0=x_1=x_2=0\}$ in the following way

$$X\coloneqq\left\{y_\chi y_\eta=y_{\chi\eta}\cdot\prod_{g\in G}f_g^{\left\lfloor\frac{\langle\chi,g\rangle+\langle\eta,g\rangle}{p}\right\rfloor},\ \text{ for all }\chi,\eta\in G^*\setminus 1\right\}.$$

The action of G naturally extends to X and $\mathrm{pr}_{|X}\colon X\longrightarrow\mathbb{P}^2$ is by construction a (p,k)-plane with building data $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ and $\{D_g\}_{g\in G}$.

We refer to Example 7.1 for an explicit application, where we find the equations in a suitable weighted projective space for a (p,k)-plane X belonging to one of the families H-(2,3) in Table 18.

The equations for the other SNC (p,k)-planes satisfying Property ★ and belonging to one of the families of our classification can be obtained similarly.

Remark 9. We point out that using these equations we are able to give X defined over $\mathbb{Q}$.

2. The classification of SNC (p,k)-planes with Property ★

We present here the tools and constraints that will be used in the classification of SNC (p,k)-planes

$$\pi\colon X\longrightarrow\mathbb{P}^2$$

with p prime number and $k\geq 1$ satisfying the extra condition: Property ★.

In particular, we derive conditions on the branch divisors and the associated numerical invariants that will play a central role in the classification.

The number of subgroups in an elementary abelian p-group of rank k is the same as the number of subspaces of a k-dimensional vector space over the finite field with p elements. This number can be computed using the Gaussian binomial coefficient

$$\begin{bmatrix}k\\m\end{bmatrix}_p=\prod_{i=0}^{m-1}\frac{p^k-p^i}{p^m-p^i},$$

where m is the dimension of the subspaces to be counted. In particular, subgroups of order p correspond to subspaces of dimension 1, and their number is

$$\phi_{p,k}\coloneqq\frac{p^k-1}{p-1}.$$

Moreover, the number of subgroups of index p in $G\cong(\mathbb{Z}/p)^k$ is equal to the number of subgroups of order p simply by projective duality, hence also $\phi_{p,k}$.

For each subgroup $H \leq G$ there is an intermediate quotient $Y_H \coloneqq X/H$. We focus on the $\phi_{p,k}$ maximal subgroups $H \cong (\mathbb{Z}/p)^{k-1} \subset G$ and their intermediate covers $Y_H \longrightarrow \mathbb{P}^2$ with Galois group $G/H \cong \mathbb{Z}/p$. For these intermediate covers it holds the following lemma.

Lemma 2.1. *Let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane with building data $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ and $\{D_g\}_{g\in G}$, so branched along the divisor $D = \sum_{g\in G} D_g$. Let $H \leq G$ be a subgroup of index p, and let $\epsilon \in G^*$ be a nontrivial character such that $\ker\epsilon = H$. Thus, we identify G/H with $\mathbb{Z}/p$ via the character ϵ. Then the induced quotient map*

$$\pi_H\colon X/H = Y_H \longrightarrow \mathbb{P}^2$$

is a cyclic $G/H \cong (\mathbb{Z}/p)$-cover whose branch divisor is again SNC and can be expressed as

$$D^{(H)} \coloneqq \sum_{\substack{g\in G\\ \langle\epsilon,g\rangle\neq 0}} D_g.$$

Moreover, the building data $\{\mathcal{L}^{(H)}_{\overline{j}}\}_{\overline{j}\in\mathbb{Z}/p}$ and $\{D^{(H)}_{\overline{j}}\}_{\overline{j}\in\mathbb{Z}/p}$ of such intermediate cover are

$$L^{(H)}_{\overline{j}} \equiv_{lin} L_{\epsilon^j}, \qquad \text{and} \qquad D^{(H)}_{\overline{j}} = \sum_{\substack{g\in G\\ \langle\epsilon,g\rangle\equiv j \pmod p}} D_g, \qquad \overline{j}\in\mathbb{Z}/p.$$

Proof. Since H has index p in G and G is elementary abelian, there exists a nontrivial character $\epsilon \in G^*$ with $\ker\epsilon = H$, and the quotient group G/H is naturally identified with $\mathbb{Z}/p$ via the homomorphism $\epsilon\colon G \longrightarrow \mathbb{C}^\times$. The quotient map $\pi_H\colon X/H \longrightarrow \mathbb{P}^2$ is therefore a cyclic $(\mathbb{Z}/p)$-cover. Moreover the branch divisor of this cover is given by a subset of the components of the divisor D, hence is again SNC.

A component D_g contributes to the branch locus of π_H if and only if the image of g in G/H is nontrivial, equivalently if $\epsilon(g) \neq 1$, namely $\langle\epsilon, g\rangle \neq 0$. Hence the branch divisor of π_H is the divisor $D^{(H)}$ defined above. Similarly, a straightforward local computation shows that the divisor D_g has local monodromy $\overline{j} \in (\mathbb{Z}/p)$ for the intermediate cover π_H if and only if $\langle\epsilon, g\rangle \equiv j \mod p$. Thus, the divisor $D^{(H)}_{\overline{j}}$ with local monodromy $\overline{j} \in \mathbb{Z}/p$ is exactly that of the statement.

Finally, applying Pardini's equation for the character $\overline{i} \in (\mathbb{Z}/p)$ yields

$$\begin{aligned} pL^{(H)}_{\overline{i}} &\equiv_{lin} \sum_{\overline{j}\in\mathbb{Z}/p} [i\cdot j]_p D^{(H)}_j \\ &\equiv_{lin} \sum_{\overline{j}\in\mathbb{Z}/p} \sum_{\substack{g\in G\\ \langle\epsilon,g\rangle\equiv j \pmod p}} [i\cdot j]_p D_g \\ &\equiv_{lin} \sum_{g\in G} \langle\epsilon^i, g\rangle D_g \\ &\equiv_{lin} pL_{\epsilon^i}, \end{aligned}$$

where the latter is obtained from Pardini's equation for the character $\epsilon^i \in G^*$. Thus, $pL^{(H)}_{\overline{i}} \equiv_{lin} pL_{\epsilon^i}$ and since $\mathbb{P}^2$ has a Picard group without torsion, then $L^{(H)}_{\overline{i}} \equiv_{lin} L_{\epsilon^i}$. □

By considering the Hasse diagram of subgroups of G and reversing it to reflect the covering structure, we obtain Diagram 1. In what follows we analyze the bottom row, i.e. the covers lying immediately above $\mathbb{P}^2$.

First of all, we need to translate the geometric Property ★ in terms of the building data $\{D_g\}_{g\in G}$ and $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ of the cover π.

In the projective plane $\mathbb{P}^2$, we denote the degrees of the line bundles $\mathcal{L}_\chi$ and divisors D_g by

$$l_\chi \coloneqq \deg \mathcal{L}_\chi, \qquad \chi \in G^*, \qquad \text{and} \qquad d_g \coloneqq \deg D_g, \qquad g \in G.$$

Proposition 2.2. *Let $\pi\colon X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane with $k \geq 1$ and building data $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ and $\{D_g\}_{g\in G}$. Then Property ★ holds if and only if the degrees $\{l_\chi\}_{\chi\in G^*}$ and $\{d_g\}_{g\in G}$ verify the following conditions:*

(1) *$1 \leq l_\chi \leq 3$ for each non trivial character $\chi \in G^*$;*
(2) *there exists $\chi \in G^*$ such that $l_\chi = 3$;*
(3)
- *if $p = 2$, for each non-trivial character $\chi \in G^*$ with $l_\chi = 3$, we have $\sum_{g\notin\ker\chi} d_g = 6$;*
- *if $p = 3$, for each (proper) cyclic subgroup $\langle\chi\rangle \leq G^*$ there exists at most one χ such that $l_\chi = 3$, in which case $l_{\chi^2} = 2$, and $\sum_{g\notin\ker\chi} d_g = 5$;*
- *if $p \geq 5$, for each (proper) cyclic subgroup $\langle\chi\rangle \leq G^*$ there exists at most one χ such that $l_\chi = 3$, in which case $l_{\chi^{p-1}} = 1$, $l_{\chi^i} = 2$ for any $i \neq 0, 1, p-1$, and $\sum_{g\notin\ker\chi} d_g = 4$.*

Proof. Let $\chi \in G^*$ be a non-trivial character of G, and let $H \coloneqq \ker\chi$ be its subgroup of index p in G. By Lemma 2.1 the intermediate quotient $\pi_H\colon Y_{\ker\chi} \longrightarrow \mathbb{P}^2$ is a SNC $(p,1)$-plane with building data $\{D^{(H)}_{\bar{j}}\}_{\bar{j}\in\mathbb{Z}/p}$ and $\{\mathcal{L}_{\chi^i}\}_{i\in\{0,\dots,p-1\}}$. Therefore, we can apply Theorem 1.14 to π_H:

$$p_g(Y_{\ker\chi}) = \sum_{i=1}^{p-1} h^0\left(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3 + l_{\chi^i})\right) = \sum_{i=1}^{p-1} \frac{(l_{\chi^i}-1)(l_{\chi^i}-2)}{2}. \tag{20}$$

Assuming Property ★, for any nontrivial character $\chi \in G^*$ the induced $(p,1)$-plane $Y_{\ker\chi}$ necessarily has geometric genus at most one. Equation (20) implies $l_{\chi^i} \leq 3$ for all i. Specifically, if $Y_{\ker\chi}$ has geometric genus one, exactly one l_{χ^j} equals to 3, while the others satisfy $l_{\chi^i} \leq 2$, for $j \neq i$. Furthermore, $l_\chi \geq 1$ as X is connected, see Remark 2. This proves (1). Moreover, (2) directly follows from Property ★, which requires that at least one of the intermediate $(p,1)$-planes $Y_{\ker\chi}$ has geometric genus equal to one.

Let us prove point (3) with a case-by-case analysis depending on the prime p.

First, assume $p = 2$, and $l_\chi = 3$. Then, by Lemma 1.5 applied for the character χ, we have

$$\sum_{g\notin\ker\chi} d_g = 2l_\chi = 6.$$

Next, assume $p = 3$ and $l_\chi = 3$, so $l_{\chi^2} \leq 2$. From Pardini's equations (3), we have

$$l_\chi + l_{\chi^2} = \sum_{g\notin\ker\chi} d_g, \qquad \text{and}$$

$$6 = l_\chi + l_\chi = l_{\chi^2} + \sum_{g\in G} \left\lfloor \frac{2\langle\chi, g\rangle}{3} \right\rfloor d_g \leq l_{\chi^2} + \sum_{g\notin\ker\chi} d_g.$$

Combining these, we obtain

$$6 \leq l_{\chi^2} + \sum_{g\notin\ker\chi} d_g = 3 + 2l_{\chi^2},$$

which forces $l_{\chi^2} \geq \frac{3}{2}$, hence $l_{\chi^2} = 2$, and consequently $\sum_{g\notin\ker\chi} d_g = 5$.

Finally, assume $p \geq 5$ and $l_\chi = 3$, so $l_{\chi^i} \leq 2$ for each $i \neq 1$. By contradiction, if $l_{\chi^{p-1}} = 2$, then Pardini's equations 3 imply

$$l_{\chi^2} + l_{\chi^{p-2}} = \sum_{g\notin\ker\chi} d_g = l_\chi + l_{\chi^{p-1}} \geq 5,$$

which is not possible as both $l_{\chi^2}, l_{\chi^{p-2}} \leq 2$. Thus, $l_{\chi^{p-1}} = 1$ and $\sum_{g \notin \ker \chi} d_g = 4$. For $2 \leq i \leq p-2$, the characters χ^i and χ^{p-i} are distinct from χ, χ^{p-1}. Using Pardini equations again, we find $l_{\chi^i} + l_{\chi^{p-i}} = \sum_{g \notin \ker \chi} d_g = 4$, which forces $l_{\chi^i} = l_{\chi^{p-i}} = 2$. This proves point (3).

Conversely, by Equation 20, points $(1) - (3)$ imply Property ★. □

Let us discuss the classification, distinguishing between the cases $k = 1$ and $k \geq 2$.

Remark 10. For $k = 1$, we observe that SNC $(p, 1)$-planes $\pi \colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★ have geometric genus $p_g(X) = 1$. In general, the surface X need not be smooth; indeed, as discussed in the previous section, it is a normal surface with only cyclic quotient singularities.

In this situation, the minimal model of X is a K3 surface, as will be shown in Section 3, Theorem 3.7. Accordingly, with a slight abuse of notation, we shall refer to X itself as a *K3 surface* whenever its minimal model is a K3 surface. Such covers will be also classified in Section 3, Proposition 3.2

From now on we assume $k \geq 2$. Let $\pi \colon X \longrightarrow \mathbb{P}^2$ be a SNC (p, k)-plane. By Remark 4, the set of local monodromies

$$\{\, g \in G \mid d_g \neq 0 \,\}$$

generates the whole group $G \cong (\mathbb{Z}/p)^k$.

We denote from now on by $e_1, \ldots, e_k$ a set of generators of G such that $d_{e_i} \geq 1$ for each $i = 1, \ldots, k$, and by $\epsilon_1, \ldots, \epsilon_k$ the corresponding dual set of generators of G^*, i.e. those characters for which $\langle \epsilon_i, e_j \rangle = \delta_{ij}$, for all $i, j = 1, \ldots, k$, where δ_{ij} is the Kronecker delta.

Since the rank of G is k, this immediately yields the rough inequalities

$$2 \leq k \leq \big|\{\, g \in G \mid d_g \neq 0 \,\}\big| \leq \sum_{g \in G} d_g. \tag{21}$$

We now show that the numerical constraints (21) strongly restrict the possible building data of a SNC (p, k)-plane. In particular, either the cover belongs to a very special and explicit class, or the total degree of the branch locus must satisfy stronger lower bounds than (21).

Proposition 2.3. *Let $\pi \colon X \longrightarrow \mathbb{P}^2$ be a SNC (p, k)-plane with building data $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ and $\{D_g\}_{g \in G}$. Then one of the following holds:*

(1) *X is a smooth (p, k)-plane whose building data satisfy*

$$d_{e_1} = \cdots = d_{e_k} = d_{(p-1)\sum_{i=1}^k e_i} = 1, \qquad \text{and} \qquad d_g = 0 \text{ for all other } g \in G.$$

In this case, X satisfies Property ★ if and only if

$$(p, k) \in \{(2, 5), (2, 6), (3, 4), (5, 3)\}.$$

(2) *The branch data satisfy the numerical inequalities*

$$k + 2 \leq \sum_{g \in G} d_g \qquad \text{and} \qquad k \leq \big|\{\, g \in G \mid d_g \neq 0 \,\}\big|.$$

Proof. We first consider the extremal situation where the rough inequality (21) is sharp, that is, when $\sum_{g \in G} d_g = k$. Then (21) forces

$$k = \big|\{\, g \in G \mid d_g \neq 0 \,\}\big| = \sum_{g \in G} d_g.$$

In particular, we must have

$$d_{e_1} = \cdots = d_{e_k} = 1, \qquad d_g = 0 \text{ for all } g \in G \setminus \{e_1, \ldots, e_k\}.$$

By Lemma 1.5 applied to ϵ_1, we obtain

$$p \cdot L_{\epsilon_1} \equiv_{lin} \sum_{g \in G} \langle \epsilon_1, g \rangle \, D_g = D_{e_1},$$

Passing to degrees on $\mathbb{P}^2$ yields

$$pl_{\epsilon_1} = d_{e_1} = 1,$$

a contradiction.

Thus, by (21), we must have $k+1 \leq \sum_{g \in G} d_g$. We now consider the case $\sum_{g \in G} d_g = k+1$. Again from (21), there are two possibilities for the cardinality $\big|\{\, g \in G \mid d_g \neq 0 \,\}\big|$, namely either k or $k+1$. If it was equal to k, we would obtain a contradiction by the same argument as in the previous case.

We therefore assume that

$$\big|\{\, g \in G \mid d_g \neq 0 \,\}\big| = k+1,$$

that is, there exists exactly another element h other than $e_1, \ldots, e_k$ such that

$$d_{e_1} = \cdots = d_{e_k} = d_h = 1, \qquad \text{and} \qquad d_g = 0 \text{ for all } g \in G \setminus \{e_1, \ldots, e_k, h\}.$$

By Lemma 1.5, for each $1 \leq j \leq k$ we have

$$p \, L_{\epsilon_j} = \sum_{i=1}^{k} \langle \epsilon_j, e_i \rangle \, D_{e_i} + \langle \epsilon_j, h \rangle \, D_h,$$

which implies passing to the degrees

$$pl_{\epsilon_j} = 1 + \langle \epsilon_j, h \rangle \leq p,$$

thus

$$l_{\epsilon_j} = 1 \qquad \text{and} \qquad \langle \epsilon_j, h \rangle = p-1, \quad \text{for each } j = 1, \ldots, k.$$

It follows that there is only one element h for which Pardini's equations (5) hold, and it is equal to

$$h = \sum_{j=1}^{k} \langle \epsilon_j, h \rangle \, e_j = (p-1) \sum_{j=1}^{k} e_j,$$

as claimed. This yields case (1) of Proposition 2.3. If this situation does not occur, then necessarily $\sum_{g \in G} d_g \geq k+2$, which is case (2).

Notice that in case (1), the surface X is smooth, since the branch divisor consists of $k+1$ distinct lines in $\mathbb{P}^2$ with simple normal crossings satisfying Condition (10).

We now turn to a detailed analysis of the exceptional situation described in case (1), under the additional assumption that X satisfies Property $\bigstar$. In order to exploit the numerical constraints, let us consider the following character

$$\chi \coloneqq (p-1) \sum_{i=1}^{k} \epsilon_i.$$

A direct computation of Pardini's equation (5) for the character χ yields

$$pl_\chi = k(p-1) + [k]_p. \tag{22}$$

Moreover, since X satisfies Property $\bigstar$, we have $l_\chi \leq 3$, and hence

$$3p \geq p \, l_\chi = k(p-1) + [k]_p \geq k(p-1),$$

which immediately implies

$$k \leq 3 + \frac{3}{p-1}. \tag{23}$$

Let $\eta \in G^*$ be a character such that $l_\eta = 3$. By the bound (23) on k and Proposition 2.2 (3), we obtain

$$4 + \frac{3}{p-1} \;\geq\; k+1 = \sum_{g \in G} d_g \;\geq\; \sum_{g \notin \ker \eta} d_g = \begin{cases} 6 & \text{if } p = 2, \\ 5 & \text{if } p = 3, \\ 4 & \text{if } p \geq 5. \end{cases}$$

It follows that the only possible pairs (p,k) are

$$(p,k) \in \{(2,5),(2,6),(3,4)\} \cup \{(p,3) \mid p \geq 5\}.$$

One can directly verify that $(2,5)$, $(2,6)$, $(3,4)$ satisfy Property ★.

Consider now the case $(p,3)$ with $p \geq 5$. In this situation, the building data are given by

$$d_{e_1} = d_{e_2} = d_{e_3} = d_{(p-1)\sum_{i=1}^3 e_i} = 1, \qquad d_g = 0 \text{ for all remaining } g \in G.$$

Thus, from Equation (22), we have for $\chi = (p-1)\sum_{i=1}^3 \epsilon_i$ that

$$l_\chi = \frac{3(p-1) + [3]_p}{p} = 3,$$

so that $p_g(Y_H) = 1$, where $H := \ker \chi$.

Now, using the notation introduced in Lemma 2.1 we have that the building data of the intermediate $(\mathbb{Z}/p)$-cover $\pi_H \colon Y_H \longrightarrow \mathbb{P}^2$ are

$$D_{\bar{j}}^{(H)} = \sum_{\langle \chi, g \rangle \equiv j \,(\mathrm{mod}\, p)} D_g, \qquad \text{for} \quad \bar{j} \in \mathbb{Z}/p.$$

In the present case, $\langle \chi, e_i \rangle = p-1$ for $i = 1,2,3$, while $\langle \chi, (p-1)\sum_{i=1}^3 e_i \rangle = 3$. Therefore,

$$D_{\overline{p-1}}^{(H)} = D_{e_1} + D_{e_2} + D_{e_3}, \qquad \text{and} \qquad D_{\bar{3}}^{(H)} = D_{(p-1)\sum_{i=1}^3 e_i},$$

and $D_{\bar{j}}^{(H)} = 0$ for all other $\bar{j} \in \mathbb{Z}/p$. According to Proposition 3.2, the unique SNC $(p,1)$-plane of geometric genus one with these building data occurs for $p = 5$. More precisely, this is the case $A-(5,1)(3)$ of Table 11. Consequently, the only pair $(p,3)$ with $p \geq 5$ satisfying Property ★ is $(5,3)$. □

The (p,k)-planes appearing in the previous Proposition 2.3(1) form a distinguished class that was extensively studied by Hirzebruch [**26**]. Therefore, we introduce the following terminology.

Definition 2.4. *A (p,k)-plane $\pi \colon X \longrightarrow \mathbb{P}^2$ is said to be* of Hirzebruch type *if its building data satisfy the conditions of Proposition 2.3(1).*

The next proposition provides a decisive bound on the parameters (p,k) and the degrees of the components of the branch divisor. In particular, it implies that only finitely many families of (p,k)-planes can satisfy Property ★, making a complete classification possible.

Proposition 2.5. *Let $\pi \colon X \longrightarrow \mathbb{P}^2$ be an SNC (p,k)-plane with $k \geq 2$ and building data $\{\mathcal{L}_\chi\}_{\chi \in G^*}$ and $\{D_g\}_{g \in G}$, not of Hirzebruch type. If π satisfies Property ★, then the following conditions hold:*

(1) *The parameters (p,k) satisfy*

$$2 \leq p \leq 19 \qquad \textit{and} \qquad 2 \leq k \leq 3 + \frac{3}{p-1}.$$

(2) *The number of nonzero branch divisors D_g satisfies*

$$\big|\{\, g \in G \mid d_g \neq 0 \,\}\big| \geq k.$$

(3) *For every $g \in G$, one has*

$$d_g \leq 3 + \frac{3}{p-1}.$$

Moreover, if $(p,k) = (2,6)$, then $d_g \in \{0,1\}$ for all $g \in G$.

(4) *The total degree of the branch divisor satisfies the following bounds, depending on p:*

$p = 2$: *either $(p,k) = (2,2)$ and*

$$d_{e_1} = d_{e_2} = d_{e_1+e_2} = 3,$$

*in which case X is a SNC degeneration of the Garbagnati surface of type G3, see [**28**], or else*

$$\max\{6, k+2\} \leq \sum_{g \in G} d_g \leq 12 - 2^{4-k}.$$

$p = 3$:

$$\max\{5, k+2\} \leq \sum_{g \in G} d_g \leq \frac{5}{2}\left(3 - \frac{1}{3^{k-1}}\right).$$

$p \geq 5$: $\sum_{g \in G} d_g = 4$ *and* $k = 2$.

Proof. The bound $2 \leq p \leq 19$ follows directly from Property ★ since the (p,k)-plane would admit an intermediate SNC $(p,1)$-plane with geometric genus one, and we classified them in Proposition 3.2.

Next, we establish a bound on k. Applying Lemma 1.5 to the character $(p-1)\sum_{i=1}^{k} \epsilon_i$, we obtain

$$\begin{aligned} 3p &\geq p\, l_{(p-1)\sum_{i=1}^{k} \epsilon_i} \\ &\geq (p-1)\sum_{i=1}^{k} d_{e_i} + \sum_{g \notin \ker(\sum_{i=1}^{k} \epsilon_i)} d_g \\ &\geq (p-1)k. \end{aligned} \tag{24}$$

It follows that

$$2 \leq k \leq 3 + \frac{3}{p-1}.$$

This proves (1). We have already observed (2) in (21). So let us work on (3).

We first address the exceptional subcase $(p,k) = (2,6)$. In this case the inequality (24) reads

$$6 \geq 2\, l_{\sum_{i=1}^{6} \epsilon_i} = \sum_{g \notin \ker(\sum_{i=1}^{6} \epsilon_i)} d_g \geq \sum_{i=1}^{6} d_{e_i} \geq 6,$$

hence all inequalities are equalities and in particular $d_{e_1} = \cdots = d_{e_k} = 1$, and if $d_h \neq 0$, then either h is one among $e_1, \ldots, e_k$, or we have

$$\Big(\sum_{i=1}^{6} \epsilon_i\Big)(h) = 0.$$

Equivalently, h is the sum of an even number of basis elements: $h = e_{i_1} + \cdots + e_{i_{2j}}$.

Fix such a $h \neq 0$ and choose an index i_1 appearing in its expression. Set

$$\epsilon \coloneqq \epsilon_{i_1} + \sum_{j=1}^{6} \epsilon_j \in G^*.$$

Then $\epsilon(e_{i_1}) = 1$, while $\epsilon(e_i) = -1$ for every $i \neq i_1$. Moreover, since h is a sum of an even number of basis elements containing e_{i_1}, we have $\epsilon(h) = -1$, hence $h \notin \ker(\epsilon)$. Applying Lemma 1.5 to ϵ we obtain

$$2\, l_\epsilon = \sum_{h \notin \ker \epsilon} d_h \geq \sum_{i \neq i_1} d_{e_i} + d_h \geq 5 + d_h.$$

Since Property ★ gives $l_\epsilon \leq 3$, we conclude

$$6 \geq 2\, l_\epsilon \geq 5 + d_h,$$

and hence $d_h \leq 1$. In particular, for $(p, k) = (2, 6)$ every nonzero branch component has degree $d_g \in \{0, 1\}$.

Let us turn to the proof of (3). We now bound the degrees d_g by a similar argument. Fix $g \in G$, and choose a character $\chi \in G^*$ such that $\langle \chi, g \rangle = p - 1$. By Lemma 1.5, we have

$$3p \geq p\, l_\chi = \sum_{h \in G} \langle \chi, h \rangle\, d_h \geq (p-1) d_g,$$

and therefore

$$d_g \leq 3 + \frac{3}{p-1}.$$

To prove (4), let us assume first that $l_\eta = 3$ for any non trivial character $\eta \in G^*$. For any non-zero $g \in G$, Corollary 1.8 yields

$$\begin{aligned} \frac{p^{k-1}(p-1)}{2} \sum_{h \in \langle g \rangle} d_h &= 3|G \setminus \ker g| - (p-1)\,(0 + 3(|\ker g| - 1)) \\ &= 3p^k - 3p^{k-1} - 3(p-1)p^{k-1} + 3(p-1) \\ &= 3(p-1), \end{aligned}$$

which implies

$$\sum_{h \in \langle g \rangle} d_h = \frac{6}{p^{k-1}}.$$

Since the left-hand side is an integer and $k \geq 2$, this forces $(p, k) = (2, 2)$ or $(3, 2)$. The case $(2, 2)$ gives $d_g = 3$ for all $g \neq 0$, which are the building data defining either the Garbagnati surface of type G3 or its degenerations. Conversely, $(p, k) = (3, 2)$ is automatically excluded by Proposition 2.2 because for $p = 3$ any character χ with $l_\chi = 3$ satisfies $l_{\chi^2} = 2$ whilst we were assuming that all non-trivial characters $\eta \in G^*$ have $l_\eta = 3$.

Let us conclude the proof of (4) by assuming the converse, namely that there exists at least one non-trivial character $\chi \in G^*$ such that $l_\chi \leq 2$.

For $p = 2$, we have

$$\sum_{\chi \in G^*} l_\chi \leq 0 + 2 + 3(2^k - 2) = 4\big(3 \cdot 2^{k-2} - 1\big).$$

For $p = 3$, Proposition 2.2 point (3) provides more information: for each proper cyclic subgroup $\langle \chi \rangle \leq G^*$ with $l_\chi = 3$, we have $l_{\chi^2} = 2$. Thus,

$$\sum_{t=0}^{2} l_{\chi^t} \leq 0 + 2 + 3 = 5,$$

and we can conclude

$$\sum_{\chi \in G^*} l_\chi = 0 + \sum_{\langle \chi \rangle \in \mathbb{P}((\mathbb{Z}/3^k)} \left(\sum_{\eta \in \langle \chi \rangle} l_\eta \right) \leq 5 \cdot \phi_{3,k} = 5 \cdot \frac{3^k - 1}{2}.$$

Similarly, for $p \geq 5$, Proposition 2.2 point (3) implies that each proper subgroup $\langle\chi\rangle \leq G^*$ with $l_\chi = 3$ has $l_{\chi^{p-1}} = 1$ and $l_{\chi^i} = 2$ otherwise; hence

$$\sum_{i=0}^{p-1} l_{\chi^i} \leq 0 + 1 + 3 + 2(p-3) = 2(p-1).$$

Consequently,

$$\sum_{\chi \in G^*} l_\chi = 0 + \sum_{\langle\chi\rangle \in \mathbb{P}((\mathbb{Z}/3)^k)} \left(\sum_{i=0}^{p-1} l_{\chi^i} \right) \leq 2 \cdot \phi_{p,k} \cdot (p-1) = 2(p^k - 1).$$

Finally, since X is assumed to be not of Hirzebruch type, Proposition 2.3 implies $k+2 \leq \sum_g d_g$ and $k \leq |\{g \in G | d_g \neq 0\}|$. Applying Theorem 1.7(7), we obtain

$$\sum_{g \in G} d_g = \frac{2}{p^{k-1}(p-1)} \sum_{\chi \in G^*} l_\chi \leq \begin{cases} \dfrac{2}{2^{k-1}} \cdot 4\big(3 \cdot 2^{k-2} - 1\big) & \text{if } p = 2, \\ \dfrac{2}{3^{k-1} \cdot 2} \cdot 5 \cdot \dfrac{3^k - 1}{2} & \text{if } p = 3, \\ \dfrac{2}{p^{k-1}(p-1)} \cdot 2(p^k - 1) & \text{if } p \geq 5. \end{cases} \tag{25}$$

In particular,

$$k + 2 \leq \sum_{g \in G} d_g \leq \begin{cases} 12 - 2^{\,4-k} & \text{if } p = 2, \\ \dfrac{5}{2}\left(3 - \dfrac{1}{3^{k-1}}\right) & \text{if } p = 3, \\ 4 + \dfrac{4}{p-1} - \dfrac{4}{p^{k-1}(p-1)} & \text{if } p \geq 5. \end{cases} \tag{26}$$

Note that for $p \geq 5$, since $k \geq 2$, then $\sum_g d_g$ is forced to be equal to 4, which implies directly $k = 2$. Combining (26) with the constraints from Proposition 2.2 on $\sum_g d_g$, we obtain point (4). □

This completes the numerical analysis of SNC (p,k)-planes satisfying Property ★. We are now in a position to carry out a classification.

A classification of SNC $(p,1)$-planes satisfying Property ★, so those with geometric genus one, will be achieved in Section 3, Proposition 3.2. Thus, we only consider the case $k \geq 2$. In this case, Proposition 2.5 provides explicit numerical constraints on p, k, and $\sum_{g \in G} d_g$ that reduce the problem to a finite and computationally manageable set of cases.

We emphasize that Proposition 2.5 plays a crucial role in controlling the computational complexity. Without these bounds, for large values of p, k, and $\sum_g d_g$, there could be millions of distinct ways to label the irreducible components of the branch locus of a cover so that the reduced Pardini equations of Theorem 1.4 are satisfied. Moreover, many of these combinatorial configurations define isomorphic SNC (p,k)-planes. Indeed, an automorphism of the group $(\mathbb{Z}/p)^k$ may send the building data of one cover to that of another, yielding isomorphic $(\mathbb{Z}/p)^k$-coverings of $\mathbb{P}^2$.

The classification is implemented algorithmically through a single fundamental routine, which generates all admissible building data. We used the computational algebra system `MAGMA` [**5**]. All routines, functions, auxiliary scripts used in the implementation of the algorithm, and a commented version of the outputs are available online here

`MAGMA` scripts and results repository

where the full source code and documentation can be found.

The algorithm proceeds as follows. The function

```
Building_Data(card,d,G)
```

takes as input a elementary p-abelian group $G \cong (\mathbb{Z}/p)^k$ and two positive integers `card`, and `d`. It returns a list of vectors

$$[(g, d_g) \colon g \in G,\ d_g \neq 0]$$

satisfying the following conditions:

(1) for every character $\chi \in G^*$, the numbers

$$l_\chi \coloneqq \frac{1}{p} \sum_{g \in G} \langle \chi, g \rangle d_g$$

are integers;

(2) the collections $\{l_\chi\}_{\chi \in G^*}$ and $\{d_g\}_{g \in G}$ satisfy the numerical conditions (1), (2), and (3) of Proposition 2.2;

(3) $\big|\{\, g \in G \mid d_g \neq 0 \,\}\big| =$ `card`;

(4) $\sum_{g \in G} d_g =$ `d`.

As an explicit example, we apply the function `Building_Data(card,d,G)` to reproduce cases $G - (2,3)1)$ and $H - (2,3)1)$ of Table 18, corresponding to $(p,k) = (2,3)$.

```
G:=AbelianGroup([2,2,2]);
Building_Data(5,8,G);
{
    [
                <G.3, 3>,
                <G.1 + G.2, 1>,
                <G.2, 2>,
                <G.1, 1>,
                <G.2 + G.3, 1>
    ]
    [
               <G.3, 1>,
               <G.1 + G.2 + G.3, 1>,
               <G.2, 1>,
               <G.1, 1>,
               <G.2 + G.3, 4>
    ]
}
```

We refer to Section 7 where we study the example $H - (2,3)1)$ in full detail.

Given such an output vector, one then chooses a collection of (possibly not prime) divisors $\{D_g\}_{g \in G}$ on $\mathbb{P}^2$ such that $\deg D_g = d_g$ and the divisor

$$D \coloneqq \sum_{g \in G} D_g$$

is reduced, has smooth irreducible components, and has simple normal crossings. By the properties (1)–(4) imposed above on the degrees $\{d_g\}_{g \in G}$, the divisors $\{D_g\}_{g \in G}$ satisfy the reduced Pardini's equations (cf. Theorem 1.4 and the Remark 3), and therefore define a SNC (p,k)-plane for which Property ★ holds, from Proposition 2.2.

To avoid redundancy, the function filters out vectors that define isomorphic coverings. Consequently, the output consists of vectors defining SNC (p,k)-planes that are pairwise non-isomorphic as $(\mathbb{Z}/p)^k$-coverings of $\mathbb{P}^2$.

Remark 11. The relation of being isomorphic as $(\mathbb{Z}/p)^k$-coverings of $\mathbb{P}^2$ is stronger than the standard notion of isomorphism between surfaces. Consequently, it is possible for two SNC (p,k)-planes to be isomorphic as surfaces even if they are not isomorphic as $(\mathbb{Z}/p)^k$-coverings of $\mathbb{P}^2$.

The complete classification of SNC (p,k)-planes satisfying Property ★ and with $k \geq 2$ is attained using the function `Building_Data(card,d,G)` for each possible `G`, `card`, and `d` according to the finite list of possibilities established in Proposition 2.5.
More precisely, we have that $2 \leq p \leq 19$, and for $5 \leq p \leq 19$, then Proposition 2.5(2) implies that $k = 2$ and $\sum_{g\in G} d_g = 4$. Consequently, there are only three possible values for

$$\texttt{card} = \big|\{\, g \in G \mid d_g \neq 0 \,\}\big|,$$

namely 2, 3, or 4. Running the function `Building_Data(card,4,AbelianGroup([p,p]))` for each p and each $\texttt{card} \in \{2,3,4\}$, we find that admissible cases occur only for $p = 5$ and $p = 7$.

For the rest of cases, namely $p \in \{2,3\}$, there is greater flexibility in the choice of the parameters k, $\big|\{g \in G \mid d_g \neq 0\}\big|$, and $\sum_g d_g$, although they remain bounded by Proposition 2.5, point (2). We therefore ran the function for all valid combinations of these parameters, obtaining a more extensive list of cases.

The resulting SNC (p,k)-planes, including those of Hirzebruch type (cf. Proposition 2.3, point (1)) and the Garbagnati surfaces of type $G3$ together with their SNC degenerations (cf. Proposition 2.5, point (4) for $p = 2$), are studied in Section 6.

For the sake of readability, all cases produced by the algorithm are collected in Appendix 8, where they are organized into several tables according to the values of (p,k).

3. SNC $(p,1)$-planes with geometric genus one

We discuss SNC $(p,1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 1$. The reader can find the proof of the main Theorem 0.2, which is Theorem 3.7 in this section.

First of all, we study automorphisms of prime order on a projective surface S. In particular, we analyze the induced action on the second cohomology group and, more precisely, on the Néron–Severi group and on the transcendental lattice.

Proposition 3.1. *Let S be a smooth projective surface. Let $\rho(S)$ be the Picard number of S and let $T(S) := \mathrm{NS}(S)^\perp \subset H^2(S,\mathbb{Z})$ be the transcendental lattice of S. Then*

$$\rho(S) \leq 10\chi(S) + 2q(S) - K_S^2 \quad \textit{and} \quad \mathrm{rk}\, T(S) \leq 12\chi(S) + 4q(S) - K_{S_{min}}^2 - 3, \tag{27}$$

where S_{min} is the minimal model of S.

Furthermore, if $g \in \mathrm{Aut}(S)$ has prime order p and acts non-trivially on $H^{2,0}(S)$, then

$$p \leq 12\chi(S) + 4q(S) - K_{S_{min}}^2 - 2. \tag{28}$$

Proof. By Lefschetz Theorem on $(1,1)$-classes, then $\rho(S) \leq h^{1,1}(S)$. From Hodge theory, we have

$$b_1(S) = h^{1,0}(S)+h^{0,1}(S) = 2q(S), \quad \text{and} \quad b_2(S) = h^{0,2}(S)+h^{1,1}(S)+h^{2,0}(S) = 2p_g(S)+h^{1,1}(S),$$

so applying Noether's formula we obtain

$$\begin{aligned} 10\chi(S) - K_S^2 &= e(S) - 2\chi \\ &= 2 - 2b_1(S) + b_2(S) - 2 + 2q(S) - 2p_g(S) \\ &= h^{1,1}(S) - 2q(S). \end{aligned} \tag{29}$$

This proves $\rho(S) \leq h^{1,1}(S) = 10\chi(S)+2q(S)-K_S^2$. Furthermore, S is projective, hence $\rho(S) \geq 1$ and by the definition of trascendental lattice we obtain

$$\begin{aligned} \text{rk } T(S) &= b_2(S) - \rho(S) \\ &\leq b_2(S) - 1 \\ &= 2p_g(S) + h^{1,1}(S) - 1 \\ &= 2p_g(S) + 2q(S) + 10\chi(S) - K_S^2 - 1 \\ &= 12\chi(S) + 4q(S) - K_S^2 - 3. \end{aligned} \tag{30}$$

Let S_{min} be the minimal model of S, so there exists a blow-up $b\colon S \longrightarrow S_{\min}$ at finitely many points. Blowing up a smooth projective surface does not change its transcendental lattice, so $T(S_{min}) \cong T(S)$ as integral lattices and in particular

$$\text{rk } T(S) = \text{rk } T(S_{min}) \leq 12\chi(S) + 4q(S) - K_{S_{min}}^2 - 3.$$

Finally, let $g \in \text{Aut}(S)$ has prime order p and acts non-trivially on $H^{2,0}(S)$. Then g^* preserves $\text{NS}(S)$ and the cup-product form, hence it preserves $T(S)$. Since the irreducible representations of $\mathbb{Z}/p$ over $\mathbb{Q}$ are only the trivial one and the cyclotomic representation $\mathbb{Q}(\zeta_p)$ of dimension $p-1$, then $T(S)_{\mathbb{Q}}$ splits as a direct sum of two isotypic components

$$T(S)_{\mathbb{Q}} \simeq 1^{\oplus n} \oplus \mathbb{Q}(\zeta_p)^{\oplus m}, \qquad \text{for some} \qquad n, m \geq 0. \tag{31}$$

However, the representation of $\mathbb{Z}/p$ on $T(S)_{\mathbb{Q}}$ is not trivial otherwise g would act trivially on $H^{2,0}(S)$, so $m \geq 1$. In particular,

$$p - 1 \leq \text{rk}\, T(S), \qquad \text{and hence} \qquad p \leq \text{rk}\, T(S) + 1 \leq 12\chi(S) + 4q(S) - K_{S_{min}}^2 - 2.$$

This concludes the proof. □

Remark 12. The assumption that the action of g on $H^{2,0}(S)$ is non-trivial is essential. If $g^*|_{H^{2,0}(S)} = \text{id}$, the automorphism is cohomologically symplectic, and the above representation-theoretic argument on the transcendental lattice no longer applies. In this situation, additional phenomena may occur; see for instance the next Section 4 for examples of SNC $(p, 1)$ planes with $p_g(S) = 0$ and unbounded p, and Cai [**6**] for examples and a detailed analysis in the elliptic case.

Moreover, by the result of Morrison [**34**] on the structure of the transcendental lattice of projective surfaces with $p_g = 1$, the transcendental lattice is an irreducible Hodge structure of K3 type. Consequently, once the non-trivial action on $H^{2,0}(S)$ is assumed, the above argument specialized to surfaces with $p_g = 1$ reduces immediately to the well-known theory of non-symplectic automorphisms of K3 surfaces, for which is known (see for instance [**4**] and [**24**]) that $n = 0$ in Equation (31) and then we would get the stronger condition $p-1 \mid \text{rk } T(S)$ instead of $p - 1 \leq \text{rk } T(S)$.

Remark 13. Assume that S is a smooth projective surface not ruled, and that $g \in \text{Aut}(S)$ has prime order p and acts non-trivially on $H^{2,0}(S)$. From Enriques-Kodaira classification, then $K_{S_{min}}^2 \geq 0$, so Equation 28 gives

$$p \leq 12\chi(S) + 4q(S) - 2.$$

We are ready to provide a classification of SNC $(p, 1)$-planes with $p_g(Y) = 1$.

Proposition 3.2. *Let $\pi\colon Y \longrightarrow \mathbb{P}^2$ be a SNC $(p, 1)$-plane with $p_g(Y) = 1$. Then $p \in \{2, 3, 5, 7, 11, 13, 17, 19\}$, and the building data of the cover is one among those listed in tables 9 to 16 of the Appendix 8.*

Proof. From Lemma 1.15, then the minimal resolution of singularities $\widetilde{Y}$ of Y has $q(\widetilde{Y}) = 0$, so $\chi(\widetilde{Y}) = 2$.

The action of $\mathbb{Z}/p$ on Y lifts naturally to an action on $\widetilde{Y}$. The action on $\widetilde{Y}$ induces an action on $H^{2,0}(\widetilde{Y})$, which is not trivial. Otherwise, an invariant global holomorphic 2-form would descend to a global holomorphic 2-form on $Y/(\mathbb{Z}/p) \cong \mathbb{P}^2$, a contradiction.

Since $\widetilde{Y}$ is a smooth projective surface that is not ruled, then from Proposition 3.1 and Remark 13 we have $p \leq 12\chi - 2 = 22$. Finally, from Proposition 2.2, then $\sum_g \deg(D_g) \leq 6, 5, 4$ accordingly if p is equal either to $2, 3$ or $5 \leq p \leq 19$, respectively. Thus, for any prime number $p \leq 19$, one can compute either by hand or using a computer station, via the commented function `Building_Data(card,d,G)` explained in Section 2, all the few possibilities for the degrees of the divisors D_g for which the reduced Pardini's equations of Theorem 1.4 are satisfied; the obtained cases are those listed in tables 9 to 16. □

Let us study more in detail the geometry of the obtained SNC $(p, 1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 1$. In particular, we prove that the minimal model of Y is a K3 surface.

The strategy to prove this is as follows.

Remark 14. By Proposition 3.2, Y falls in one of the cases listed in tables 9 to 16. We show for each of those cases that the numerical canonical class of the minimal resolution of singularities $\rho\colon \widetilde{Y} \longrightarrow Y$ is supported on a divisor whose irreducible components are precisely the curves contracted in the passage from $\widetilde{Y}$ to the minimal model $\widetilde{Y}_{min}$.
Thus, we would obtain a minimal surface $\widetilde{Y}_{min}$ with trivial numerical canonical class that has $p_g(\widetilde{Y}_{min}) = 1$, $q(\widetilde{Y}_{min}) = 0$. Thus, it is a K3 surface from Enriques-Kodaira classification.

The cases from tables 9 to 11, so those with $p \in \{2, 3, 5\}$, can be treated together and are discussed in the following lemma.

Lemma 3.3. *Let $\pi\colon Y \longrightarrow \mathbb{P}^2$ be a SNC $(p, 1)$-plane with $p_g(Y) = 1$, and $p \in \{2, 3, 5\}$. Then the minimal model of Y is a K3 surface.*

Proof. From Proposition 3.2, then Y falls in one of the cases in tables 9 to 11. The case $p = 2$ is already well-understood and known in the literature: a double cover of $\mathbb{P}^2$ ramified over a (possibly singular with normal crossings) sextic curve has a minimal resolution of singularities that is a K3 surface. Indeed, from Formula (16), we have

$$2K_{\widetilde{Y}} \equiv_{lin} \widetilde{\pi}^* \left(-2K_{\mathbb{P}^2} + \sum_{g \in G} D_g \right) = \widetilde{\pi}^*(-6H + 6H) = 0,$$

so $K_{\widetilde{Y}}$ is numerically trivial, and then $\widetilde{Y}$ is a K3 surface, as explained in Remark 14.

The remaining cases for $p = 3$ and $p = 5$ are quite similar. We consider the line ℓ of the branch locus of π (cf. Tables 10 to 11) passing through the points of $\mathbb{P}^2$ corresponding to the points of Y which are non-canonical singularities. It turns out that the strict transform $\widetilde{\ell}$ of $\widetilde{\pi}^*\ell$ in $\widetilde{Y}$ is a (-1)-curve and that $K_{\widetilde{Y}} \equiv \widetilde{\ell}$. Thus, after contracting $\widetilde{\ell}$ we obtain a surface with trivial canonical class, hence a K3 surface by the discussion in Remark 14.

In particular, for $p = 3$, looking at the local monodromies of Table 10, the line ℓ is the branch divisor $\ell = D_{2e_1}$ which intersects transversally in four points the (possibly singular with normal crossings) quartic D_{e_1}. The singularities of Y lying over those four points are of type $\frac{1}{3}(1, 1)$. From Proposition 1.16, we have

$$\widetilde{\pi}^*\ell = 3\widetilde{\ell} + \sum_{i=1}^{4} E_i^1,$$

and from Formulae (17) and Remark 8, we have $g(\widetilde{\ell}) = 0$ and

$$\widetilde{\ell}^2 = \frac{1}{3} \cdot 1 - 4 \cdot \frac{1}{3} = -1,$$

so $\widetilde{\ell}$ is a (-1)-curve of $\widetilde{Y}$. Furthermore, from Formula 16 we also have

$$K_{\widetilde{Y}} \equiv \frac{1}{3} \left(\widetilde{\pi}^*(3K_{\mathbb{P}^2} + 2\sum_{g \in G} D_g) - \sum_{i=1}^{4} E_i^1 \right) = \frac{1}{3} \left(\widetilde{\pi}^*(H) - \sum_{i=1}^{4} E_i^1 \right) \equiv \widetilde{\ell},$$

as we wanted to prove.

Instead, for $p = 5$, looking at Table 11, the line ℓ has local monodromy $\ell = D_{3e_1}$, and it intersects transversally in three points the (possibly singular with normal crossings) cubic D_{4e_1}. The singularities of $\widetilde{Y}$ over those three points are of type $\frac{1}{5}(1,2)$, so that we have

$$\widetilde{\pi}^*(\ell) = 5\widetilde{\ell} + \sum_{i=1}^{3} (2E_{1,i} + E_{2,i}),$$

and from Formula (17) and Remark 8, we have $g(\widetilde{\ell}) = 0$ and

$$\widetilde{\ell}^2 = \frac{1}{5} \cdot 1 - 3 \cdot \frac{2}{5} = -1.$$

In other words, $\widetilde{\ell}$ is a (-1)-curve of $\widetilde{Y}$. Furthermore, Formula (16) gives

$$K_{\widetilde{Y}} \equiv \frac{1}{5} \left(\widetilde{\pi}^*(5K_{\mathbb{P}^2} + 4\sum_{g \in G} D_g) - \sum_{i=1}^{3} (2E_i^1 + E_i^2) \right) \equiv \frac{1}{5} \left(\widetilde{\pi}^*(H) - \sum_{i=1}^{3} (2E_i^1 + E_i^2) \right) \equiv \widetilde{\ell},$$

so the thesis follows. □

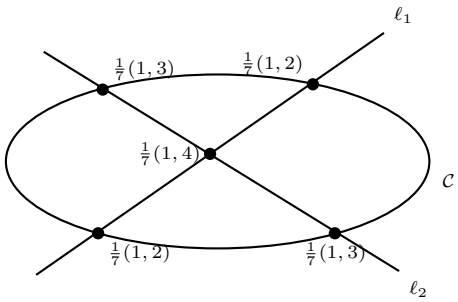
Let us discuss the cases for $p = 7$.

Lemma 3.4. *Let $\pi \colon Y \longrightarrow \mathbb{P}^2$ be a SNC $(7,1)$-plane with $p_g(Y) = 1$. Then the minimal model of Y is a K3 surface.*

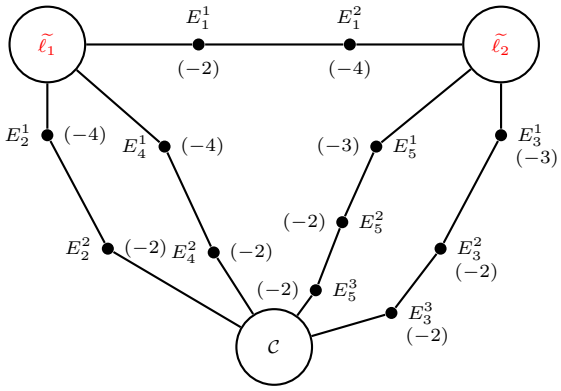
Proof. By Table 12, the branch divisor of π consists of two lines ℓ_1, ℓ_2 and a conic $\mathcal{C}$ with local monodromies $4e_1, 5e_1$ and $6e_1$, respectively. We have two possibilities: either $\mathcal{C}$ is smooth or it decomposes as two distinct lines. We only treat the first case, which corresponds to the case $A - (7,1)(1)$ of Table 12. The second case $A - (7,1)(2)$ can be done in a similar way. The corresponding cyclic quotient singularities of Y in case $A - (7,1)(1)$ are shown in Figure 1a. We now proceed with the calculation of the self-intersection of the strict transform of the lines ℓ_1, ℓ_2 and the conic $\mathcal{C}$ on $\widetilde{Y}$. Using the Formula (17) we see that

$$\widetilde{\ell_1}^2 = -1, \quad \widetilde{\ell_2}^2 = -1, \quad \widetilde{C}^2 = -2.$$

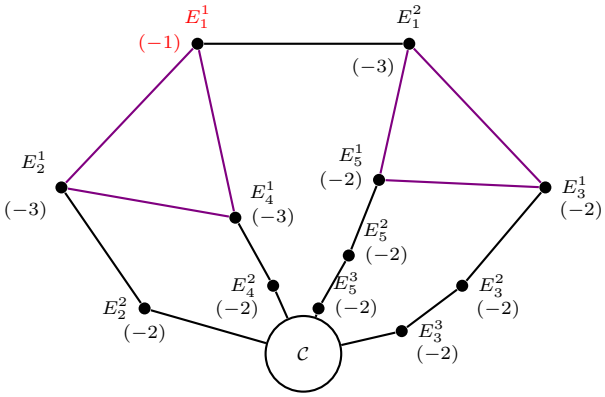
Clearly the curves $\tilde{\ell_1}$, $\tilde{\ell_2}$ are rational from Remark 8, so they are (-1)-curves. We can therefore determine the dual graph of the exceptional configuration in the minimal resolution $\widetilde{Y}$, shown in Figure 1b. To facilitate the minimalization process, we highlight the (-1)-curves in red. Moreover, at each step of the minimalization process, we shall denote the strict transforms of the curves by the same symbols as the original curves in order to avoid unnecessarily heavy notation. We proceed by contracting the two (-1)-curves $\widetilde{\ell_1}$ and $\widetilde{\ell_2}$. The diagram that we obtain is given in Figure 1c. After this contraction, we have only another (-1)-curve E_1^1 to contract and no others occurs in the exceptional configuration, see Figure 1d. Notice that the new exceptional (-2)-curves E_2^1 and E_4^1 become tangent. In this case, we use the convention that a double edge indicates that the corresponding curves are tangent.



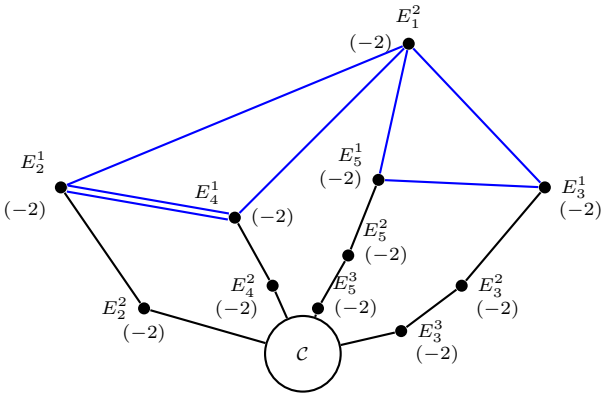
(a) Branch configuration of $A-(7,1)(1)$.



(b) Dual graph of the minimal resolution of singularities $\widetilde{Y}$.



(c) First contraction of the (-1)-curves $\widetilde{\ell_1}$ and $\widetilde{\ell_2}$.



(d) Final contraction of the (-1)-curve E_1^1.

FIGURE 1. Complete contraction process for the case $A-(7,1)(1)$.

As explained in Remark 14 and already done in Lemma 3.3, we proceed by giving the numerical canonical class of the minimal resolution $\widetilde{Y}$ using Formula (16):

$$7K_{\widetilde{Y}} \equiv_{lin} \widetilde{\pi}^*(3H) - \left(2E_1^1 + 4E_1^2\right) - \left(4E_2^1 + 2E_2^2\right) - \left(4E_4^1 + 2E_4^2\right) - \sum_{j=3,5} \left(3E_j^1 + 2E_j^2 + E_j^3\right).$$

However, using Formula 19, then

$$\widetilde{\pi}^*\ell_1 = 7\widetilde{\ell_1} + (4E_1^1 + E_1^2) + (2E_2^1 + E_2^2) + (2E_4^1 + E_4^2)$$
$$\widetilde{\pi}^*\ell_2 = 7\widetilde{\ell_2} + (E_1^1 + 2E_1^2) + \sum_{j=3,5}(3E_j^1 + 2E_j^2 + E_j^3),$$

so that the numerical class of $K_{\widetilde{Y}}$ can be written as

$$K_{\widetilde{Y}} \equiv 2\widetilde{\ell_1} + \widetilde{\ell_2} + E_1^1.$$

Thus, we see that the support of the canonical divisor of $\widetilde{Y}$ is on contracted curves. Hence the minimal model $\widetilde{Y}_{min}$ is a K3 surface, as explained in Remark 14. □

The remaining cases $p = 11, 13, 17, 19$ can be studied by the same computation.

Lemma 3.5. *Let $\pi\colon Y \longrightarrow \mathbb{P}^2$ be a SNC $(p,1)$-plane with $p_g(Y) = 1$, and $p \in \{11, 13, 17, 19\}$. Then the minimal model of Y is a K3 surface.*

Proof. The argument is essentially identical for $p = 11, 13, 17$ and 19 and it follows the strategy presented in Remark 14 already used to prove lemmas 3.3 and 3.4. Therefore, we present only the case $p = 11$ in full detail.

By Table 13, for $p = 11$ the branch divisor of π consists of four lines with monodromies $6e_1, 8e_1, 9e_1$ and $10e_1$. The corresponding cyclic quotient singularities of Y are shown in Figure 2a.

We now proceed with the calculation of the self intersection of the strict transform of the lines $\ell_1, \ldots, \ell_4$. Using Formula (17) we see that

$$\tilde{\ell_1}^2 = -1, \quad \tilde{\ell_2}^2 = -1, \quad \tilde{\ell_3}^2 = -1, \quad \tilde{\ell_4}^2 = -2.$$

Clearly the curves $\tilde{\ell_1}$, $\tilde{\ell_2}$ and $\tilde{\ell_3}$ are all rational, by Remark 8. We can therefore determine the dual graph of the exceptional configuration in the minimal resolution of singularities $\widetilde{Y}$, shown in Figure 2b. To facilitate the minimalization process, we highlight the (-1)-curves in red.

Moreover, at each step of the minimalization process, we shall denote the strict transforms of the curves by the same symbols as the original curves in order to avoid unnecessarily heavy notation.

The surface obtained after the first contraction of $\widetilde{\ell}_1, \widetilde{\ell}_2, \widetilde{\ell}_3$ is still not minimal. Indeed, it contains two further (-1)-curves shown in red in Figure 2c. We therefore proceed by contracting these curves.

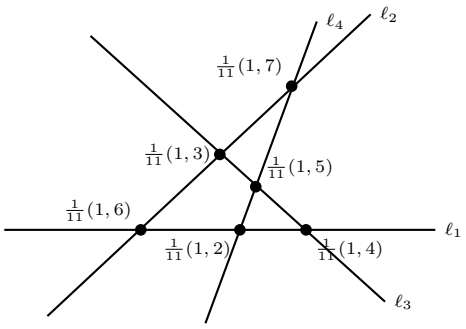
In Figure 2d, we adopt the convention that a double edge indicates that the corresponding curves are tangent, rather than merely intersecting transversely, as done for the Figure 1d in Lemma 3.4. We have one last curve E_2^1 to contract, this is done in the last Figure 2e. We point out a subtle phenomenon that does not occur in the cases $p \leq 7$, but appears in all the cases with $p \geq 11$. Indeed, during the minimalization process we contract a (-1)-curve that is tangent to other curves in the configuration.

Consequently, the strict transforms of the tangent curves become singular and their self-intersection must be computed with particular care. In the present case, the self-intersection of the strict transform of E_3^1 increases by 4 due to the tangential intersection with the contracted curve.

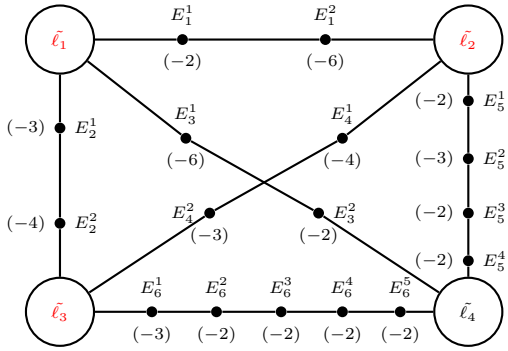
We finally give the numerical canonical class of the minimal resolution using Theorem 1.14:

$$K_{\widetilde{Y}} \equiv 4\widetilde{\ell}_1 + 2\widetilde{\ell}_2 + \widetilde{\ell}_3 + 2\widetilde{E}_1^1 + \widetilde{E}_2^1 + \widetilde{E}_5^1.$$

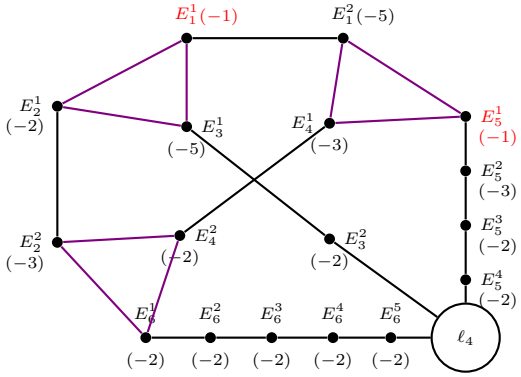
Proceeding with the contraction as explaind above we see that the support of the canonical divisor $K_{\widetilde{Y}}$ was on contracted curves. Hence the minimal model $\widetilde{Y}_{min}$ is a K3 surface, as explained in Remark 14.



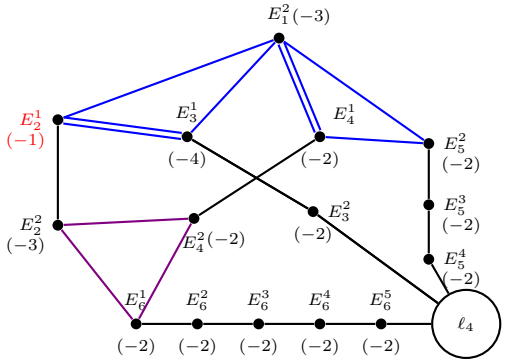
(a) Branch configuration for the case $A-(11,1)$.



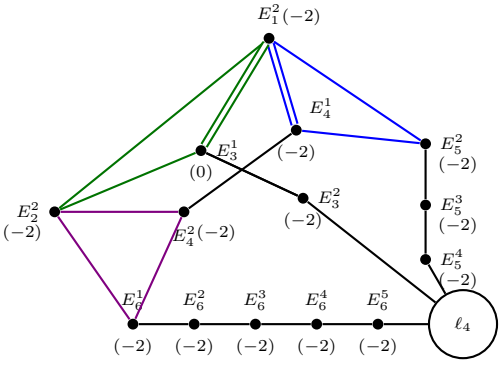
(b) Dual graph of the minimal resolution of singularities $\widetilde{Y}$.



(c) First contraction of the (-1)-curves $\widetilde{\ell}_1$, $\widetilde{\ell}_2$, $\widetilde{\ell}_3$.



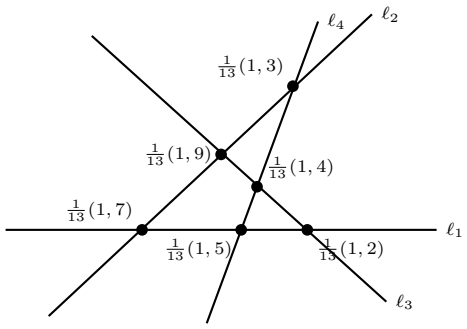
(d) Second contraction of the (-1)-curves E_1^1 and E_5^1.



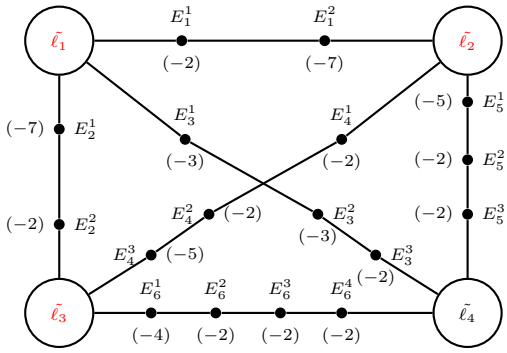
(e) Final contraction of the (-1)-curve E_2^1.

FIGURE 2. Complete contraction process for the case $A-(11,1)$.

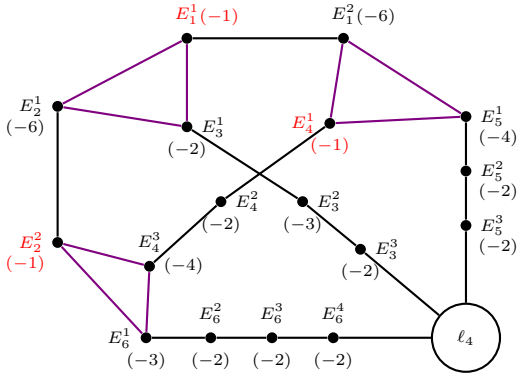
We conclude the proof showing the numerical canonical class of each remaining case $p = 13, 17, 19$ together with its minimalization process. In this way, we exhibit the numerical canonical class of $\widetilde{Y}$ as supported on the contracted curves of the minimalization, so that the minimal model of Y turns out to be a K3 surface, still by Remark 14.



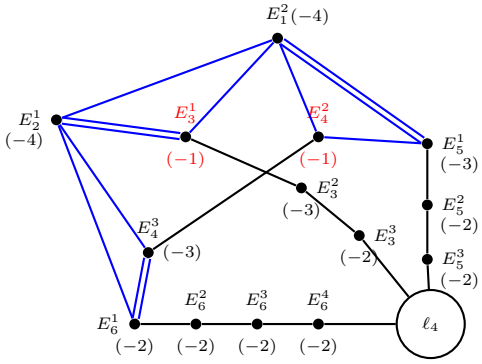
(a) Branch configuration for the case $A-(13,1)$.



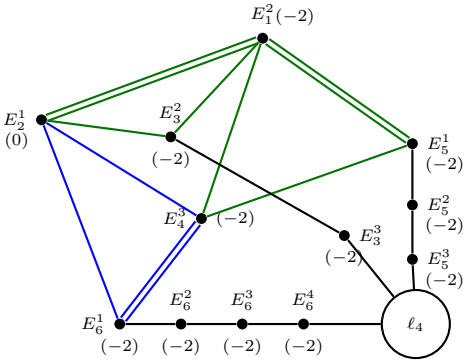
(b) Dual graph of the minimal resolution of singularities $\widetilde{Y}$.



(c) First contraction of the (-1)-curves $\widetilde{\ell}_1$, $\widetilde{\ell}_2$, $\widetilde{\ell}_3$.

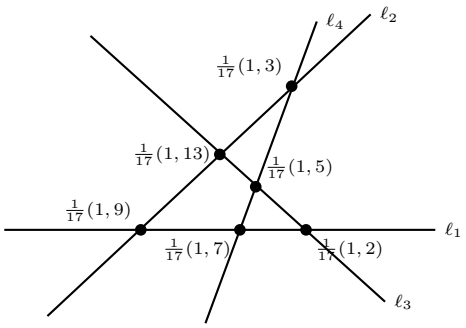


(d) Second contraction of the (-1)-curves E_1^1, E_2^2, and E_4^1.

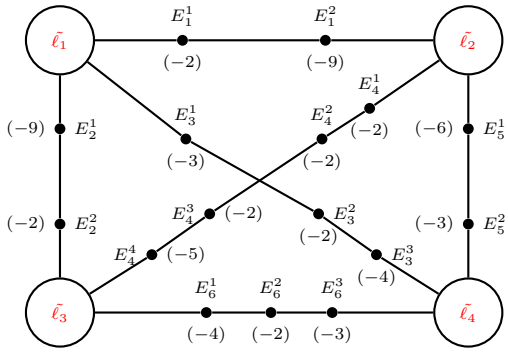


(e) Final contraction of the (-1)-curves E_3^1 and E_4^2.

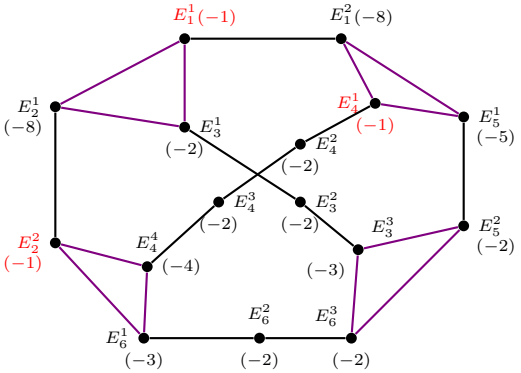
FIGURE 3. Complete contraction process for the case $A-(13,1)$.



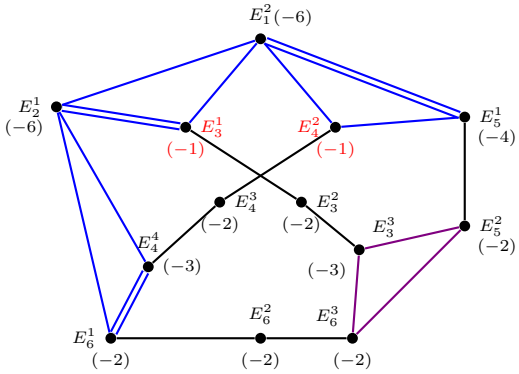
(a) Branch configuration for the case $A-(17,1)$.



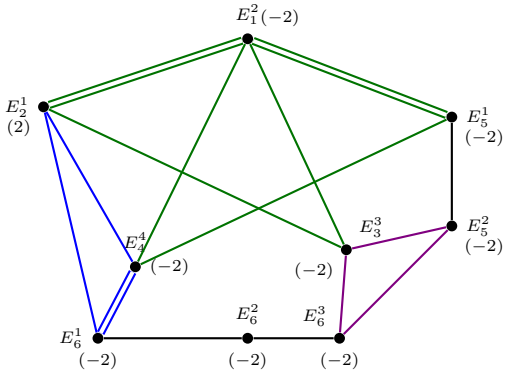
(b) Dual graph of the minimal resolution of singularities $\widetilde{Y}$.



(c) First contraction of the (-1)-curves $\widetilde{\ell}_1$, $\widetilde{\ell}_2$, $\widetilde{\ell}_3$ and $\widetilde{\ell}_4$.

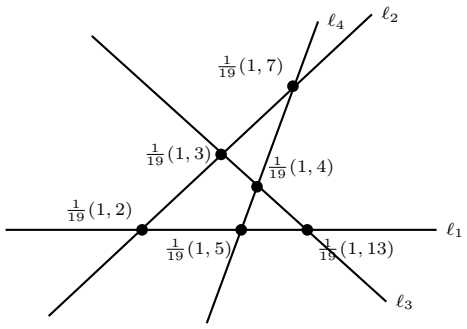


(d) Second contraction of the (-1)-curves E_1^1, E_2^2, and E_4^1.

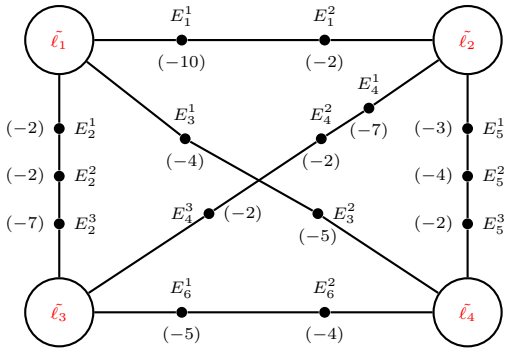


(e) Final contraction of the (-1)-curves E_3^1, E_4^2, and then E_3^2,E_4^3.

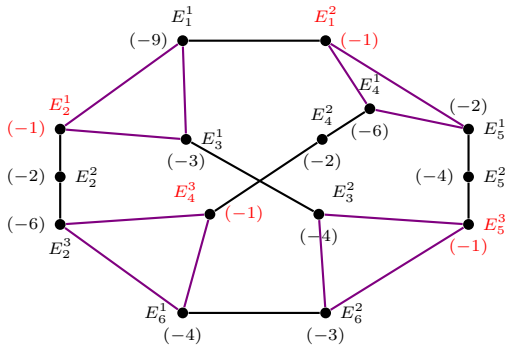
FIGURE 4. Complete contraction process for the case $A-(17,1)$.



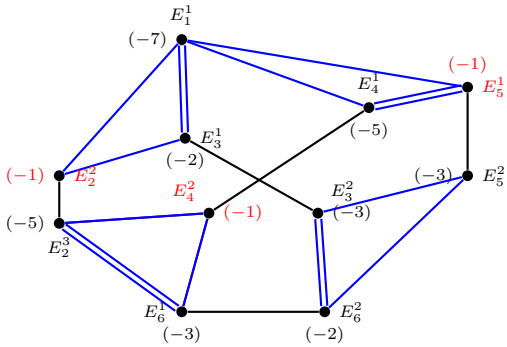
(a) Branch configuration for the case $A-(19,1)$.



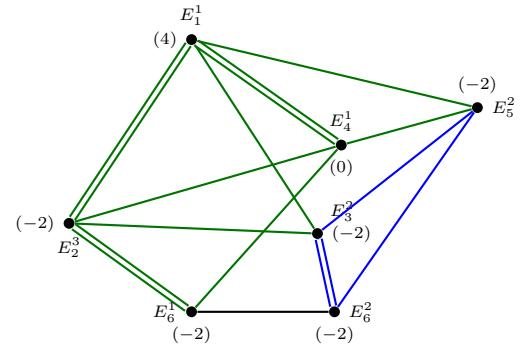
(b) Dual graph of the minimal resolution of singularities $\widetilde{Y}$.



(c) First contraction of the (-1)-curves $\widetilde{\ell}_1$, $\widetilde{\ell}_2$, $\widetilde{\ell}_3$ and $\widetilde{\ell}_4$.



(d) Second contraction of the (-1)-curves E_1^2, E_2^1, E_4^3, and E_5^3.



(e) Final contraction of the (-1)-curves E_2^2, E_4^2, E_5^1, and then E_3^1.

FIGURE 5. Complete contraction process for the case $A-(19,1)$.

For the case $A-(13,1)$, we have that the canonical numerical class is

$$K_{\widetilde{Y}_{min}} \equiv 4\widetilde{\ell}_1 + 3\widetilde{\ell}_2 + 2\widetilde{\ell}_3 + 2E_1^1 + E_2^2 + E_3^1 + 2E_4^1 + E_4^2,$$

and the minimalization process is Figure 3.

For the case $A-(17,1)$, we have that the canonical numerical class is

$$K_{\widetilde{Y}_{min}} \equiv 6\widetilde{\ell}_1 + 4\widetilde{\ell}_2 + 2\widetilde{\ell}_3 + \widetilde{\ell}_4 + 3E_1^1 + E_2^2 + 2E_3^1 + E_3^2 + 3E_4^1 + 2E_4^2 + E_4^3,$$

and the minimalization process is Figure 4.

Finally, for the case $A-(19,1)$, we have that the canonical numerical class is

$$K_{\widetilde{Y}_{min}} \equiv 6\widetilde{\ell}_1 + 4\widetilde{\ell}_2 + 3\widetilde{\ell}_3 + 2\widetilde{\ell}_4 + 2E_1^2 + 4E_2^1 + 2E_2^2 + E_3^1 + E_4^2 + 2E_4^3 + E_5^1 + E_5^3,$$

and the minimalization process is Figure 5. □

To conclude the section, we determine the Picard number and the transcendental part of the Hodge structure of a very general SNC $(p,1)$-plane with geometric genus one.

We denote by $\mathcal{M}_{K3}^p$ the moduli space of K3 surfaces admitting a non-symplectic automorphism of order p. For each family of $(p,1)$-planes, let η denote the action on the K3 lattice induced by the covering automorphism on the minimal K3 model, up to conjugacy. We shall study the corresponding moduli locus $\mathcal{M}_\eta \subseteq \mathcal{M}_{K3}^p$, parametrizing K3 surfaces equipped with a non-symplectic automorphism whose cohomological action is conjugate to η.

Proposition 3.6. *Let $\pi : Y \longrightarrow \mathbb{P}^2$ be a very general member of one of the families of SNC $(p,1)$-planes with $p_g(Y)=1$ with building data $\{D_g\}_{g\in G}$. Let $\widetilde{Y}_{\min}$ be the minimal K3 model of the minimal resolution of singularities $\rho\colon \widetilde{Y} \longrightarrow Y$, and let σ be the non-symplectic automorphism of order p on $\widetilde{Y}_{min}$ given by a generator of the Galois group of π. Let $S(\sigma) := \{x \in H^2(\widetilde{Y}_{min},\mathbb{Z})\colon \sigma^*(x)=x\}$. Then*

- $\mathrm{NS}(\widetilde{Y}_{min})_{\mathbb{Q}} = S(\sigma)_{\mathbb{Q}}$;
- *the Picard number of the minimal model $\widetilde{Y}_{min}$ of Y is*

$$\rho(\widetilde{Y}_{min}) = 22 + (6+[p]_2)(p-1) - (p-1)\sum_{\substack{g\in G\\ \Delta_g \leq D_g}} \left(\frac{(\deg \Delta_g + 2)(\deg \Delta_g + 1)}{2} - 1\right), \tag{32}$$

  where $[p]_2 \in \{0,1\}$ is the residue of p modulo 2;
- *the Picard number of Y is $\rho(Y)=1$;*
- *the transcendental part of the Hodge structure of Y is of K3 type, and precisely is equal to*

$$(h^{2,0}(Y), h^{1,1}(Y), h^{0,2}(Y))_{tra} = (1, 20-\rho(\widetilde{Y}_{min}), 1).$$

Moreover, the corresponding family of pairs $(\widetilde{Y}_{\min},\sigma)$ is dense in the irreducible moduli locus $\mathcal{M}_\eta$ associated with the cohomological action $\eta=\sigma^*$.

Proof. Since σ is non-symplectic, we have

$$S(\sigma) \subseteq \mathrm{NS}(\widetilde{Y}_{min}),$$

see e.g., [**4**, Thm. 2.1(a)]. We prove the equality of their ranks for the very general member of each family of $(p,1)$-planes in tables 9 to 16.

The pushforward of the strict transforms of the ramification components of π together with the exceptional curves of the resolution, which are invariant divisor classes on the minimal K3 model $\widetilde{Y}_{min}$, generate a sublattice V of $S(\sigma)$. For $p=2$, the pullback of a line and the N exceptional curves give $1+N$ independent invariant classes; for $p=3$, the preceding computations give $4+2N$ independent invariant classes, where N denotes the number of nodes of the branch quartic. For $5 \leq p \leq 19$, using `SAGE` [**44**] (see `SAGE` scripts and results repository), we computed the intersection pair matrix among those invariant divisor classes, which has been shown case-by-case that the rank of such a sublattice is $\mathrm{rk}(V) = 1 + |\mathrm{Exc}(\rho)| + K_{\widetilde{Y}}^2$, where $\mathrm{Exc}(\rho)$ is the set

of irreducible curves contracted by ρ. Note that the same formula agrees with $1+N$ for $p=2$ and $4+2N$ for $p=3$. Thus, for any p, it gives then the inequality

$$1+|\mathrm{Exc}(\rho)|+K_{\widetilde{Y}}^2=\mathrm{rk}(V)\leq \mathrm{rk}(S(\sigma)). \tag{33}$$

For the case when the branch locus of π consists of four lines, namely one of the cases among $A-(5,1)3), A-(7,1)2), A-(11,1)1), A-(13,1)1), A-(17,1)1), A-(19,1)1)$, we have $\mathrm{rk}(V)=1+|\mathrm{Exc}(\rho)|+K_{\widetilde{Y}}^2=23-p$. Furthermore, as shown in the proof of Proposition 3.1, we have $p-1\leq \mathrm{rk}\, T(\widetilde{Y}_{min})$, which implies $\rho(\widetilde{Y}_{min})\leq 23-p$. Hence

$$23-p\leq \mathrm{rk}\, S(\sigma)\leq \rho(\widetilde{Y}_{min})\leq 23-p,$$

and the equality of the ranks follows. Note that we did not use arguments involving a very general member of the family. Indeed, in such cases the corresponding family of the $(p,1)$-cover π consists only of one point, which is the cover itself. The period-domain dimension formula gives $\dim \mathcal{M}_\eta=0$; hence the nonempty family coincides with $\mathcal{M}_\eta$.

For the remaining families of $(p,1)$-planes to be studied, let $\mathcal{B}$ denote the parameter space of the corresponding branch configurations modulo projectivity, and consider the period map

$$\Psi:\mathcal{B}\longrightarrow \mathcal{M}_\eta,\qquad [\pi: Y\longrightarrow \mathbb{P}^2]\longmapsto [\widetilde{Y}_{min},\sigma],$$

where $\widetilde{Y}_{min}$ is the minimal K3 model of Y and σ is the induced automorphism.

Claim. Ψ has zero-dimensional fibres.

Once the claim is proved, we can compare the dimension of $\mathcal{B}$ with the dimension of $\mathcal{M}_\eta$, computed using the period-domain dimension formula [**4**, Sec. 9]. What we obtain is the inequality

$$\dim \mathcal{B}=\sum_{\substack{g\in G\\ \Delta_g\leq D_g}}\dim|\Delta_g|-8\leq \dim M_\eta=\frac{22-\mathrm{rk}(S(\sigma))}{p-1}-2+[p]_2,$$

which gives the upper-bound

$$\mathrm{rk}(S(\sigma))\leq 22-\left(\sum_{\substack{g\in G\\ \Delta_g\leq D_g}}\dim|\Delta_g|-6-[p]_2\right)(p-1). \tag{34}$$

Surprisingly, one can check that the left-hand side of (33) and the right hand side of (34) are equal case-by-case. This directly implies (34) is an equality. Since Ψ has zero-dimensional fibres and $\dim\mathcal{B}=\dim\mathcal{M}_\eta$, it is dominant. A very general member of our family therefore avoids the proper Noether–Lefschetz loci where additional classes become algebraic, yielding the desired equality of the ranks of $S(\sigma)$ and $\mathrm{NS}(\widetilde{Y}_{min})$. Thus, $\rho(\widetilde{Y}_{min})=\mathrm{rk}\, S(\sigma)$ and the equality (34) gives Formula (32).

Moreover, being also (33) an equality, we also have $\rho(\widetilde{Y}_{min})=1+|\mathrm{Exc}(\rho)|+K_{\widetilde{Y}}^2$. We observe that since $\widetilde{Y}_{min}$ is a K3 surface, then the number of (-1)-curves to contract to pass from $\widetilde{Y}$ to the minimal model equals exactly $-K_{\widetilde{Y}}^2$. Thus, the Picard number of $\widetilde{Y}$ is equal to $\rho(\widetilde{Y})=1+|\mathrm{Exc}(\rho)|$. Consequently, the equality $\rho(Y)=1$ follows by subtracting from $\rho(\widetilde{Y})$ the number of curves contracted by the resolution ρ.

Finally, resolutions and blow-ups only add algebraic classes, so the transcendental Hodge structure of Y is isomorphic to that of $\widetilde{Y}_{\min}$.

proof of the Claim. Let us consider the case $p=2$. Note that in this case, the preceding computations show that $\widetilde{Y}$ is already minimal, so $\widetilde{Y}=\widetilde{Y}_{\min}$. On a fixed K3 surface $\widetilde{Y}_{min}$, the pullback $\mathcal{L}:=\widetilde{\pi}^*\mathcal{O}_{\mathbb{P}^2}(1)$ satisfies $h^0(\widetilde{Y}_{min},\mathcal{L})=3$ and $L^2=2$, so the complete linear system

$|L|$ determines the degree 2 morphism $\widetilde{\pi}$ up to projective transformations of the target. Since $\mathrm{Pic}^0(\widetilde{Y}_{\min}) = 0$, then $\mathrm{Pic}(\widetilde{Y}_{min})$ is a countable set, so there are only countably many line bundles $\mathcal{L}$ on the fixed K3 surface that can arise as $\widetilde{\pi}^*\mathcal{O}_{\mathbb{P}^2}(1)$ from such $(2,1)$-plane presentations. Thus the fibres of Ψ are countable and hence zero-dimensional.

For $p \in \{3,5\}$ we proceed in a similar way. Note that the resolution $\widetilde{Y}$ is recovered by blowing up an isolated fixed point of $\widetilde{Y}_{min}$ (see Lemma 3.3). For the fixed pair $(\widetilde{Y}_{min}, \sigma)$, there are only finitely many such points. On each resulting surface $\widetilde{Y}$, the possible pullback line bundles form a countable set, as in the case $p = 2$, and the three-dimensional space of invariant sections for the natural linearization determines a degree 3 (resp. 5) morphism $\widetilde{\pi}$ up to projective transformations. Thus the fibres of the map Ψ to moduli are zero-dimensional.

It only remains to discuss the case $A-(7,1)1)$. Here we reverse the contractions from $\widetilde{Y}$ to $\widetilde{Y}_{min}$ described in Lemma 3.4, see Figure 1a. Thus, first, we blow up the point q on the K3 surface obtained by contracting E_1^1, and then a point a on the resulting exceptional curve, recovering the ramification curve $\widetilde{\ell}_1$. Since $\widetilde{\ell}_1$ is fixed pointwise, the tangent action at a is scalar and nontrivial. The local blow-up calculation therefore shows that the tangent eigenvalues at q are λ, λ^2, with λ a nontrivial seventh root of unity. Thus q is an isolated fixed point, and the action on its exceptional curve is nontrivial, with exactly two fixed points from which a can be chosen. Finally, $\widetilde{\ell}_2$ is recovered by blowing up another isolated fixed point of the K3 surface, away from q. All blow-up centres therefore have only finitely many possibilities. The argument is then the same as in the previous cases, which implies that the fibres of the map Ψ to moduli are zero-dimensional. □

Remark 15. The rank of the invariant lattice determines the dimension of the moduli space with the prescribed non-symplectic action. The Picard number may nevertheless increase on Noether–Lefschetz loci, where additional non-invariant classes become algebraic. A family contained in such a locus may therefore have $\mathrm{NS}(S)_{\mathbb{Q}} \supsetneq S(\sigma)_{\mathbb{Q}}$ even for its very general member. The density established in Proposition 3.6

Remark 16. For a very general member of each family in Tables 11–16, the invariant lattice, and hence the Néron–Severi lattice of $\widetilde{Y}_{\min}$ by Proposition 3.6 is as follows:

case	$\mathrm{rk}\, S(\sigma)$	$S(\sigma)$
$A-(5,1)1)$	6	$H_5 \oplus A_4$
$A-(5,1)2)$	14	$H_5 \oplus A_4 \oplus E_8$
$A-(5,1)3)$	18	$H_5 \oplus E_8^{\oplus 2}$
$A-(7,1)1)$	10	$U(7) \oplus E_8$
$A-(7,1)2)$	16	$U \oplus E_8 \oplus A_6$
$A-(11,1)1)$	12	$U \oplus A_{10}$
$A-(13,1)1)$	10	$H_{13} \oplus E_8$
$A-(17,1)1)$	6	$U \oplus L_{17}$
$A-(19,1)1)$	4	$U \oplus K_{19}$

The intersection matrices and their lattice identifications are recorded in the `SAGE` script and its output log `cyclic-cover-lattices.log`. The resulting invariant lattices agree with the classification of Artebani–Sarti–Taki [**4**].

These families describe the moduli loci associated with their respective cohomological actions; our $\mathcal{M}_\eta$ need not be maximal irreducible components of the moduli space $\mathcal{M}_{K3}^p$.

For $p = 3$, write $Q = D_{e_1}$ and let N and s denote the number of nodes and irreducible components of Q, respectively. The fixed locus on the minimal K3 model consists of the s ramification curves lying over the components of Q, one isolated point over each node of Q, and the isolated point obtained by contracting the ramification curve over D_{2e_1}. Thus the pairs

(n,k) (in the notation of [**3**, Table 2]) are, respectively, $(1,1)$, $(4,2)$, $(5,2)$, $(6,3)$ and $(7,4)$; the following lattice identifications follow from [**3**, Table 2].

case	$S(\sigma)$
$A-(3,1)1)$	$U(3)\oplus A_2$
$A-(3,1)2)$	$U\oplus A_2^{\oplus 4}$
$A-(3,1)3)$	$U\oplus A_2^{\oplus 5}$
$A-(3,1)4)$	$U\oplus E_6\oplus A_2^{\oplus 3}$
$A-(3,1)5)$	$U\oplus E_6^{\oplus 2}\oplus A_2$

For $p=2$, the invariant lattices can be identified using the classification of 2-elementary lattices; see for example [**19**] and [**23**] for computations of this kind and the works of Nikulin [**36**–**38**]. We do not pursue these identifications here, since the ranks established in Proposition 3.6 suffice for our purposes.

We conclude the section with the following theorem, which summarizes the results obtained above.

Theorem 3.7. *Let $\pi\colon Y\longrightarrow\mathbb{P}^2$ be a SNC $(p,1)$-plane with $p_g(Y)=1$. Then the minimal model of Y is a K3 surface and the building data of the cover falls in one of the cases listed in tables 9 to 16 in Appendix 8. Furthermore, a very general member of the family has Picard number one, i.e. $\rho(Y)=1$, and we also provide for such member the Picard number $\rho(\widetilde{Y}_{min})$ of the minimal model of Y together with the transcendental part of its Hodge structure.*

Proof. The cases listed in tables 9 to 16 are attained in Proposition 3.2. The proof that each obtained case has a minimal model that is a K3 surface follows by lemmas 3.3, 3.4, and 3.5.

Finally, the proof that a very general member of the family has Picard number one and the computation of the Picard number of its minimal model together with its transcendental Hodge structure can be found in tables 9 to 16 and are computed case-by-case using the formulae in Proposition 3.6. □

4. SNC $(p,1)$-planes with geometric genus zero

In this section, we classify SNC $(p,1)$-planes $\pi\colon Y\longrightarrow\mathbb{P}^2$ with $p_g(Y)=0$. The main theorem of the section is the following

Theorem 4.1. *Let $\pi\colon Y\longrightarrow\mathbb{P}^2$ be a SNC $(p,1)$-plane. Let $\rho\colon\widetilde{Y}\longrightarrow Y$ be the minimal resolution of singularities of Y. Then $\widetilde{Y}$ is a rational surface if and only if $p_g(Y)=0$. In this case, the building data of the cover falls in one of the cases listed in tables 2 to 5, where we also provide the Picard number $\rho(\widetilde{Y})=h^{1,1}(\widetilde{Y})$ of $\widetilde{Y}$ together with the Picard number $\rho(Y)$ of Y.*

Proof. The cases listed from tables 2 to 5 are attained in Proposition 4.2. The rationality of the obtained cases is proved in Proposition 4.3 below. Furthermore, rational surfaces have $\chi(\widetilde{Y})=1$, $q(\widetilde{Y})=0$ and maximal Picard number $\rho(\widetilde{Y})=h^{1,1}(\widetilde{Y})$, so from Equation (29) we have

$$\rho(\widetilde{Y})=h^{1,1}(\widetilde{Y})=10-K_{\widetilde{Y}}^2.$$

The self-intersection of the canonical class of $\widetilde{Y}$ can be easily computed with Formula (16) once we determine the type of cyclic quotient singularities of Y, using Proposition 1.12.

Finally, the Picard group of Y is obtained from $\widetilde{Y}$ by contracting all the exceptional curves of the resolution of Y, so $\rho(Y)=\rho(\widetilde{Y})-\sum_{x\in\mathrm{Sing}(Y)}l_x$, where l_x is the length of the Hirzebruch-Jung string of the resolution of the singular point x. □

case	D_1	Sing(Y)	$K^2_{\widetilde{Y}}$	$\rho(\widetilde{Y})$	$\rho(Y)$	Surface
1	conic	$\emptyset$	8	2	2	$\mathbb{P}^1 \times \mathbb{P}^1$
2	2 lines	$1/2$	8	2	1	$\mathbb{F}_2$
3	quartic	$\emptyset$	2	8	8	del Pezzo of degree 2
4	line + cubic	$1/2^3$	2	8	5	Weak del Pezzo of degree 2
5	2 conics	$1/2^4$	2	8	4	Weak del Pezzo of degree 2
6	2 lines + conic	$1/2^5$	2	8	3	Weak del Pezzo of degree 2
7	4 lines	$1/2^6$	2	8	2	Weak del Pezzo of degree 2

TABLE 2. SNC $(2,1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 0$.

case	D_1	D_2	Sing(Y)	$K^2_{\widetilde{Y}}$	$\rho(\widetilde{Y})$	$\rho(Y)$	Surface
1	line	line	$1/3$	8	2	1	$\mathbb{F}_3$
2	cubic	0	$\emptyset$	3	7	7	del Pezzo of degree 3
3	line + conic	0	$2/3^2$	3	7	3	Weak del Pezzo of degree 3
4	3 lines	0	$2/3^3$	3	7	1	Weak del Pezzo of degree 3
5	conic	conic	$1/3^4$	-1	11	7	Rational
6	2 lines	conic	$2/3, 1/3^4$	-1	11	5	Rational
7	2 lines	2 lines	$2/3^2, 1/3^4$	-1	11	3	Rational

TABLE 3. SNC $(3,1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 0$.

case	D_1	D_2	D_3	D_4	Sing(Y)	$K^2_{\widetilde{Y}}$	$\rho(\widetilde{Y})$	$\rho(Y)$	Surface
1	line	0	0	line	$1/5$	8	2	1	$\mathbb{F}_5$
2	line	conic	0	0	$3/5^2$	1	9	5	Rational
3	line	2 lines	0	0	$3/5^2, 4/5$	1	9	1	Rational
4	conic	0	0	conic	$1/5^4$	-7	17	13	Rational
5	2 lines	0	0	conic	$1/5^4, 4/5$	-7	17	9	Rational
6	2 lines	0	0	2 lines	$1/5^4, 4/5^2$	-7	17	5	Rational
7	line	line	line	line	$1/5^2, 2/5^2, 3/5^2$	-5	15	5	Rational

TABLE 4. SNC $(5,1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 0$.

case	D_1	D_g	$D_{\frac{p-1}{2}}$	D_{p-1-g}	D_{p-g}	D_{p-1}	Sing(Y)
1	line	0	0	0	0	line	$1/p$
2	conic	0	0	0	0	conic	$(1/p)^4$
3	2 lines	0	0	0	0	conic	$(1/p)^4, (p-1)/p$
4	2 lines	0	0	0	0	2 lines	$(1/p)^4, ((p-1)/p)^2$
5	line	0	conic	0	0	0	$(((p+1)/2)/p)^2$
6	line	0	2 lines	0	0	0	$(((p+1)/2)/p)^2$, $(p-1)/p$
7 $2 \le g < \frac{p-1}{2}$	line	line	0	line	0	0	$(p-g)/p, (g+1)/p$, $(g^{-1}+1)/p$
8 $2 \le g \le \frac{p-1}{2}$	line	line	0	0	line	line	$(p-g)/p, g/p, (1/p)^2$, $g^{-1}/p, (p-g^{-1})/p$

TABLE 5. SNC $(p,1)$-planes $\pi\colon Y \longrightarrow \mathbb{P}^2$ with $p_g(Y) = 0$, and $p \ge 7$.

case	$K^2_{\widetilde{Y}}$	$\rho(\widetilde{Y})$	$\rho(Y)$	Surface
1	8	2	1	$\mathbb{F}_p$
2	$8-3p$	$2+3p$	$3p-2$	Rational
3	$8-3p$	$2+3p$	$2p-1$	Rational
4	$8-3p$	$2+3p$	p	Rational
5	$6-p$	$4+p$	p	Rational
6	$6-p$	$4+p$	1	Rational
7 $2\leq g<\frac{p-1}{2}$	$9-\left(l_{\frac{p-g}{p}}+l_{\frac{g+1}{p}}+l_{\frac{g^{-1}+1}{p}}\right)$	$1+\left(l_{\frac{p-g}{p}}+l_{\frac{g+1}{p}}+l_{\frac{g^{-1}+1}{p}}\right)$	1	Rational
8 $2\leq g\leq\frac{p-1}{2}$	$8-p-2\left(l_{\frac{g}{p}}+l_{\frac{p-g}{p}}\right)$	$2+p+2\left(l_{\frac{g}{p}}+l_{\frac{p-g}{p}}\right)$	p	Rational

TABLE 6. Invariants of SNC $(p,1)$-planes with $p_g(Y)=0$, and $p\geq 7$ of Table 5.

The rest of this section is devoted to give a proof of the two propositions 4.2 and 4.3 used to prove Theorem 4.1.

Proposition 4.2. *Let $\pi\colon Y\longrightarrow\mathbb{P}^2$ be a SNC $(p,1)$-plane. Then $p_g(Y)=0$ if and only if we have one of the following possibilities for the degrees of the building data $\{\mathcal{L}_j\}_{j\in\mathbb{Z}/p}$ and $\{D_j\}_{j\in\mathbb{Z}/p}$ of the cover, depending on p:*

$p=2$: (1) $l_1=1$, *and* $d_1=2$; *or*
(2) $l_1=2$, *and* $d_1=4$.

$p=3$: (1) $l_1=l_2=1$, *and* $d_1=d_2=1$;
(2) $l_1=1, l_2=2$, *and* $d_1=3$;
(3) $l_1=l_2=2$, *and* $d_1=d_2=2$.

$p=5$: (1) $l_j=1$, *for* $j\neq 0$, *and* $d_1=d_4=1$;
(2) $l_1=l_3=1$, $l_2=l_4=2$, *and* $d_1=1$, $d_2=2$;
(3) $l_j=2$, *for* $j\neq 0$, *and* $d_1=d_4=2$;
(4) $l_j=2$, *for* $j\neq 0$,*and* $d_1=d_2=d_3=d_4=1$.

$p\geq 7$ (1) $l_j=1$, *for* $j\neq 0$, *and* $d_1=d_{p-1}=1$;
(2) $l_j=2$, *for* $j\neq 0$, *and* $d_1=d_{p-1}=2$;
(3) $l_j=2-[j]_2$, *for* $j\neq 0$, *and* $d_1=1$, $d_{\frac{p-1}{2}}=2$;
(4) *for each chosen element* $g\in\mathbb{Z}/p$ *with* $2\leq g<\frac{p-1}{2}$, *we have* $l_j=1+\left\lfloor\frac{[jg]_p+j-1}{p}\right\rfloor\in\{1,2\}$, *for* $j\neq 0$, *and* $d_1=d_g=d_{p-1-g}=1$;
(5) *for each chosen element* $g\in\mathbb{Z}/p$ *with* $2\leq g\leq\frac{p-1}{2}$, *we have* $l_j=2$, *for* $j\neq 0$, *and* $d_1=d_g=d_{p-g}=d_{p-1}=1$.

Proof. From Equation (11) in Theorem 1.14, we have

$$0=p_g(Y)=\sum_{j=0}^{p-1}h^0(\mathbb{P}^2,\mathcal{O}_{\mathbb{P}^2}(-3+l_j))=\sum_{j=1}^{p-1}\frac{(l_j-1)(l_j-2)}{2},$$

so that $l_j\in\{1,2\}$ for all $j\neq 0$. However, from Theorem 1.7(7), we also have

$$\frac{p-1}{2}\sum_{j=1}^{p-1}d_g=\sum_{j=1}^{p-1}l_j\leq 2(p-1),\qquad\text{so}\qquad\sum_{j=1}^{p-1}d_j\leq 4. \tag{35}$$

We analyse case-by-case according to the possible values of $\sum_{j=1}^{p-1}d_j\in\{1,2,3,4\}$. We automatically discard the case $\sum_{j=1}^{p-1}d_j=1$ because otherwise from Lemma 1.5 we would have $pl_1=\sum_{j=1}^{p-1}jd_j<p$, so $l_1=0$, a contradiction.

Up to automorphisms of $\mathbb{Z}/p$, we can always assume from the rest of the proof that $d_1 \neq 0$. Let us consider the case where $\sum_{j=1}^{p-1} d_j = 2$. If $d_1 = 2$, then from Lemma 1.5 we would have $pl_1 = 2$, which forces $p = 2$, $l_1 = 1$ and $d_1 = 2$. This is the case $p = 2(1)$.

Instead, if $d_1 = 1$, then there is only another $g \neq 0, 1$ such that $d_g = 1$. In this case, still by Lemma 1.5, we would have $pl_1 = 1 + g \cdot 1 \leq 1 + (p-1) = p$, so $l_1 = 1$ and $g = p-1$. Furthermore, from Lemma 1.5, we have $pl_j = j + (p-j) \cdot 1 = p$, so $l_j = 1$ for each $j \neq 0$. This is $p = 3(1)$, $p = 5(1)$, and $p \geq 7(1)$.

We completely covered all the possibilities for $\sum_{j=1}^{p-1} d_j = 2$, so let us consider the next case $\sum_{j=1}^{p-1} d_j = 3$.

If $d_1 = 3$, then we would simply have $pl_1 = 3$, which forces $p = 3$, $l_1 = 1$, and $d_1 = 3$. This is the case $p = 3(2)$.

If $d_1 = 1$ and $d_g = 2$ for some other $g \neq 0, 1$, then $pl_1 = 1 + g \cdot 2 \leq 1 + 2(p-1) = 2p - 1$. This implies $l_1 = 1$ and $g = \frac{p-1}{2}$; moreover

$$pl_j = j + \left[\frac{p-1}{2} j\right]_p \cdot 2 = 2p - p[j]_2, \qquad \text{so} \qquad l_j = 2 - [j]_2.$$

This is the case $p = 5(2)$ and $p \geq 7(3)$. Note that the case $d_1 = 2$ and $d_g = 1$ can be skipped because we would get a corresponding covering isomorphic to the case just studied, via the automorphism g^{-1}.

Thus, let us discuss the case $d_1 = d_{g_1} = d_{g_2} = 1$ for some $1 < g_1 < g_2 \leq p-1$. From Lemma 1.5, we have $pl_1 = 1 + g_1 \cdot 1 + g_2 \cdot 1 \leq 1 + (p-1) + (p-1) = 2p - 1$, so $l_1 = 1$ and then $g_2 = p - 1 - g_1$. For simplicity, we denote $g := g_1$, so now the building data are $d_1 = d_g = d_{p-1-g} = 1$. In this case, we have

$$pl_j = j + [jg]_p + [(p-1-g)j]_p = j + [jg]_p + \left(p - j - [jg]_p + \left\lfloor \frac{[jg]_p + j - 1}{p} \right\rfloor \cdot p\right),$$

so $l_j = 1 + \left\lfloor \frac{[jg]_p + j - 1}{p} \right\rfloor$; we obtained the case $p \geq 7(4)$. Note that from the symmetry of the local monodromies g and $p - 1 - g$, all the possible cases are obtained choosing $2 \leq g \leq \frac{p-1}{2}$. Furthermore, the choice $g = \frac{p-1}{2}$ must be discarded otherwise we would have $g_1 = g_2 = \frac{p-1}{2}$.

We continue with the last case $\sum_{j=1}^{p-1} d_j = 4$. In this case, the left and right hand side of Equation (35) are the same, that is equal to $2(p-1)$, and this can happen only if

$$l_j = 2 \qquad \text{for each} \qquad j \neq 0. \tag{36}$$

Assume that $d_1 = 4$. Then from Lemma 1.5 we have $pl_1 = 4$, which forces $p = 2$, $l_1 = 2$ and $d_1 = 4$. This is $p = 2(2)$ case.

Let us consider now the case $d_1 = 1$ and $d_g = 3$ for some $g \neq 0, 1$. Then $2p = pl_1 = 1 + g \cdot 3$ so we must have $p \equiv 2 \mod 3$ and $g = \frac{2p-1}{3}$. However,

$$pl_3 = 3 + [3g]_p \cdot 3 = 3 + (p-1)3 = 3p, \qquad \text{so} \qquad l_3 = 3,$$

a contradiction. We can exclude also the symmetric case $d_1 = 3$ and $d_g = 1$, $g \neq 0, 1$, since it is isomorphic to the previous case by applying the automorphism g^{-1}.

Let us study what happens for $d_1 = d_g = 2$, for some $g \neq 0, 1$. In this case, still from Lemma 1.5, we have that $2p = pl_1 = 2 + g \cdot 2$, so $g = p - 1$. This is the case $p = 3(3)$, $p = 5(3)$, and $p \geq 7(2)$.

Let us consider the case $d_1 = d_{g_1} = 1$ and $d_{g_2} = 2$, where $g_1, g_2 \neq 0, 1$ and $g_1 \neq g_2$. Pardini equations (4) are satisfied when

$$2p = pl_j = j + [jg_1]_p + [jg_2]_p \cdot 2 \qquad \text{for all} \qquad j = 1, \ldots, p-1. \tag{37}$$

This would imply that the zero-sum sequence (cf. [**30**, Def. 1.1]) $s = (1, g_1, g_2, g_2)$ over the cyclic group $\mathbb{Z}/p$ of prime order p has index $ind(s) = 2$. However, by the Index Theorem for length 4 zero-sum sequences [**30**, Thm. 1.2], any minimal zero-sum sequence of length 4 have to be of index $ind(s) = 1$. This forces our s to not be a minimal zero-sum sequence, so there exists a proper subsequence of s that is a zero-sum sequence. This proper subsequence could not have length 1 or 3, otherwise one of the elements of s would be a multiple of p, because from (37) we have $2p = 1 + g_1 + 2g_2$. Thus, it there exists a zero-sum length 2 subsequence of s. If this subsequence is not containing 1, then the condition $2p = 1 + g_1 + 2g_2$ would implies that the complementary length 2 subsequence of s, which is containing 1, has zero-sum. Thus, one between g_1 and g_2 has to be equal to $p - 1$, forcing either $2g_2 = p$, if $g_1 = p - 1$, or $g_1 = 1$ if $g_2 = p - 1$. In both cases, we have a contradiction, so we can exclude the case $d_1 = d_{g_1} = 1$ and $d_{g_2} = 2$.

It only remains to study the case $d_1 = d_{g_1} = d_{g_2} = d_{g_3} = 1$. Here we adopt a similar strategy as in the previous case. Indeed, remembering (36) and Pardini equations (4) we have

$$2p = pl_j = j + [jg_1]_p + [jg_2]_p + [jg_3]_p \qquad \text{for all} \qquad j = 1, \dots, p-1,$$

so $s = (1, g_1, g_2, g_3)$ is a zero-sum length 4 sequence of index $ind(s) = 2$, which then must be not minimal, by [**30**, Thm. 1.2]. Hence, such as in the previous case, it there exists a zero-sum length 2 subsequence of s, which is containing 1 (use the condition $2p = 1 + g_1 + g_2 + g_3$). Thus, one among g_1, g_2, g_3 has to be equal to $p - 1$, let us say g_3, and so this also forces $g_2 = p - g_1$. We have obtained the building data $d_1 = d_g = d_{p-g} = d_{p-1} = 1$, which corresponds to the cases $p = 5(4)$ and $p \geq 7(5)$. □

We are ready to prove that any surface belonging to one of the families obtained in Proposition 4.2 is rational.

Proposition 4.3. *Let $\pi\colon Y \longrightarrow \mathbb{P}^2$ be a SNC $(p,1)$-plane with $p_g(Y) = 0$. Then the minimal resolution of singularities $\widetilde{Y}$ of Y is a rational surface.*

Proof. From Lemma 1.15, then $\widetilde{Y}$ is a regular surface, so $q(\widetilde{Y}) = 0$. Thus, if we prove that $h^0(\widetilde{Y}, 2K_{\widetilde{Y}}) = 0$, then $\widetilde{Y}$ is a rational surface from Castelnuovo Rationality criterion.

The strategy is to find a nef divisor D on $\widetilde{Y}$ such that $K_{\widetilde{Y}}D < 0$. Indeed, in this case no multiples of $K_{\widetilde{Y}}$ would be effective divisors and so $h^0(\widetilde{Y}, 2K_{\widetilde{Y}}) = 0$.

From Equation (16), then we have

$$pK_{\widetilde{Y}} \equiv \widetilde{\pi}^*((-3p + (p-1)\sum_g d_g)H) - E, \tag{38}$$

where E is a divisor supported on the Hirzebruch-Jung strings of the resolution of Y. In particular, if $-3p + (p-1)\sum_g d_g < 0$, which means $\sum_g d_g < 3 + \frac{3}{p-1}$, then $D := \widetilde{\pi}^*(H)$ is a nef divisor with

$$K_{\widetilde{Y}}D = -3p + (p-1)\sum_g d_g < 0,$$

so $\widetilde{Y}$ is a rational surface. This automatically proves the rationality of the cases $p = 2(1)$ and (2), $p = 3(1)$,(2), and (3), $p = 5(1)$ and (2), and finally $p \geq 7(1)$, (3), and (4).

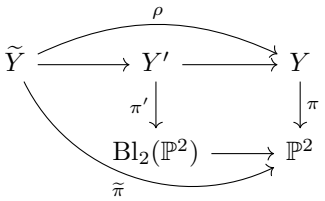
Let us discuss the remain cases $p = 5(3)$ and (4), and $p \geq 7(2)$ and (5). For these cases, the divisor D is constructed as follows. We prove that there exist two (-1)-curves E_1 and E_2 of a birational model S of $\widetilde{Y}$ intersecting each other. Indeed, in this case $D := E_1 + E_2$ is a nef divisor with $K_S D = -2 < 0$. Thus, S is rational and then $\widetilde{Y}$ is rational too.

All of these remain cases can be treated in an analogue way by observing that one can select a line $\ell \subseteq \mathbb{P}^2$ not belonging to the branch locus of π but passing through exactly two intersection

points of the branch locus, the points whose inverse images p_1 and p_2 on Y are two singularities of type $\frac{1}{p}(1,1)$; see Figure 6 where we consider the case $p \geq 7(2)$.

We claim that $\widetilde{\pi}^*(\ell)$ consists of p disjoint (-1)-curves on $\widetilde{Y}$, each of them intersecting transversally in one point two exceptional curves E_1 and E_2 of self-intersection $-p$, those obtained by the resolution of the two singularities p_1 and p_2. If this would be true, then after contracting $p-1$ of these (-1)-curves, we would obtain two (-1)-curves, the images of E_1 and E_2, intersecting each other. This would prove the rationality of $\widetilde{Y}$.

To prove the claim, let us blowing-up $\mathrm{Bl}_2(\mathbb{P}^2) \longrightarrow \mathbb{P}^2$ the two intersection points of the branch locus of π where ℓ is passing from. One can verify via Pardini equations (3) that we can construct another $(\mathbb{Z}/p)$-cover $\pi' \colon Y' \longrightarrow \mathrm{Bl}_2(\mathbb{P}^2)$ having building data that are the strict transforms on $\mathrm{Bl}_2(\mathbb{P}^2)$ of the components of the branch locus of π. We point out that Y' could still be singular but it is resolving the singularities of the two singular points p_1 and p_2 of Y. Indeed, denoted by T_1 and T_2 the two (-1)-curves of the blow-up $\mathrm{Bl}_2(\mathbb{P}^2)$, then $E_i := (\pi')^*(T_i)$ is a smooth rational curve of self-intersection $-p$ by projection formula, for $i = 1, 2$. The contraction of these exceptional curves E_1 and E_2 is the resolution $Y' \longrightarrow Y$ of the singular points p_1 and p_2. We have the following commutative diagram



Let us denote by $\widetilde{\ell}$ the strict transform of ℓ on $\mathrm{Bl}_2(\mathbb{P}^2)$, which is a (-1)-curve. Then $\widetilde{\ell}$ is intersecting transversally in one point T_1 and T_2 and it is disjoint from the branch locus of π'. ,

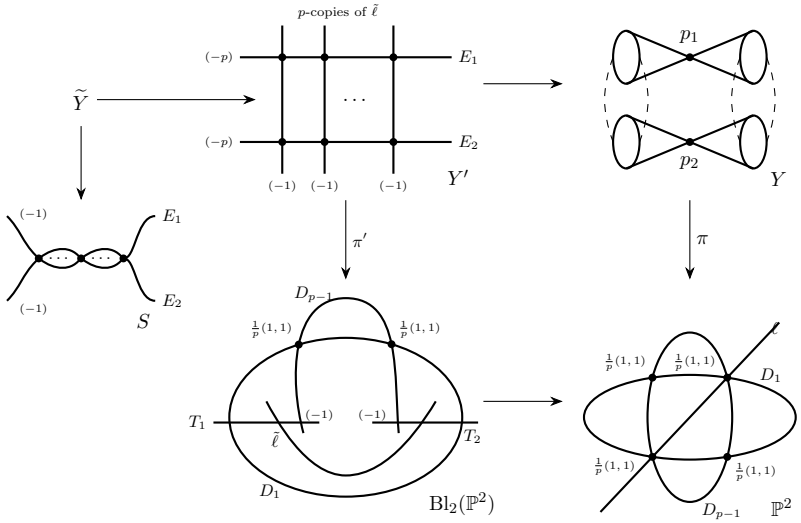


FIGURE 6. The figure shows the case $p \geq 7(2)$. As one can see, $(\pi')^*(\widetilde{\ell})$ is the disjoint union of p copies of $\widetilde{\ell}$. There is the same configuration of curves on $\widetilde{Y}$. After contracting $p-1$ of those (-1)-curves on $\widetilde{Y}$, we obtain a surface S admitting two (-1) curves E_1 and E_2 intersecting to each other.

Thus, $(\pi')^*(\widetilde{\ell})$ consists of p copies of $\widetilde{\ell}$ on Y', that are hence (-1)-curves intersecting transversally each E_i in one point, see Figure 6. However, these (-1)-curves together with E_1 and E_2 are not passing through other singular points of Y', so we would obtain the same configuration of exceptional curves on the minimal resolution $\widetilde{Y}$. This proves the claim and hence the thesis. □

5. The Hodge structures of SNC (p,k)-planes satisfying Property ★

In this section, we shall analyze the Hodge structure of smooth projective surfaces S endowed with an action of an elementary abelian p-group $G \cong (\mathbb{Z}/p)^k$, where p is a prime number. Since G is finite and abelian, the induced complex representation on $H^2(S,\mathbb{C})$ decomposes into character eigenspaces:

$$H^2(S,\mathbb{C}) = \bigoplus_{\chi \in G^*} H^2(S,\mathbb{C})_\chi, \tag{39}$$

where

$$H^2(S,\mathbb{C})_\chi := \left\{ \alpha \in H^2(S,\mathbb{C}) \;\middle|\; g^*\alpha = \chi(g)\alpha \text{ for every } g \in G \right\}$$

is the isotypic component of the character $\chi \in G^*$.

Since the action of G on S is holomorphic, every $g \in G$ preserves the Hodge decomposition. Hence each character eigenspace decomposes according to Hodge type:

$$H^2(S,\mathbb{C})_\chi = H^{2,0}(S)_\chi \oplus H^{1,1}(S)_\chi \oplus H^{0,2}(S)_\chi.$$

Moreover, complex conjugation exchanges the eigenspaces corresponding to conjugate characters:

$$\overline{H^{r,s}(S)_\chi} = H^{s,r}(S)_{\overline{\chi}}.$$

For every nontrivial character $\chi \in G^*$, the subgroup $H := \ker(\chi)$ has index p in G. Conversely, every subgroup $H \leq G$ of index p arises as the kernel of a nontrivial character. Moreover, two nontrivial characters have the same kernel if and only if they generate the same cyclic subgroup of G^*. Equivalently, if $\ker(\chi) = H$, then the nontrivial characters with kernel H are precisely $\chi, \chi^2, \ldots, \chi^{p-1}$.

For every subgroup $H \leq G$ of index p, choose a nontrivial character $\chi \in G^*$ such that $\ker(\chi) = H$, and define

$$V_H := \bigoplus_{j=1}^{p-1} H^2(S,\mathbb{C})_{\chi^j}. \tag{40}$$

Clearly, this definition is independent of the choice of χ. We also set

$$V_0 := H^2(S,\mathbb{C})^G = H^2(S,\mathbb{C})_1.$$

Proposition 5.1. *Let S be a smooth projective surface endowed with a G-action, $G \cong (\mathbb{Z}/p)^k$. With the notation introduced above, one has a decomposition*

$$H^2(S,\mathbb{C}) = V_0 \oplus \bigoplus_{\substack{H \leq G \\ [G:H]=p}} V_H. \tag{41}$$

Each summand is compatible with the Hodge decomposition of $H^2(S,\mathbb{C})$. Moreover, for every subgroup $H \leq G$ of index p, one has

$$H^2(S,\mathbb{C})^H = V_0 \oplus V_H.$$

Proof. The decomposition (41) follows directly replacing (40) into (39) for each subgroup $H \leq G$ of index p. Finally, we observe that

$$H^2(S,\mathbb{C})^H = H^2(S,\mathbb{C})_1 \oplus \bigoplus_{j=1}^{p-1} H^2(S,\mathbb{C})_{\chi^j} = V_0 \oplus V_H,$$

and the thesis follows. □

We now turn to the rational structure underlying the decomposition 39. Although the individual isotypic components $H^2(S,\mathbb{C})_\chi$ need not be defined over $\mathbb{Q}$, the blocks V_H introduced above admit a natural rational structure.

For every subgroup $H \leq G$, consider the operator

$$P_H := \frac{1}{|H|} \sum_{h \in H} h^* \tag{42}$$

defined on $H^2(S,\mathbb{Q})$. We also define $Q_H := P_H - P_G$.

Since every g^*, $g \in G$, is a morphism of rational Hodge structures, the operators P_H, P_G, and Q_H are morphisms of rational Hodge structures. In particular, we have

$$\mathrm{Im}(P_H) = H^2(S,\mathbb{Q})^H.$$

For this equality, notice that P_H is indeed the projector onto $H^2(S,\mathbb{Q})^H$. We set $W_0 := \mathrm{Im}(P_G) = H^2(S,\mathbb{Q})^G$, and for every subgroup $H \leq G$ of index p,

$$W_H := \mathrm{Im}(Q_H).$$

The rational analogue of Proposition 5.1 is the following.

Proposition 5.2. *Let S be a smooth projective surface endowed with a G-action, $G \cong (\mathbb{Z}/p)^k$. Then one has a decomposition of rational Hodge structures*

$$H^2(S,\mathbb{Q}) = W_0 \oplus \bigoplus_{\substack{H \leq G \\ [G:H]=p}} W_H.$$

Moreover, $W_0 \otimes_{\mathbb{Q}} \mathbb{C} = V_0$ and, for every subgroup $H \leq G$ of index p, $W_H \otimes_{\mathbb{Q}} \mathbb{C} = V_H$.

Proof. Let $H \leq G$ be a subgroup of index p. On a isotypic component $H^2(S,\mathbb{C})_\eta$ of character $\eta \in G^*$, the operator P_H (seen as an operator of $H^2(S,\mathbb{C})$) acts as the multiplication by the complex number

$$\frac{1}{|H|} \sum_{h \in H} \eta(h).$$

Therefore,

$$P_H\big|_{H^2(S,\mathbb{C})_\eta} = \begin{cases} \mathrm{id} & \text{if } \eta|_H = 1, \\ 0 & \text{otherwise.} \end{cases}$$

It follows that $\mathrm{Im}(P_H) \otimes_{\mathbb{Q}} \mathbb{C} = V_0 \oplus V_H$, and $\mathrm{Im}(P_G) \otimes_{\mathbb{Q}} \mathbb{C} = V_0$. Hence $W_H \otimes_{\mathbb{Q}} \mathbb{C} = \mathrm{Im}(Q_H) \otimes_{\mathbb{Q}} \mathbb{C} = V_H$.

Let now H and K be two distinct subgroups of G of index p. Since they are distinct maximal proper subgroups, one has $HK = G$. As G is abelian, the corresponding operators satisfy $P_H \circ P_K = P_G$. Moreover, $P_H \circ P_G = P_G$ and $P_G^2 = P_G$.

Consequently, $Q_H \circ Q_K = (P_H - P_G) \circ (P_K - P_G) = 0$ for $H \neq K$, whereas $Q_H^2 = (P_H - P_G)^2 = Q_H$. Thus the Q_H are pairwise orthogonal idempotents. Furthermore, one has

$$P_G + \sum_{\substack{H \leq G \\ [G:H]=p}} Q_H = id_{H^2(S,\mathbb{Q})}.$$

Therefore,

$$H^2(S,\mathbb{Q}) = W_0 \oplus \bigoplus_{\substack{H\leq G\\ [G:H]=p}} W_H.$$

After tensoring with $\mathbb{C}$, their images, together with the image of P_G, give precisely the decomposition (41).

Finally, since all the projectors involved are morphisms of rational Hodge structures, each W_H and W_0 is a rational Hodge substructure of $H^2(S,\mathbb{Q})$. □

Remark 17. When $p = 2$, the above discussion simplifies considerably. Indeed, every character of G takes values in $\{\pm 1\} \subset \mathbb{Q}$, and every subgroup $H \leq G$ of index 2 is the kernel of a unique nontrivial character χ. Hence the individual isotypic component $H^2(S,\mathbb{Q})_\chi$ is already defined over $\mathbb{Q}$, and no grouping of characters is necessary. In this case, $W_H \otimes_{\mathbb{Q}} \mathbb{C} = H^2(S,\mathbb{C})_\chi$.

We now give a geometric interpretation of the rational Hodge substructures introduced above. Let $H \leq G$ be any subgroup and consider the quotient map

$$\pi_H : S \longrightarrow Y_H := S/H.$$

The surface Y_H may be singular. Nevertheless, since π_H is a finite morphism by a finite group, its rational cohomology is naturally identified with the H-invariant part of the cohomology of S.

Proposition 5.3. *Let S be a smooth projective surface endowed with a G-action, $G \cong (\mathbb{Z}/p)^k$, and let $H \leq G$ be a subgroup. Then the pullback map*

$$\pi_H^* : H^2(Y_H,\mathbb{Q}) \longrightarrow H^2(S,\mathbb{Q})$$

induces an isomorphism

$$H^2(Y_H,\mathbb{Q}) \xrightarrow{\sim} H^2(S,\mathbb{Q})^H.$$

This is an isomorphism of rational Hodge structures.

In particular, we have

$$H^2(Y_H,\mathbb{Q}) \cong W_0 \oplus W_H. \tag{43}$$

Proof. Since the quotient map $\pi_H : S \longrightarrow Y_H$ is finite, its fibers are finite and hence $R^i\pi_{H*}\mathbb{Q}_S = 0$ for all $i > 0$. Therefore, $H^2(S,\mathbb{Q}) \cong H^2(Y_H, \pi_{H*}\mathbb{Q}_S)$.

The group H acts naturally on the sheaf $\pi_{H*}\mathbb{Q}_S$. Since every fiber of π_H is an H-orbit, its invariant subsheaf is the constant sheaf: $(\pi_{H*}\mathbb{Q}_S)^H = \mathbb{Q}_{Y_H}$. Moreover, since we work with rational coefficients, taking H-invariants is an exact functor. It follows that

$$H^2(Y_H,\mathbb{Q}) \cong H^2(S,\mathbb{Q})^H.$$

The isomorphism is induced by the algebraic morphism π_H, hence it is compatible with the natural mixed Hodge structures. Since $H^2(S,\mathbb{Q})^H$ is a pure Hodge structure of weight 2, the same is true for $H^2(Y_H,\mathbb{Q})$, and the above isomorphism is therefore an isomorphism of rational Hodge structures.

Finally, $P_G + Q_H = P_H$ and $P_G \circ Q_H = 0$ gives

$$H^2(S,\mathbb{Q})^H = W_0 \oplus W_H,$$

and the last assertion follows. □

Applying Proposition 5.3 to the action of G itself, we obtain

$$H^2(Y_G,\mathbb{Q}) \cong H^2(S,\mathbb{Q})^G = W_0.$$

Hence, under the identification

$$H^2(Y_H, \mathbb{Q}) \cong W_0 \oplus W_H,$$

the summand W_0 is precisely the part coming from the total quotient Y_G, whereas W_H is the complementary rational Hodge substructure.

Thus W_H may be regarded as the new cohomological contribution associated with the $G/H \cong \mathbb{Z}/p$ -intermediate quotient $Y_H \longrightarrow Y_G$ of the G-quotient $\pi_G \colon S \longrightarrow Y_G$.

We now study how the second cohomology changes when passing from a quotient surface to a resolution of its singularities. In our setting, all quotient surfaces have only quotient singularities, and therefore their exceptional divisors consist of configurations of smooth rational curves.

Lemma 5.4. *Let Y be a normal projective surface with quotient singularities, and let $\rho : \widetilde{Y} \longrightarrow Y$ be a resolution of singularities. Denote by*

$$E := \bigcup_{x \in \mathrm{Sing}(Y)} E_x$$

the exceptional divisor, and let

$$\mathcal{E} := \langle [E_{x,j}] \mid x \in \mathrm{Sing}(Y),\ E_{x,j} \text{ an irreducible component of } E_x \rangle_{\mathbb{Q}} \subset H^2(\widetilde{Y}, \mathbb{Q}).$$

Then

$$H^2(\widetilde{Y}, \mathbb{Q}) = \rho^* H^2(Y, \mathbb{Q}) \oplus \mathcal{E}.$$

Moreover,

$$\mathcal{E} \subseteq \mathrm{NS}(\widetilde{Y}) \otimes \mathbb{Q}.$$

In particular, the resolution introduces only algebraic classes of type $(1,1)$.

Proof. This is standard for resolutions of quotient surface singularities; see, for instance, [**35**]. The exceptional fiber over each singular point is a tree of smooth rational curves, and its intersection matrix is negative definite.

The long exact sequence of the pair $(\widetilde{Y}, E)$, together with excision, gives

$$0 \longrightarrow H^2(Y, \mathbb{Q}) \xrightarrow{\rho^*} H^2(\widetilde{Y}, \mathbb{Q}) \longrightarrow H^2(E, \mathbb{Q}) \longrightarrow 0.$$

Since the intersection matrix of the exceptional curves is nondegenerate, their classes map isomorphically onto $H^2(E, \mathbb{Q})$. Hence $H^2(\widetilde{Y}, \mathbb{Q}) = \rho^* H^2(Y, \mathbb{Q}) \oplus \mathcal{E}$.

Finally, the classes generating $\mathcal{E}$ are classes of algebraic curves, so $\mathcal{E} \subset \mathrm{NS}(\widetilde{Y}) \otimes \mathbb{Q}$. □

We can now apply the previous lemma to the intermediate quotient $Y_H = S/H$ and compare the transcendental Hodge structures of its resolution with the H-invariant part of the transcendental cohomology of S.

Lemma 5.5. *Let S be a smooth projective surface endowed with a G-action, $G \cong (\mathbb{Z}/p)^k$. Let $H \leq G$, set $Y_H := S/H$, and let $\rho_H : \widetilde{Y}_H \longrightarrow Y_H$ be a resolution of singularities. Then there is a natural isomorphism of the transcendental part of the rational Hodge structures*

$$T(\widetilde{Y}_H)_{\mathbb{Q}} \cong T(S)^H_{\mathbb{Q}}.$$

Proof. By Proposition 5.3 and Lemma 5.4, we have a decomposition of rational Hodge structures

$$H^2(\widetilde{Y}_H, \mathbb{Q}) \cong H^2(S, \mathbb{Q})^H \oplus \mathcal{E}_H, \tag{44}$$

where $\mathcal{E}_H$ is the subspace generated by the classes of the irreducible components of the exceptional divisor of ρ_H. In particular, $\mathcal{E}_H \subseteq \mathrm{NS}(\widetilde{Y}_H)_{\mathbb{Q}}$.

On the other hand, the action of H preserves both $\mathrm{NS}(S)_{\mathbb{Q}}$ and the intersection pairing, hence it also preserves the transcendental Hodge structure $T(S)_{\mathbb{Q}} = \mathrm{NS}(S)^{\perp}_{\mathbb{Q}}$.

Thus the orthogonal decomposition $H^2(S,\mathbb{Q}) = \mathrm{NS}(S)_{\mathbb{Q}} \oplus T(S)_{\mathbb{Q}}$ is H-equivariant, so

$$H^2(S,\mathbb{Q})^H = \mathrm{NS}(S)^H_{\mathbb{Q}} \oplus T(S)^H_{\mathbb{Q}}.$$

Furthermore, note that

$$\mathrm{NS}(\widetilde{Y}_H)_{\mathbb{Q}} \cong \mathrm{NS}(S)^H_{\mathbb{Q}} \oplus \mathcal{E}_H.$$

Indeed, the rational $(1,1)$-classes in the summand $H^2(S,\mathbb{Q})^H$ are precisely $\mathrm{NS}(S)^H_{\mathbb{Q}}$ by the Lefschetz theorem on $(1,1)$-classes. Moreover, all classes belonging to $\mathcal{E}_H$ are algebraic. Conversely, if a rational algebraic class on $\widetilde{Y}_H$ is decomposed according to (44), then its component in $H^2(S,\mathbb{Q})^H$ is again of Hodge type $(1,1)$, and therefore belongs to $\mathrm{NS}(S)^H_{\mathbb{Q}}$. Consequently,

$$\frac{H^2(\widetilde{Y}_H,\mathbb{Q})}{\mathrm{NS}(\widetilde{Y}_H)_{\mathbb{Q}}} \cong \frac{H^2(S,\mathbb{Q})^H}{\mathrm{NS}(S)^H_{\mathbb{Q}}}.$$

Using the above orthogonal decompositions, the left-hand side is isomorphic to $T(\widetilde{Y}_H)_{\mathbb{Q}}$, while the right-hand side is isomorphic to $T(S)^H_{\mathbb{Q}}$. □

Remark 18. The isomorphism of Lemma 5.5 is compatible with the natural intersection pairings up to a scalar factor. More precisely, let

$$\Phi_H : T(\widetilde{Y}_H)_{\mathbb{Q}} \longrightarrow T(S)^H_{\mathbb{Q}}$$

denote the isomorphism induced by the quotient map π_H. Then, for every $\alpha, \beta \in T(\widetilde{Y}_H)_{\mathbb{Q}}$, one has

$$\big\langle \Phi_H(\alpha), \Phi_H(\beta) \big\rangle_S = |H| \, \langle \alpha, \beta \rangle_{\widetilde{Y}_H}.$$

Thus Φ_H is a similitude of polarized rational Hodge structures with multiplier $|H|$. Equivalently,

$$\Big(T(\widetilde{Y}_H)_{\mathbb{Q}}, |H| \langle\ ,\ \rangle_{\widetilde{Y}_H}\Big) \cong \big(T(S)^H_{\mathbb{Q}}, \langle\ ,\ \rangle_S\big)$$

as polarized rational Hodge structures.

Corollary 5.6. *Let S be a smooth projective surface endowed with a G-action, $G \cong (\mathbb{Z}/p)^k$ and assume that the quotient S/G is a rational surface.*

For every subgroup $H \leq G$ of index p, let $Y_H := S/H$ and let $\widetilde{Y}_H$ be a resolution of singularities of Y_H. Then there is an isomorphism of rational Hodge structures

$$T(S)_{\mathbb{Q}} \cong \bigoplus_{\substack{H \leq G \\ [G:H]=p}} T(\widetilde{Y}_H)_{\mathbb{Q}}, \tag{45}$$

where the decomposition is orthogonal with respect to the intersection pairing, and induces a similitude as polarized rational Hodge structures.

Proof. Applying Lemma 5.5 to the subgroup G itself give $T(S)^G_{\mathbb{Q}} \cong T(\widetilde{Y_G})_{\mathbb{Q}}$. Since Y_G is rational, its resolution is rational as well, and hence $T(S)^G_{\mathbb{Q}} = 0$.

The rational decomposition of Proposition 5.2 is preserved by the transcendental part, and therefore

$$T(S)_{\mathbb{Q}} = T(S)^G_{\mathbb{Q}} \oplus \bigoplus_{\substack{H \leq G \\ [G:H]=p}} (T(S)_{\mathbb{Q}} \cap W_H) = \bigoplus_{\substack{H \leq G \\ [G:H]=p}} (T(S)_{\mathbb{Q}} \cap W_H).$$

For a subgroup H of index p, we have

$$T(S)^H_{\mathbb{Q}} = T(S)^G_{\mathbb{Q}} \oplus (T(S)_{\mathbb{Q}} \cap W_H) = T(S)_{\mathbb{Q}} \cap W_H.$$

Combining this together with Lemma 5.5, we obtain

$$T(\widetilde{Y}_H)_{\mathbb{Q}} \cong T(S)_{\mathbb{Q}} \cap W_H.$$

Finally, the projectors P_H, P_G, and Q_H are self-adjoint with respect to the intersection pairing. Hence the different summands $T(S)_{\mathbb{Q}} \cap W_H$ are mutually orthogonal. Since $|H| = p^{k-1}$ for every subgroup $H \leq G$ of index p, the last assertion follows from Remark 18. □

We can now extend the decomposition (45) to SNC (p,k)-planes.

Theorem 5.7. *Let $\pi\colon X \longrightarrow Y$ be a SNC (p,k)-cover of a rational surface Y. Let $\rho : \widetilde{X} \longrightarrow X$ be the minimal resolution of singularities, and let $\widetilde{X}_{min}$ be its minimal model.*

For every subgroup $H \leq G$ of index p, set $Y_H := X/H$, and let $\widetilde{Y}_H$ be the minimal resolution of singularities of Y_H. Then there is an isomorphism of rational Hodge structures

$$T(\widetilde{X}_{min})_{\mathbb{Q}} \cong \bigoplus_{\substack{H \leq G \\ [G:H]=p}} T(\widetilde{Y}_H)_{\mathbb{Q}}.$$

Moreover, this decomposition is orthogonal with respect to the intersection pairing, and induces a similitude as polarized rational Hodge structures.

Proof. The action of G on X lifts to its minimal resolution $\widetilde{X}$. Moreover, the G-equivariant morphism ρ induces a birational morphism $\widetilde{X}/G \longrightarrow X/G \cong Y$. Hence $\widetilde{X}/G$ is rational.

We can therefore apply Corollary 5.6 to $\widetilde{X}$ and obtain

$$T(\widetilde{X})_{\mathbb{Q}} \cong \bigoplus_{\substack{H \leq G \\ [G:H]=p}} T(\widetilde{\widetilde{X}/H})_{\mathbb{Q}},$$

where $\widetilde{\widetilde{X}/H}$ denotes any smooth resolution of $\widetilde{X}/H$. For every subgroup $H \leq G$, the morphism $\widetilde{X} \longrightarrow X$ induces a birational morphism $\widetilde{X}/H \longrightarrow Y_H$. Thus $\widetilde{\widetilde{X}/H}$ and $\widetilde{Y}_H$ are birational smooth projective surfaces. Since the transcendental Hodge structure is a birational invariant of smooth projective surfaces, $T(\widetilde{\widetilde{X}/H})_{\mathbb{Q}} \cong T(\widetilde{Y}_H)_{\mathbb{Q}}$. Similarly, $\widetilde{X}$ and $\widetilde{X}_{min}$ are birational smooth projective surfaces, hence $T(\widetilde{X})_{\mathbb{Q}} \cong T(\widetilde{X}_{min})_{\mathbb{Q}}$.

Finally, birational transformations between smooth projective surfaces add or remove only algebraic classes generated by exceptional curves, and therefore preserve the intersection form on the transcendental part. The statement concerning the polarization follows from Corollary 5.6. □

We now specialize Theorem 5.7 to SNC (p,k)-planes satisfying Property ★.

Corollary 5.8. *Let X be a SNC (p,k)-plane satisfying Property ★. For every subgroup $H \leq G$ of index p, let $Y_H := X/H$, and denote by $\widetilde{Y_H}$ the minimal resolution of singularities of Y_H and with Y_H° its minimal model. Then*

$$T(X_{\min})_{\mathbb{Q}} \cong \bigoplus_{\substack{H \leq G,\ [G:H]=p \\ p_g(Y_H)=1}} T(Y_H^\circ)_{\mathbb{Q}}.$$

In particular, every nonzero summand on the right-hand side is the transcendental Hodge structure of a K3 surface. Moreover, it induces a similitude as polarized rational Hodge structures.

Proof. By Property ★, every intermediate $(p,1)$-plane Y_H has geometric genus either 0 or 1.

By Proposition 4.3, if $p_g(Y_H) = 0$, then the minimal resolution of Y_H is rational. Hence $T(\widetilde{Y}_H)_{\mathbb{Q}} = 0$. On the other hand, by Theorem 3.7, if $p_g(Y_H) = 1$, then its minimal model Y_H° is a K3 surface. Since $\widetilde{Y}_H$ and Y_H° are birational smooth projective surfaces, $T(\widetilde{Y}_H)_{\mathbb{Q}} \cong T(Y_H^\circ)_{\mathbb{Q}}$. The result now follows immediately from Theorem 5.7. □

Corollary 5.9. *Let X be a SNC (p,k)-plane, and let $\rho : \widetilde{X} \longrightarrow X$ be its minimal resolution of singularities. For every subgroup $H \leq G$ of index p, let $\widetilde{Y}_H$ be the minimal resolution of $Y_H := X/H$. Then*

$$\operatorname{rk} T(\widetilde{X}) = \sum_{\substack{H \leq G \\ [G:H]=p}} \operatorname{rk} T(\widetilde{Y}_H),$$

and

$$\rho(\widetilde{X}) = 12\chi(\mathcal{O}_{\widetilde{X}}) - K_{\widetilde{X}}^2 - 2 - \sum_{\substack{H \leq G \\ [G:H]=p}} \operatorname{rk} T(\widetilde{Y}_H). \tag{46}$$

Furthermore,

$$\rho(X) = 12\chi(\mathcal{O}_{\widetilde{X}}) - K_{\widetilde{X}}^2 - 2 - \sum_{\substack{H \leq G \\ [G:H]=p}} \operatorname{rk} T(\widetilde{Y}_H) - \sum_{x \in \operatorname{Sing}(X)} \ell_x,$$

where ℓ_x is the number of irreducible components of the exceptional divisor over the singular point x.

Proof. The first equality follows directly from Theorem 5.7. While Equation 46 is standard. Finally, the exceptional curves of the resolution of the quotient singularities of X are linearly independent in $\operatorname{NS}(\widetilde{X})_{\mathbb{Q}}$. Hence resolving a singular point x increases the Picard number by precisely ℓ_x. Thus

$$\rho(X) = \rho(\widetilde{X}) - \sum_{x \in \operatorname{Sing}(X)} \ell_x,$$

which gives the last formula. □

5.1. **The Mumford-Tate conjecture for SNC (p,k)-planes satisfying Property ★.** We conclude this section explaining briefly how to address the Mumford-Tate conjecture for SNC (p,k)-planes constructed in this paper.

A detailed and comprehensive treatment of the Mumford-Tate groups arising from Hodge structures of K3 type associated with abelian covers of rational surfaces can be found in [**23**]. We start with some facts on categories of motives and fix the notation that will be used here. For a more detailed introduction to the subject, we refer to [**32**, **33**].

Let $K \subseteq \mathbb{C}$ be a number field. We consider the category of André motives over K, that we denote with Mot_K, introduced in [**2**] to which we refer for all the definition and fact about this category. Therefore a *motive* is a triple (X, t, m), where X is a smooth projective variety over K, t is an idempotent motivated cycle on $X \times X$, and $m \in \mathbb{Z}$. Given a smooth projective variety X over K, we denote by $\operatorname{H}_{\mathrm{mot}}(X)$ the object $(X, \Delta_X, 0)$, where Δ_X is the class of the diagonal in $X \times X$.

This category of motives has several good properties, for example is Tannakian, semisimple, graded and polarized, see [**2**, Thm 0.2]. Moreover the Künneth decomposition always exists

$$\operatorname{H}_{\mathrm{mot}}(X) = \bigoplus_i \operatorname{H}^i_{\mathrm{mot}}(X),$$

where $\operatorname{H}^i_{\mathrm{mot}}(_)$ is the *i-th Künneth component* and is a contravariant functor from the category of smooth projective varieties over K to Mot_K, see [**2**, Proposition 2.2].

We denote with AbMot_K the Tannakian subcategory of *abelian motives* that are generated by the motives of the form $\operatorname{H}_{\mathrm{mot}}(X)$, where X is either an abelian variety or $X = \operatorname{Spec}(L)$ for some finite extension L/K with $L \subset \mathbb{C}$.

To conclude with the notation, for a given Weil cohomology $\mathcal{H}$ (here we restrict only to the Betti and the étale cohomology), we introduce the *realization functor* $H_{\mathcal{H}}(_)$ from Mot_K to $\mathrm{Vect}_{\mathbb{Q}}$ for the Betti cohomology and to $\mathrm{Vect}_{\mathbb{Q}_\ell}$ for the étale cohomology, such that $H_{\mathcal{H}} \circ \mathrm{H}_{\mathrm{mot}} = \mathcal{H}$. The Betti realization functor on Mot_K is faithful, see [**2**, §6].

We now want to consider the case of SNC (p,k)-planes.

Throughout the remainder of this subsection, we work over a number field $K \subset \mathbb{C}$ over which the covering, the action of $G \cong (\mathbb{Z}/p)^k$, and all the intermediate models and maps used below are defined. We also assume that $\mathrm{NS}(S_{\mathbb{C}})_{\mathbb{Q}}$ is generated by classes of divisors defined over K. These conditions can be achieved after a single finite extension of the original field of definition, see Subsection 1.3.

Let $\pi : X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane, and let $\rho : \widetilde{X} \longrightarrow X$ be its minimal resolution. We denote $S := \widetilde{X}$ for short. We shall work motivically with S, rather than with the singular model X. The action of G on X lifts naturally to S. So it makes sense to ask if the decomposition in (45) is induced by motivated correspondences.

Following the notation introduced in Proposition 5.2, for every subgroup $H \leq G$ of index p, we define

$$P_H := \frac{1}{|H|} \sum_{h \in H} \Gamma_h, \qquad P_G := \frac{1}{|G|} \sum_{g \in G} \Gamma_g, \qquad Q_H := P_H - P_G,$$

where $\Gamma_g \subset S \times S$ denotes the graph of the automorphism induced by g.

Lemma 5.10. *Let $S = \widetilde{X}$ be the minimal resolution of singularities of a SNC (p,k)-plane. Then P_G and the Q_H's define mutually orthogonal idempotent endomorphisms of $H^2_{\mathrm{mot}}(S)$, and*

$$P_G + \sum_{\substack{H \leq G \\ [G:H]=p}} Q_H = \mathrm{id}_{H^2_{\mathrm{mot}}(S)}\,.$$

Consequently, defined $M_0 := \mathrm{Im}(P_G)$ and $M_H := \mathrm{Im}(Q_H)$, we have

$$H^2_{\mathrm{mot}}(S) \simeq M_0 \oplus \bigoplus_{\substack{H \leq G \\ [G:H]=p}} M_H.$$

Moreover, under Betti realization, we have $H_B(M_0) = W_0$ and $H_B(M_H) = W_H$, where W_0 and W_H are the $\mathbb{Q}$-vector spaces that appear in Proposition 5.2.

Proof. The proof goes as the proof of Proposition 5.2 once one observes that P_H, P_G and Q_H are motived. □

The summands M_H may contain algebraic classes, including those arising from the resolution of singularities. We therefore separate their algebraic and transcendental parts and identify the latter with the transcendental motives of the intermediate K3 surfaces, whenever the SNC (p,k)-plane satisfies the further Property ★.

Proposition 5.11. *Let $\pi : X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane satisfying Property ★, and let $S = \widetilde{X}$ be its minimal resolution of singularities. For every subgroup $H \leq G$ of index p, let M_H be the motivated summand of $H^2_{\mathrm{mot}}(S)$ defined by Q_H.*

There exist motivated projectors π_{alg} and π_{tr} on $H^2_{\mathrm{mot}}(S)$ whose Betti realizations project onto $\mathrm{NS}(S)_{\mathbb{Q}}$ and $T(S)_{\mathbb{Q}}$, respectively.

Setting

$$M_H^{\mathrm{alg}} := \mathrm{Im}(Q_H \circ \pi_{\mathrm{alg}}), \qquad M_H^{\mathrm{tr}} := \mathrm{Im}(Q_H \circ \pi_{\mathrm{tr}}),$$

one obtains

$$M_H \simeq M_H^{\mathrm{alg}} \oplus M_H^{\mathrm{tr}}.$$

If $p_g(Y_H) = 1$, then there exists a motivated direct summand

$$T_{\mathrm{mot}}(Y_H^\circ) \subset H^2_{\mathrm{mot}}(Y_H^\circ)$$

whose Betti realization is $T(Y_H^\circ)_{\mathbb{Q}}$, and an isomorphism

$$M_H^{\mathrm{tr}} \simeq T_{\mathrm{mot}}(Y_H^\circ)$$

in Mot_K. *Moreover, M_H^{alg} is a direct sum of Lefschetz motives.*

Proof. Choose divisor classes $D_1, \dots, D_\rho$ defined over K and forming a basis of $\mathrm{NS}(S)_{\mathbb{Q}}$.

Let

$$A = (D_i \cdot D_j)_{i,j}, \qquad A^{-1} = (a^{ij})_{i,j}.$$

Since the intersection form on $\mathrm{NS}(S)_{\mathbb{Q}}$ is nondegenerate, the algebraic correspondence

$$\pi_{\mathrm{alg}} := \sum_{i,j=1}^{\rho} a^{ij} D_i \times D_j$$

induces the orthogonal projector $H^2(S, \mathbb{Q}) \longrightarrow \mathrm{NS}(S)_{\mathbb{Q}}$.

Thus, π_{alg} is a motivated projector on $H^2_{\mathrm{mot}}(S)$. We set $\pi_{\mathrm{tr}} := \mathrm{id}_{H^2_{\mathrm{mot}}(S)} - \pi_{\mathrm{alg}}$. Its Betti realization is the orthogonal projector onto $T(S)_{\mathbb{Q}}$.

Since the action of G preserves both $\mathrm{NS}(S)_{\mathbb{Q}}$ and the intersection pairing, these projectors commute with every Q_H, $H \leq G$. Consequently,

$$M_H \simeq M_H^{\mathrm{alg}} \oplus M_H^{\mathrm{tr}},$$

with

$$H_B(M_H^{\mathrm{tr}}) = W_H \cap T(S)_{\mathbb{Q}}.$$

Suppose now that $p_g(Y_H) = 1$, so from Theorem 3.7 the minimal model Y_H° of Y_H is a K3 surface. Looking at the proof of Corollary 5.6, we have

$$W_H \cap T(S)_{\mathbb{Q}} \cong T(Y_H^\circ)_{\mathbb{Q}}.$$

Consider the rational map $S \dashrightarrow Y_H^\circ$ induced by the quotient map $\pi_H \colon S \longrightarrow Y_H$ and the corresponding birational transformations. We may resolve its indeterminacies over K:

$$\begin{array}{ccc} Z_H & \xrightarrow{f_H} & Y_H^\circ \\ {\scriptstyle \mu_H}\downarrow & \nearrow & \\ S & & \end{array}$$

where μ_H is birational and f_H is generically finite of degree $d_H = |H|$.

Let C_H be the algebraic correspondence induced by this diagram, acting on Betti cohomology as

$$(C_H)_B = \mu_{H*} \circ f_H^*.$$

Since C_H is algebraic, it is motivated. Define also

$$\alpha_H := (\pi_{\mathrm{tr}} \circ Q_H) \circ C_H : H^2_{\mathrm{mot}}(Y_H^\circ) \longrightarrow M_H^{\mathrm{tr}}$$

and

$$\beta_H := \frac{1}{d_H}\, {}^t C_H \circ (Q_H \circ \pi_{\mathrm{tr}}) : M_H^{\mathrm{tr}} \longrightarrow H^2_{\mathrm{mot}}(Y_H^\circ).$$

By the transcendental decomposition established above, the Betti realization of α_H restricts to an isomorphism

$$(\alpha_H)_B : T(Y_H^\circ)_{\mathbb{Q}} \xrightarrow{\sim} W_H \cap T(S)_{\mathbb{Q}}.$$

We claim that β_H induces its inverse. Indeed, for $u \in T(Y_H^\circ)_{\mathbb{Q}}$, the class $f_H^* u$ belongs to $T(Z_H)_{\mathbb{Q}}$. Since μ_H is birational, one has

$$\mu_H^* \circ \mu_{H*} \circ f_H^* u = f_H^* u.$$

Therefore, using the projection formula,

$$\begin{aligned}({}^tC_H \circ C_H)u &= (f_{H*} \circ \mu_H^* \circ \mu_{H*} \circ f_H^*)u \\ &= (f_{H*} \circ f_H^*)u \\ &= d_H u.\end{aligned}$$

Since $Q_H \pi_{\mathrm{tr}}$ acts as the identity on the transcendental image of C_H, it follows that $(\beta_H \circ \alpha_H)_B = \mathrm{id}_{T(Y_H^\circ)_{\mathbb{Q}}}$. As $(\alpha_H)_B$ is an isomorphism onto $H_B(M_H^{\mathrm{tr}})$, we also obtain $(\alpha_H \circ \beta_H)_B = \mathrm{id}_{H_B(M_H^{\mathrm{tr}})}$.

By the faithfulness of the Betti realization functor on motivated motives,

$$\alpha_H \circ \beta_H = \mathrm{id}_{M_H^{\mathrm{tr}}}.$$

Setting

$$e_H := \beta_H \circ \alpha_H \in \mathrm{End}_{\mathrm{Mot}_K}\big(H^2_{\mathrm{mot}}(Y_H^\circ)\big),$$

note that $e_H^2 = e_H$. Since C_H maps algebraic classes to algebraic classes, the Betti realization of α_H vanishes on $\mathrm{NS}(Y_H^\circ)_{\mathbb{Q}}$. Together with $(\beta_H \circ \alpha_H)_B|_{T(Y_H^\circ)_{\mathbb{Q}}} = \mathrm{id}_{T(Y_H^\circ)_{\mathbb{Q}}}$, this shows that $(e_H)_B$ is the orthogonal projector onto $T(Y_H^\circ)_{\mathbb{Q}}$.

Defining

$$T_{\mathrm{mot}}(Y_H^\circ) := \mathrm{Im}(e_H),$$

we obtain mutually inverse motivated isomorphisms

$$M_H^{\mathrm{tr}} \simeq T_{\mathrm{mot}}(Y_H^\circ).$$

Finally, M_H^{alg} is a direct summand of the motivated algebraic part of $H^2(S)$, which is a direct sum of Lefschetz motives. □

Theorem 5.12. *Let $\pi : X \longrightarrow \mathbb{P}^2$ be a SNC (p,k)-plane satisfying Property ★, defined over a number field $K \subset \mathbb{C}$, and let $S := \widetilde{X}$ be its minimal resolution of singularities. Then both the Mumford–Tate conjecture and the Tate conjecture hold for S. Consequently, they also hold for its minimal model $\widetilde{X}_{\min}$.*

Proof. By Lemma 5.10, there is a decomposition in the category of motivated motives

$$H^2_{\mathrm{mot}}(S) \simeq M_0 \oplus \bigoplus_{\substack{H \leq G \\ [G:H]=p}} M_H.$$

By Proposition 5.11, each summand admits a decomposition

$$M_H \simeq M_H^{\mathrm{alg}} \oplus M_H^{\mathrm{tr}}.$$

Since the total quotient $X/G \cong \mathbb{P}^2$ is rational, the G-invariant transcendental part vanishes. Hence M_0 is entirely algebraic.

If $p_g(Y_H) = 0$, then Y_H is rational by Proposition 4.3, Consequently, $M_H^{\mathrm{tr}} = 0$, and the corresponding summand is algebraic.

If $p_g(Y_H) = 1$, Proposition 5.11 gives

$$M_H^{\mathrm{tr}} \simeq T_{\mathrm{mot}}(Y_H^\circ),$$

where Y_H° is a K3 surface, by Theorem 3.7.

Combining these observations, we obtain

$$H^2_{\mathrm{mot}}(S) \simeq M_{\mathrm{alg}} \oplus \bigoplus_{\substack{H\leq G,\ [G:H]=p \\ p_g(Y_H)=1}} T_{\mathrm{mot}}(Y_H^\circ),$$

where M_{alg} is an algebraic motive which is a direct sum of Lefschetz motives.

By André's theorem, see [**2**], the motivated cohomology of every K3 surface is an abelian motive. The Mumford–Tate conjecture holds for the transcendental motive of each K3 surface. Since these are abelian motives, Commelin's result [**11**, Theorem 10.3] (see also [**10**]), implies that it also holds for their finite direct sum, together with the Lefschetz summands. It follows that the Mumford–Tate conjecture holds for $H^2(S)$.

In degree two, the Tate conjecture follows from the Mumford–Tate conjecture and the Lefschetz theorem on $(1,1)$-classes. The remaining degrees are immediate for a smooth projective regular surface. Therefore the Tate conjecture holds for S. Finally, passing from S to its minimal model $\widetilde{X}_{\min}$ only removes algebraic Lefschetz summands from the degree-two motive. Thus both conjectures hold for $\widetilde{X}_{\min}$ as well. □

6. Invariants of SNC (p,k)-planes satisfying Property ★

We now proceed to study further properties of the SNC (p,k)-planes $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★ achieved in Section 2. In particular, we determine their Kodaira dimension and discuss the minimality of the minimal resolution of singularities $\rho\colon \widetilde{X} \longrightarrow X$. Moreover, we also list their numerical invariants.

We remind that SNC $(p,1)$-planes satisfying Property ★, so those having geometric genus one, do not need to be studied: from Theorem 3.7 their minimal model is always a K3 surface. Furthermore, the minimal resolution of singularities $\widetilde{X}$ is a minimal surface only for $p=2$, while it is not minimal for the remaining cases $3 \leq p \leq 19$, and we need to contract $(-K^2_{\widetilde{X}})$-curves to obtain the minimal model.

Thus, in this section we only consider the case $k \geq 2$. The main theorem is as follows

Theorem 6.1. *Let us consider a SNC (p,k)-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★ and with $k \geq 2$. The minimal resolution of singularities $\widetilde{X}$ of X is a regular surface. It is a minimal surface with the exception of cases $C-(3,2)$, $A-(5,2)$, and $B-(7,2)$, whose minimal models are obtained by contracting a single (-1)-curve on $\widetilde{X}$.*

The surface $\widetilde{X}$ is always of general type, except in the following cases:

- *$\widetilde{X}_{min}$ is a K3 surface in cases $A-(2,2)$, $A-(2,3)$, $B-(2,3)$, $A-(2,4)$, $A-(2,5)$, $C-(3,2)$, $A-(5,2)$;*
- *$\widetilde{X}$ is a minimal properly elliptic surface (with Kodaira dimension one) in cases $B-(3,2)$ and $C-(3,3)$.*

In the following tables, we list for each case the self-intersection of a canonical divisor of the minimal model $\widetilde{X}_{min}$, along with its geometric genus, topological Euler characteristic, Kodaira dimension, and surface type. Finally, we also list in tables 17 to 29 the Picard number of $\widetilde{X}_{min}$ together with the transcendental part of the Hodge structure of a very general member of the family.

case	p_g	q	$K^2_{\widetilde{X}_{min}}$	$e(\widetilde{X}_{min})$	κ	$\widetilde{X}$ *minimal*	*Surface Name*
A-(2,2)	1	0	0	24	0	✓	*K3*
B-(2,2)	1	0	1	23	2	✓	*Special Kunev*
C-(2,2)	2	0	1	35	2	✓	*(1,2)-surface, i.e. I-surface*
D-(2,2)	2	0	4	32	2	✓	*Garbagnati Type* 2
E-(2,2)	3	0	9	39	2	✓	*Garbagnati Type* 3
A-(2,3)	1	0	0	24	0	✓	*K3*
B-(2,3)	1	0	0	24	0	✓	*K3*
C-(2,3)	3	0	2	46	2	✓	*Horikawa*
D-(2,3)	2	0	2	34	2	✓	*Special Horikawa*
E-(2,3)	3	0	8	40	2	✓	*C3 surface [**16**]*
F-(2,3)	1	0	2	22	2	✓	*Todorov surface*
G-(2,3)	3	0	8	40	2	✓	*E3 surface [**16**]*
H-(2,3)	4	0	8	52	2	✓	
I-(2,3)	5	0	18	54	2	✓	
J-(2,3)	2	0	8	28	2	✓	
K-(2,3)	4	0	18	42	2	✓	
L-(2,3)	4	0	18	42	2	✓	
M-(2,3)	6	0	32	52	2	✓	
A-(2,4)	1	0	0	24	0	✓	*K3*
B-(2,4)	4	0	4	56	2	✓	*Horikawa*
C-(2,4)	3	0	4	44	2	✓	*D4 surface [**16**]*
D-(2,4)	2	0	4	32	2	✓	
E-(2,4)	5	0	16	56	2	✓	
F-(2,4)	5	0	16	56	2	✓	
G-(2,4)	6	0	16	68	2	✓	
H-(2,4)	9	0	36	84	2	✓	
I-(2,4)	1	0	4	20	2	✓	
J-(2,4)	4	0	16	44	2	✓	
K-(2,4)	8	0	36	72	2	✓	
L-(2,4)	9	0	36	84	2	✓	
M-(2,4)	3	0	16	32	2	✓	*C4 surface [**16**]*
N-(2,4)	7	0	36	60	2	✓	
O-(2,4)	7	0	36	60	2	✓	
P-(2,4)	6	0	36	48	2	✓	
Q-(2,4)	11	0	64	80	2	✓	
R-(2,4)	10	0	64	68	2	✓	

TABLE 7. Invariants of the minimal model $\widetilde{X}_{min}$ for the resolution of singularities $\widetilde{X}$ of a SNC (p,k)-plane X.

case	p_g	q	$K^2_{\widetilde{X}_{min}}$	$e(\widetilde{X}_{min})$	κ	$\widetilde{X}$ *minimal*	*Surface Name*
A-(2,5)	1	0	0	24	0	✓	*K3, Hirzebruch type*
B-(2,5)	5	0	8	64	2	✓	
C-(2,5)	3	0	8	40	2	✓	*D5 surface [**16**]*
D-(2,5)	9	0	32	88	2	✓	
E-(2,5)	9	0	32	88	2	✓	
F-(2,5)	7	0	32	64	2	✓	
G-(2,5)	15	0	72	120	2	✓	
H-(2,5)	13	0	72	96	2	✓	
I-(2,5)	13	0	72	96	2	✓	
J-(2,5)	21	0	128	136	2	✓	
A-(2,6)	7	0	16	80	2	✓	*Hirzebruch type*
B-(2,6)	15	0	64	128	2	✓	
C-(2,6)	27	0	144	192	2	✓	
A-(3,2)	1	0	1	23	2	✓	*Special Kunev*
B-(3,2)	2	0	0	36	1	✓	*Properly elliptic*
C-(3,2)	1	0	0	24	0	×	*K3*
D-(3,2)	3	0	9	27	2	✓	
E-(3,2)	2	0	8	28	2	✓	
A-(3,3)	3	0	3	45	2	✓	
B-(3,3)	1	0	3	21	2	✓	
C-(3,3)	2	0	0	36	1	✓	*Properly elliptic*
D-(3,3)	5	0	27	45	2	✓	
A-(3,4)	5	0	9	63	2	✓	*Hirzebruch type*
A-(5,2)	1	0	0	24	0	×	*K3*
B-(5,2)	2	0	1	35	2	✓	*(1,2)-surface, i.e. I-surface*
A-(5,3)	4	0	5	55	2	✓	*Fermat quintic in* $\mathbb{P}^3$
A-(7,2)	2	0	9	27	2	✓	
B-(7,2)	3	0	2	46	2	×	*Horikawa*

TABLE 8. Invariants of the minimal model $\widetilde{X}_{min}$ for the resolution of singularities $\widetilde{X}$ of a SNC (p,k)-plane X.

Proof. Any SNC (p,k)-plane is a regular surface by Lemma 1.15. Using Theorem 1.14 one can compute the invariants of $\widetilde{X}$ from the building data of the covering. Thus, using Corollary 5.9, one can compute case-by-case the Picard number of $\widetilde{X}$ together with the transcendental part of the Hodge structure for very general member of the family, as explained in the later Example 7.1.

Finally, the remaining information such as the minimality of $\widetilde{X}$, the Kodaira dimension, and the surface type are obtained applying lemmas 6.2, 6.3, 6.4, 6.5, 6.6, 6.7, 6.8, which are proved below. □

Lemma 6.2. *Any (p,k)-plane, with $k \geq 2$, satisfying Property ★ and with at most canonical singularities is a minimal surface. Furthermore, they are of general type, with the exception of the cases $A-(2,2)$, $A-(2,3)$, $B-(2,3)$, $A-(2,4)$, $A-(2,5)$, which are $K3$ surfaces.*

Proof. According to Enriques-Kodaira classification, any surface S with $p_g(S) \geq 1$ and $K_S^2 \geq 1$ is a surface of general type.

Let us consider a SNC (p,k)-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★, with $k \geq 2$. Then from Theorem 1.14, we have

$$p_g(\widetilde{X}) = |\{\chi\colon l_\chi = 3\}|,$$

which is greater than or equal to one from Property ★ and Proposition 2.2.

Assume that X has at most canonical singularities. Then from Theorem 1.14 one has

$$pK_{\widetilde{X}} \equiv \widetilde{\pi}^* \left(\left(-3p + (p-1) \sum_g d_g \right) H \right).$$

Tables from 17 to 29 show that $-3p + (p-1)\sum_g d_g \geq 0$. Since the pullback of a nef divisor by a generically finite map is itself nef, $pK_{\widetilde{X}}$ (and thus $K_{\widetilde{X}}$) is nef.

Therefore, if $K^2_{\widetilde{X}} \geq 1$, then $\widetilde{X}$ is a minimal surface of general type.

On the other hand, if $K^2_{\widetilde{X}} = 0$, then $(p-1)\sum_g d_g = 3p$, which forces $p = 2$ and $\sum_g d_g = 6$. In this case, we would have $2K_{\widetilde{X}} \equiv 0$. Furthermore, by inspecting tables 17 through 23, the only admissible cases are $A-(2,2)$, $A-(2,3)$, $B-(2,3)$, $A-(2,4)$, $A-(2,5)$. These cases also satisfy $p_g(\widetilde{X}) = 1$, $q(\widetilde{X}) = 0$; thus, by Enriques-Kodaira classification, they are $K3$ surfaces. □

Remark 19. Lemma 6.2 already covers most of the cases of tables 17 to 29. Indeed, it remains to discuss only cases $B-(3,2)$, $C-(3,2)$, $E-(3,2)$, $C-(3,3)$, $A-(5,2)$, $B-(7,2)$.

For the subsequent lemmas, we fix the following notation. The map $\rho\colon \widetilde{X} \to X$ denotes the minimal resolution of singularities of X, and $\widetilde{\pi}$ denotes the composition $\widetilde{\pi} \coloneqq \pi \circ \rho\colon \widetilde{X} \to \mathbb{P}^2$. Let Δ_g be a smooth irreducible curve in the branch locus of π, with local monodromy$g \in G$. According to Proposition 1.16, the reduced divisor of $\widetilde{\pi}^*\Delta_g$ decomposes into the sum of an irreducible smooth curve $\widetilde{\Delta_g}$ (the strict transform of $\pi^*\Delta_g$ under ρ) and some Hirzebruch-Jung strings of the resolution. We shall repeatedly invoke Proposition 1.16 to determine the configuration of $\widetilde{\Delta_g}$ and these Hirzebruch-Jung strings, and to compute the self-intersection and genus of $\widetilde{\Delta_g}$.

The case $B-(3,2)$ in Lemma 6.3 is discussed in more details in the later Example 7.2. One can use it as a guide for all the other remaining lemmas.

Lemma 6.3. *Any $(3,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $B-(3,2)$ is a minimal properly elliptic surface of Kodaira dimension one. The canonical map $\varphi_{|K_{\widetilde{X}}|}\colon \widetilde{X} \longrightarrow \mathbb{P}^1$ is a morphism with a generic fibre that is a smooth elliptic curve.*

Proof. Let us consider the line $l \coloneqq D_{e_2}$. From Proposition 1.16, the reduced pullback of l under $\widetilde{\pi}\colon \widetilde{X} \longrightarrow \mathbb{P}^2$ consists of the three exceptional (-3)-curves E_1, E_2, and E_3 of the resolution of X, together with the strict transform $\widetilde{l}$. More precisely, $\widetilde{\pi}^* l = 3\widetilde{l} + E_1 + E_2 + E_3$. Furthermore, $\widetilde{l}$ intersects each exceptional curve E_1, E_2 and E_3 transversally at a single point. Proposition 1.16 gives the self-intersection and the genus of $\widetilde{l}$:

$$\widetilde{l}^2 = 1 - 3 \cdot \frac{1}{3} = 0, \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 3\left(-2 + 3\left(1 - \frac{1}{3}\right)\right) \implies g(\widetilde{l}) = 1.$$

Thus, $\widetilde{l}$ is a smooth elliptic curve of self-intersection zero.

From Theorem 1.14, we have

$$3K_{\widetilde{X}} \equiv \widetilde{\pi}^* H - (E_1 + E_2 + E_3) \equiv 3\widetilde{l}.$$

Consequently, $\widetilde{X}$ is a minimal surface. Indeed, a hypothetical (-1)-curve E would have $3K_{\widetilde{X}}E = -3 < 0$, which would force E to be a fixed component of $|3K_{\widetilde{X}}|$. However, this is impossible as E would have to be $\widetilde{l}$, which is instead an elliptic curve.

Therefore, $\widetilde{X}$ is a minimal surface with $K^2_{\widetilde{X}} = 0$, and, by Theorem 1.14, we have $p_g(\widetilde{X}) = 2, q(\widetilde{X}) = 0$. By the Enriques-Kodaira classification, $\widetilde{X}$ is a properly elliptic surface (Kodaira dimensione one). More precisely, we claim that the canonical map of $\widetilde{X}$ is an elliptic fibration. Indeed, $\varphi_{|K_{\widetilde{X}}|}$ is a morphism since $K^2_{\widetilde{X}} = 0$ and it has not fixed part since any divisor of the canonical linear system $|K_{\widetilde{X}}|$ is a component of the fibration $\varphi_{|K_{\widetilde{X}}|}$. Finally, by Bertini the general curve F of the linear system is smooth and irreducible and from the genus formula it has $g(F) = 1 + K^2_{\widetilde{X}} = 1$, so it is an elliptic curve. □

Lemma 6.4. *Any $(3,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $C-(3,2)$ has a minimal resolution of singularities $\widetilde{X}$ that is not a minimal surface. The minimal model of $\widetilde{X}$ is a K3 surface and it is obtained after contracting a single (-1)-curve of $\widetilde{X}$.*

Proof. Consider the line $l \coloneqq D_{e_1+e_2}$. According to Proposition 1.16, we have

$$\widetilde{\pi}^* l \coloneqq 3\widetilde{l} + E_1 + \cdots + E_6$$

where $E_1, \ldots, E_6$ are the (-3)-curves arising from the resolution of X and $\widetilde{l}$ is the strict transform of l. The curve $\widetilde{l}$ intersects transversally each (-3)-curve E_i at a single point. Applying Proposition 1.16, we compute its self-intersection and genus:

$$\widetilde{l}^2 = 1 - 3 \cdot 2 \cdot \frac{1}{3} = -1, \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 3\left(-2 + 2 \cdot \left(1 - \frac{1}{3}\right)\right) \implies g(\widetilde{l}) = 0.$$

Thus, $\widetilde{l}$ is a (-1)-curve of $\widetilde{X}$.

From Theorem 1.14, we have

$$3K_{\widetilde{X}} = \widetilde{\pi}^* H - (E_1 + \cdots + E_6) \equiv_{lin} 3\widetilde{l}.$$

After contracting the (-1)-curve $\widetilde{l}$, the divisor $3K_{\widetilde{X}'}$ of the resulting surface $\widetilde{X}'$ becomes trivial, implying $\widetilde{X}'$ is a minimal surface. Furthermore, since $K^2_{\widetilde{X}'} = 0$, $p_g(\widetilde{X}') = 1$ and $q(\widetilde{X}') = 0$ by Theroem 1.14, it follows that $\widetilde{X}'$ is a K3 surface. □

Lemma 6.5. *Any $(3,2)-$plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $E-(3,2)$ is a minimal surface of general type with $p_g(\widetilde{X}) = 2$, $q(\widetilde{X}) = 0$, and $K^2_{\widetilde{X}} = 8$.*

Proof. Consider the line $l \coloneqq D_{e_2}$. According to Proposition 1.16, the pullback is

$$\widetilde{\pi}^* l \coloneqq 3\widetilde{l} + E_1 + E_2 + E_3$$

where E_1, E_2, E_3 are the (-3)-curves of the resolution of X and $\widetilde{l}$ is the strict transform of l. The curve $\widetilde{l}$ intersects each (-3)-curve E_i transversally at a single point. Applying Proposition 1.16, we obtain

$$\widetilde{l}^2 = 1 - 3 \cdot 1 \cdot \frac{1}{3} = 0, \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 3\left(-2 + 4\left(1 - \frac{1}{3}\right)\right) \implies g(\widetilde{l}) = 2.$$

Now, let us consider the remain lines $t_1 \coloneqq D_{2e_1+e_2}$ and $t_2 \coloneqq D_{2e_1+2e_2}$ together with their strict transforms $\widetilde{t_1}$ and $\widetilde{t_2}$. We have $\widetilde{\pi}^* t_i = 3\widetilde{t_i}$, with $\widetilde{t_i}^2 = \widetilde{t_i} \cdot \widetilde{l} = \widetilde{t_1} \cdot \widetilde{t_2} = 1$. From Theorem 1.14, we have

$$3K_{\widetilde{X}} \equiv_{lin} \widetilde{\pi}^* 3H - (E_1 + E_2 + E_3) = 3(\widetilde{t_1} + \widetilde{t_2} + \widetilde{l}) \implies K^2_{\widetilde{X}} = 8.$$

Furthermore, $\widetilde{X}$ is a minimal surface, as any hypothetical $(-1)-$curve would have to be a fixed component of $|3K_{\widetilde{X}}|$, namely one among $\widetilde{t_1}$, $\widetilde{t_2}$ or $\widetilde{l}$, neither of which is a (-1)-curve. Finally, using Theoroem 1.14, we compute $q(\widetilde{X}) = 0$ and $p_g(\widetilde{X}) = 2$. □

Lemma 6.6. *Any $(3,3)-$plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $C-(3,3)$ is a minimal properly elliptic surface of Kodaira dimension one. The canonical map $\varphi_{|K_{\widetilde{X}}|}\colon \widetilde{X} \longrightarrow \mathbb{P}^1$ is a morphism with a generic fibre that is a smooth elliptic curve.*

Proof. Consider the line $l \coloneqq D_{e_2}$. Applying Proposition 1.16, we have

$$\widetilde{\pi}^* l = 3\widetilde{l} + E_1 + \cdots + E_9,$$

where the E_i are the (-3)-curves of the resolution of X. This gives

$$\widetilde{l}^2 = 3 - 9 \cdot \frac{1}{3} = 0 \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 9\left(-2 + 3 \cdot \left(1 - \frac{1}{3}\right)\right) \implies g(\widetilde{l}) = 1.$$

Furthermore, from Theorem 1.14, we have

$$3K_{\widetilde{X}} = \widetilde{\pi}^* H - (E_1 + \cdots + E_9) \equiv 3\widetilde{l}$$

Thus, $\widetilde{X}$ is a minimal surface, since any hypothetical (-1)-curve E would be a fixed component of the linear system $|3K_{\widetilde{X}}|$, namely E must be $\widetilde{l}$, which is instead an elliptic curve. Finally, we have a minimal surface with $K^2_{\widetilde{X}} = 0$. By Theorem 1.14, then $p_g(\widetilde{X}) = 2$ and $q(\widetilde{X}) = 0$, so it follows from Enriques-Kodaira classification that $\widetilde{X}$ is a properly elliptic surface. Finally, the canonical map of $\widetilde{X}$ is an elliptic fibration following the same argument in the proof of Lemma 6.3. □

Lemma 6.7. *The $(5,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $A-(5,2)$ has a minimal resolution of singularities $\widetilde{X}$ that is not a minimal surface. The minimal model of $\widetilde{X}$ is a K3 surface and it is obtained after contracting a single (-1)-curve of $\widetilde{X}$.*

Proof. Consider the line $l \coloneqq D_{e_1+e_2}$. Applying Proposition 1.16, we have

$$\widetilde{\pi}^* l \coloneqq 5\widetilde{l} + \sum_{i=1}^{5} \left(2E_1^i + E_2^i\right),$$

where E_1^i and E_2^1 are respectively the (-3) and (-2)-curves of the resolution of the i-th cyclic quotient singularity $\frac{1}{5}(1,2)$ on X, for $i = 1, \ldots, 5$. The strict transform $\widetilde{l}$ of l intersects each (-3)-curve E_1^i transversally at a single point and it is disjoint from the (-2)-curves E_2^i. Thus, we have from Proposition 1.16

$$\widetilde{l}^2 = 1 - 5 \cdot \frac{2}{5} = -1, \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 5\left(-2 + 2 \cdot \left(1 - \frac{1}{5}\right)\right) \implies g(\widetilde{l}) = 0.$$

Thus, $\widetilde{l}$ is a (-1)-curve of $\widetilde{X}$. From Theorem 1.14, we have

$$5K_{\widetilde{X}} = \widetilde{\pi}^* H - \sum_{i=1}^{5} \left(2E_1^i + E_2^i\right) \equiv_{lin} 5\widetilde{l}.$$

After contracting the (-1)-curve $\widetilde{l}$, the divisor $5K_{\widetilde{X}'}$ of the resulting surface $\widetilde{X}'$ becomes trivial, implying that $\widetilde{X}'$ is a minimal surface. Furthermore, since $p_g(\widetilde{X}') = 1$ and $q(\widetilde{X}') = 0$ by Theorem 1.14, it follows that $\widetilde{X}'$ is a K3 surface. □

Lemma 6.8. *The $(7,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ of the case $B-(7,2)$ has a minimal resolution of singularities $\widetilde{X}$ that is not a minimal surface. The minimal model $\widetilde{X}_{min}$ is obtained after contracting a single (-1)-curve of $\widetilde{X}$. Moreover, $\widetilde{X}_{min}$ is a Horikawa surface with $K^2_{\widetilde{X}_{min}} = 2$, $q(\widetilde{X}_{min}) = 0$ and $p_g(\widetilde{X}_{min}) = 3$.*

Proof. Consider the line $l \coloneqq D_{e_1+e_2}$. From Proposition 1.16, we have

$$\widetilde{\pi}^* l \coloneqq 7\widetilde{l} + \sum_{i=1}^{7} \left(2E_1^i + E_2^i\right),$$

where E_1^i and E_2^i are, respectively, the (-4) and (-2)-curves of the resolution of the i-th cyclic quotient singularity $\frac{1}{7}(1,2)$ on X, for $i = 1,\dots,7$. The strict transform $\widetilde{l}$ intersects each (-4)-curve E_1^i transversally at a single point, and is disjoint from the (-2)-curves E_2^i. Thus, we find from Proposition 1.16

$$\widetilde{l}^2 = 1 - 7\cdot\frac{2}{7} = -1, \qquad \text{and} \qquad 2g(\widetilde{l}) - 2 = 7\left(-2 + 2\cdot\left(1-\frac{1}{7}\right)\right) \implies g(\widetilde{l}) = 0.$$

Accordingly, $\widetilde{l}$ is a (-1)-curve on $\widetilde{X}$. Now, consider the line $t \coloneqq D_{e_1}$ and its strict transform $\widetilde{t}$. We have $\widetilde{\pi}^* t = 7\widetilde{t}$ and $\widetilde{t}\cdot\widetilde{l} = 1$. From Theorem 1.14, then

$$7K_{\widetilde{X}} \equiv \widetilde{\pi}^* 3H - \sum_{i=1}^{7}\left(4E_1^i + 2E_2^i\right) \equiv 7(\widetilde{t} + 2\widetilde{l}). \tag{47}$$

This implies $K^2_{\widetilde{X}} = 1 + 4(-1) + 2\cdot 2 = 1$.

Let $b\colon \widetilde{X} \longrightarrow \widetilde{X}'$ be the contraction of the (-1)-curve $\widetilde{l}$. Then (47) yields $7b^*K_{\widetilde{X}'} \equiv 7\widetilde{t} + 7\widetilde{l}$, so $7K_{\widetilde{X}'} \equiv_{lin} 7b_*\widetilde{t}$. Since $\widetilde{t}$ intersects $\widetilde{l}$ transversally at one point, we have $b^*b_*\widetilde{t} = \widetilde{t} + \widetilde{l}$, which implies $(b_*\widetilde{t})^2 = 1 + (-1) + 2 = 2$. Therefore, $b_*\widetilde{t}$ is not a (-1)-curve, and $\widetilde{X}'$ is a minimal surface with $K^2_{\widetilde{X}'} = K^2_{\widetilde{X}} + 1 = 2$. Finally, from Theorem 1.14, we have $q(\widetilde{X}') = 0$ and $p_g(\widetilde{X}') = 3$, confirming that $\widetilde{X}'$ is a Horikawa surface. □

7. Explicit examples of SNC (p,k)-planes satisfying Property ★

We present two concrete examples of (p,k)-planes that illustrate how the output of the algorithm explained in Section 2 is translated into geometry. As a byproduct, we will show some interesting phenomenons can occur.

Example 7.1. We consider the case $H-(2,3)1)$ of Table 18. So we are dealing with a smooth $(2,3)$-plane $\pi\colon S \longrightarrow \mathbb{P}^2$. This corresponds to the second output produced by the algorithm explained at the end of Section 2.

In this case, the building data $\{D_g\}_{g\in G}$ and $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ of the $(2,3)$-plane, where $G \cong (\mathbb{Z}/2)^3 = \langle e_1, e_2, e_3\rangle$, have the following degrees:

$$d_{e_2+e_3} = 4, \qquad d_{e_1} = d_{e_2} = d_{e_3} = d_{e_1+e_2+e_3} = 1, \quad \text{and} \quad d_g = 0 \quad \text{otherwise.}$$
$$l_{\epsilon_1} = l_{\epsilon_2\epsilon_3} = 1, \qquad l_{\epsilon_1\epsilon_2\epsilon_3} = 2, \qquad l_\chi = 3 \quad \text{for all other } \chi \in G^* \setminus 1.$$

Thus, the branch divisor consists of four lines and a smooth quartic curve, that we colour as follows respectively for the convenience:

- D_{e_1} (black), D_{e_2} (orange), D_{e_3} (green), $D_{e_1+e_2+e_3}$ (red), and
- $D_{e_2+e_3}$ (blue),

as shown in Figure 7.

In the picture, we also indicate the following intermediate $(2,1)$-planes and their branch locus:

$$Y_1 \coloneqq S/\langle e_2, e_3\rangle, \qquad Y_2 \coloneqq S/\langle e_1, e_3\rangle, \qquad Y_3 \coloneqq S/\langle e_1, e_2\rangle,$$
$$Y_4 \coloneqq S/\langle e_1+e_2, e_3\rangle, \qquad Y_5 \coloneqq S/\langle e_1+e_3, e_2\rangle, \quad Y_6 \coloneqq S/\langle e_2+e_3, e_1\rangle,$$
$$Y_7 \coloneqq S/\langle e_2+e_3, e_1+e_2\rangle.$$

We now describe one of these intermediate quotients, namely Y_1, that corresponds to the subgroup $\ker(\epsilon_1) = \langle e_2, e_3\rangle \leq G$. The others are treated analogously.

According to Lemma 2.1, since $\epsilon_1(e_j) = (-1)^{\delta_{1j}}$, then $\langle \epsilon_1, e_j\rangle = \delta_{1j}$ and so the branch locus of the intermediate $(2,1)$-plane $Y_1 \longrightarrow \mathbb{P}^2$ consists of the two lines $D_{e_1} + D_{e_1+e_2+e_3}$. Accordingly, in Figure 7 the branch locus of Y_1 is given by the black and the red lines.

Since the double cover is branched along exactly two lines, it follows from Theorem 4.1 (see Table 2)) that Y_1 is birational to $\mathbb{F}_2$. More precisely, the surface Y_1 has a single node, and blowing up this point yields $\mathbb{F}_2$. In this example, the intermediate quotients Y_2, Y_3, Y_4, and Y_5 are all $(2,1)$-planes branched along a reduced singular sextic curve. Consequently, each of them has geometric genus one, and their minimal models are K3 surfaces, see Theorem 3.7 and Table 9. The remaining intermediate quotients Y_6 and Y_7 can be described similarly. As for Y_6, the double cover is again branched along two lines and is therefore birational to $\mathbb{F}_2$. The remaining quotient is branched along four lines and is birational to a Weak del Pezzo surface of degree 2, see again Theorem 4.1 and Table 2.

We obtained that the intermediate $(2,1)$-planes are either rational or K3 surfaces, so S satisfies Property ★.

Hence, counting the number of the intermediate $(2,1)$-planes that are K3 surfaces, the geometric genus of S is $p_g(S) = 4$. Since in this case the branch divisor of π is simple normal crossing and $p = 2$, the surface S is smooth, see Proposition 1.10. Furthermore, S is regular from Lemma 1.15. Denoting with H the hyperplane section, the remaining numerical invariants of S can be computed directly using the formulae of Proposition 1.14 (see also Remark 6)

$$\chi(\mathcal{O}_S) = 8 + \frac{1}{2}\big(H(H-3H) + H(H-3H) + 2H(2H-3H)\big) = 8 + \frac{1}{2}(-2-2-2) = 5,$$

$$K_S \equiv \frac{1}{2}\pi^*(-6H + (1+1+1+1+4)H) = \pi^*(H), \qquad \text{and} \qquad K_S^2 = 2^{3-2}\,(-6H+8H)^2 = 8.$$

Consequently, by Noether formula we have $e(S) = 52$. We conclude that S is a minimal surface of general type with ample canonical class.

We now examine more closely the weight–2 Hodge structure of S. By Corollary 5.6, the transcendental lattice $T(S)_{\mathbb{Q}}$ decompose as direct sum of the corresponding contributions coming from the intermediate $(2,1)$-planes $Y_1, \ldots, Y_7$.

The Hodge numbers of the intermediate double covers can be found in tables 2 and 9 depending if Y_i is a rational or a K3 surface respectively. More precisely, the rational surfaces Y_i with $i = 1, 6$ have

$$\big(h^{2,0}(Y_i),\, h^{1,1}(Y_i),\, h^{0,2}(Y_i)\big) = (0,1,0),$$

while for the rational surface Y_7 we obtain

$$\big(h^{2,0}(Y_7),\, h^{1,1}(Y_7),\, h^{0,2}(Y_7)\big) = (0,2,0).$$

Since these surfaces are rational, their Hodge structures are entirely algebraic and have no transcendental part. Moreover, in each case one generator of $H^{1,1}(Y_i)$ is given by the pullback of the hyperplane class on $\mathbb{P}^2$.

We now turn to the intermediate quotients Y_2, Y_3, Y_4, Y_5, whose minimal models are K3 surfaces. From Theorem 3.7 we have that the very general member of their families have $\rho(Y_i) = 1$, and the generator of the Picard group is exactly the pull back of the hyperplane section of $\mathbb{P}^2$.

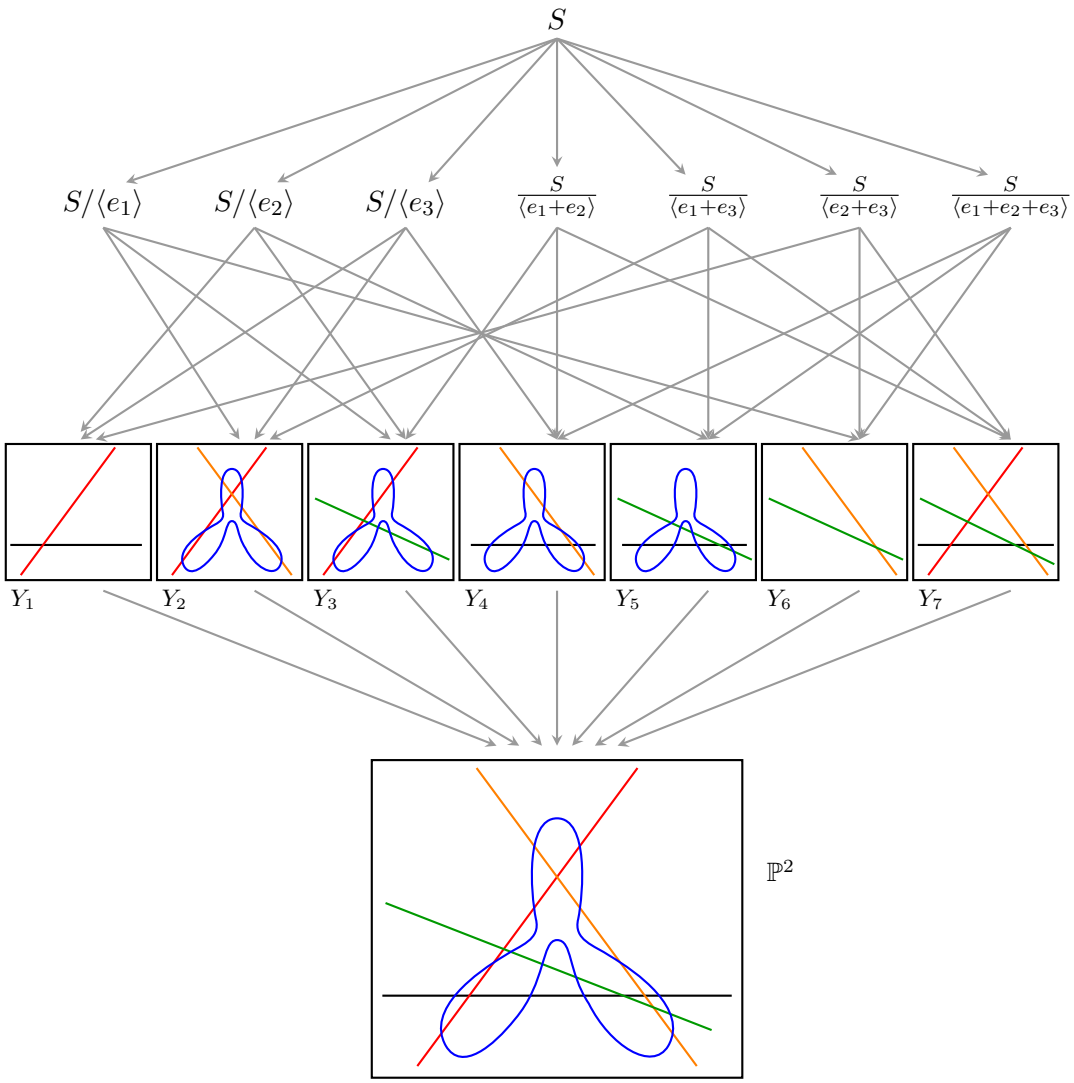


FIGURE 7. The branch locus of the case $H-(2,3)1)$ together with the branch loci of its intermediate $(2,1)$-planes $Y_1,\ldots,Y_7$.

Moreover, since all $(2,1)$-planes $Y_i \longrightarrow \mathbb{P}^2$ are ramified over a quartic and two lines, looking at Table 9 we have that the Hodge components $H^2_{tra}(Y_i)$ are of dimension $(1,10,1)$, respectively, for $i=2,3,4,5$. We have that the Hodge diamond of S is then

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 4 & & 42 & & 4 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

and $H^2_{tra}(S)=(4,40,4)$ while the rational Neron-Severi group is generated by the pullback of the hyperplane section and an algebraic class coming from Y_7.

To conclude the example we present S in a suitable weighted projective space according to the method explained in Subsection 1.3. Let $(x_0 : x_1 : x_2)$ be homogeneous coordinates on $\mathbb{P}^2$ and let

$$f_{e_1},\, f_{e_2},\, f_{e_3},\, f_e \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1)), \qquad f_{e_2+e_3} \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(4))$$

be defining equations of the branch components $D_{e_1}, D_{e_2}, D_{e_3}, D_{e_1+e_2+e_3}$ (lines) and $D_{e_2+e_3}$ (smooth quartic), respectively. Setting $\varepsilon \coloneqq \epsilon_1\epsilon_2\epsilon_3$ for short, we recall that in this case

$$\mathcal{L}_{\epsilon_1} \cong \mathcal{O}_{\mathbb{P}^2}(1), \qquad \mathcal{L}_{\epsilon_2+\epsilon_3} \cong \mathcal{O}_{\mathbb{P}^2}(1), \qquad \mathcal{L}_{\varepsilon} \cong \mathcal{O}_{\mathbb{P}^2}(2),$$

and $\mathcal{L}_\chi \cong \mathcal{O}_{\mathbb{P}^2}(3)$ for all other nontrivial $\chi \in G^*$.

We embed S into the weighted projective space $\mathbb{P}(1,1,1,1,1,2,3,3,3,3)$ with weighted homogeneous coordinates

$$(x_0 : x_1 : x_2 :\ y_{\epsilon_1} : y_{\epsilon_2+\epsilon_3} :\ y_{\varepsilon} :\ y_{\epsilon_2} : y_{\epsilon_3} : y_{\epsilon_1\epsilon_2} : y_{\epsilon_1\epsilon_3}),$$

with $\deg x_i = 1$, $\deg y_{\epsilon_1} = \deg y_{\epsilon_2\epsilon_3} = 1$, $\deg y_\varepsilon = 2$, and the remaining y_χ have degree 3.

Note from Theorem 1.3 (cf. Subsection 1.3) that one of the equations defining S is:

$$y_{\epsilon_1}\, y_{\epsilon_2\epsilon_3} = y_\varepsilon.$$

This tells us that we can eliminate the variable y_ε and consider the weighted projective space with one variable less $\mathbb{P}(1,1,1,1,1,3,3,3,3)$, by projection. Thus, S is defined by the following relations, see Subsection 1.3:

$$\operatorname{rk}\begin{pmatrix} f_{e_1}\, f_{e_2+e_3} & y_{\epsilon_1\epsilon_2} & y_{\epsilon_1\epsilon_3} & y_{\epsilon_1}\, f_{e_2+e_3} \\ y_{\epsilon_1\epsilon_2} & f_{e_2} & y_{\epsilon_2\epsilon_3} & y_{\epsilon_2} \\ y_{\epsilon_1\epsilon_3} & y_{\epsilon_2\epsilon_3} & f_{e_3} & y_{\epsilon_3} \\ y_{\epsilon_1}\, f_{e_2+e_3} & y_{\epsilon_2} & y_{\epsilon_3} & f_\varepsilon\, f_{e_2+e_3} \end{pmatrix} \le 1, \qquad y_{\epsilon_1}^2 = f_{e_1} f_\epsilon. \tag{48}$$

Note that the projection onto the first three coordinates gives the $(2,3)$-plane $\pi\colon S \longrightarrow \mathbb{P}^2$.

Finally, from Theorem 5.12 the Mumford–Tate and Tate conjectures hold or S.

Remark 20. We point out that the equations (48) defining the $(2,3)$-plane S of the case $H-(2,3)1)$ directly give a presentation for all the other cases $H-(2,3)2), \ldots, H-(2,3)5)$. Indeed, it sufficies to replace the irreducible polynomial $f_{e_2+e_3}$ with a suitable factorization of it to provide equations of the (singular) corresponding SNC $(2,3)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$.

Example 7.2. We consider case $B-(3,2)1)$ in Table 24. Thus, we are dealing with a SNC $(3,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$, where X has three cyclic quotient singularities of type $\frac{1}{3}(1,1)$. This case was roughly studied in Lemma 6.3, where it was proved that $\widetilde{X}$ is a properly elliptic surface (of Kodaira dimension one).

Hence, this case exhibits an interesting phenomenon: the singularities of a (p,k)-plane may cause the Kodaira dimension of its minimal model to jump from the expected value.

For this reason, we will study this example in greater detail and show an explicit elliptic fibration of the minimal model, which will coincide with its canonical map.

The building data $\{D_g\}_{g\in G}$ and $\{\mathcal{L}_\chi\}_{\chi\in G^*}$ of the $(3,2)$-plane have the following degrees, where $G \cong (\mathbb{Z}/3)^2 = \langle e_1, e_2\rangle$:

$$\begin{gathered} d_{e_1} = 3, \qquad d_{e_2} = d_{2e_2} = 1, \quad \text{and} \quad d_g = 0 \quad \text{otherwise.} \\ l_{\epsilon_1} = l_{\epsilon_2} = l_{\epsilon_2^2} = 1, \qquad l_{\epsilon_1^2\epsilon_2^2} = l_{\epsilon_1^2\epsilon_2} = 3, \qquad l_\chi = 2 \quad \text{for all other } \chi \in G^* \setminus 1. \end{gathered} \tag{49}$$

Thus, the branch divisor consists of two lines ℓ_1, ℓ_2 and a smooth cubic curve $\mathcal{C}$ in general position, that we colour as follows:

- $\mathcal{C} \coloneqq D_{e_1}$ (green),
- $\ell_1 \coloneqq D_{e_2}$ (blue), $\ell_2 \coloneqq D_{2e_2}$ (red),

as shown in Figure 8.

In Figure 8, we also indicate the three intermediate $(3,1)$-planes and their branch locus:

$$Y_1 \coloneqq X/\langle e_1\rangle, \quad Y_2 \coloneqq X/\langle e_2\rangle, \quad Y_3 \coloneqq X/\langle e_1+e_2\rangle, \quad \text{and} \qquad Y_4 \coloneqq X/\langle e_1+2e_2\rangle.$$

According to Lemma 2.1, the branch locus of the intermediate $(3,1)$-plane $Y_1 \longrightarrow \mathbb{P}^2$ consists of the two lines $\ell_1+\ell_2$. Since the triple cover is branched along exactly two lines, it follows from Theorem 4.1 (see Table 3)) that Y_1 is birational to $\mathbb{F}_3$. More precisely, the surface Y_1 has a single cyclic quotient singularity of type $\frac{1}{3}(1,1)$, and blowing up this point yields $\mathbb{F}_3$.

Similarly, one can say that $Y_2 \longrightarrow \mathbb{P}^2$ is branched over only a smooth cubic and still from Theorem 4.1 (see Table 3) we have Y_2 is a smooth del Pezzo of degree 3.

In addition, the remaining two cases Y_3 and Y_4 are both branched over the total branch locus of $X \longrightarrow \mathbb{P}^2$ (but with different local monodromies). Hence, from Theorem 3.7 these surfaces are K3 and correspond to the case $A-(3,1)2)$ in Table 10.

Thus, we have obtained that the intermediate quotients are either rational or K3 surfaces, so $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfies Property $\bigstar$.

Hence, counting the number of the intermediate $(3,1)$-planes that are K3 surfaces, the geometric genus of X is $p_g(X)=2$.

As mentioned above, the $(3,2)-$plane is singular only over the intersection point of the branch components l_1 and l_2, and above such a point there are 3 singularities of type $\frac{1}{3}(1,1)$, according to Proposition 1.12. The same proposition tells us that over the other intersection points of the branch locus no other singularities occur.

We denote by $\rho\colon \widetilde{X} \longrightarrow X$ the minimal resolution of singularities of X, by E_1, E_2 and E_3 the (-3)-exceptional curves, and by $\widetilde{\pi}\colon \widetilde{X} \longrightarrow \mathbb{P}^2$ the composition $\widetilde{\pi} \coloneqq \pi \circ \rho$.

The surface $\widetilde{X}$ is regular from Lemma 1.15. The remaining numerical invariants of $\widetilde{X}$ can be computed directly using the formulae of Proposition 1.14

$$\begin{aligned}
\chi(\mathcal{O}_{\widetilde{X}}) &= 9+\frac{1}{2}\big(3H(H-3H)+6H(2H-3H)+2H(3H-3H)\big) = 9+\frac{1}{2}(-6-6)=3,\\
K_{\widetilde{X}} &\equiv \frac{1}{3}\widetilde{\pi}^*(-9H+(3-1)(1+1+3)H)-(E_1+E_2+E_3) = \frac{1}{3}\pi^*(H)-(E_1+E_2+E_3),\\
K_{\widetilde{X}}^2 &= 3^{2-2}\left(-9H+10H\right)^2-3\frac{1}{3}=0.
\end{aligned}$$

Consequently, by Noether formula we have $e(\widetilde{X})=36$. Furthermore, from Proposition 1.16, the strict transforms $\widetilde{\ell_1}$ and $\widetilde{\ell_2}$ of the lines ℓ_1 and ℓ_2 on $\widetilde{X}$ are

$$\widetilde{\ell_1} \equiv \widetilde{\ell_2} \equiv \frac{1}{3}\widetilde{\pi}^*(H)-(E_1+E_2+E_3).$$

Thus, $K_{\widetilde{X}} \equiv \widetilde{\ell_1} \equiv \widetilde{\ell_2}$ and as already explained in Lemma 6.3, then $\widetilde{X}$ is minimal and properly elliptic.

We now exhibit an elliptic fibration of $\widetilde{X}$. We consider first the resolution $\widetilde{Y_1} \longrightarrow Y_1$ of Y_1, which is the Hirzebruch surface $\mathbb{F}_3$ as explained above. We note that $\widetilde{Y_1}$ is also a triple cover of the blow-up $\mathrm{Bl}_{pt}(\mathbb{P}^2)$ branched over the strict transforms ℓ_1 and ℓ_2, where pt is the intersection point of ℓ_1 and ℓ_2. From here one can easily see the natural ruling $q\colon \widetilde{Y_1} \longrightarrow \mathbb{P}^1$ of $\widetilde{Y_1}=\mathbb{F}_3$: for any line ℓ in $\mathbb{P}^2$ passing through pt there are three fibres of the ruling, that becomes a single fibre counted with multiplicity 3 when ℓ coincides with one between ℓ_1 and ℓ_2.

We now consider the strict transform $\widetilde{\mathcal{C}}$ of the cubic curve $\mathcal{C}$ on $\widetilde{Y_1}$. We observe that $\widetilde{\mathcal{C}}$ is 3-divisible in $\mathrm{Pic}(\widetilde{Y_1})$, so there is a triple cover $\pi'\colon X' \longrightarrow \widetilde{Y_1}$ branched along $\widetilde{C}$. Since $\mathcal{C}$ intersects any line passing through pt in three points, then $\widetilde{\mathcal{C}}$ intersects each fibre of q in three points.

Thus, X' inherits a natural fibration $\pi' \circ q \colon X' \longrightarrow \mathbb{P}^1$ whose general fibre is a triple cover of $\mathbb{P}^1$ branched over three points, hence is an elliptic curve.

We also observe that X' has three disjoint (-3)-curves, that are the pull-back of the unique (-3)-curve of $\widetilde{Y_1}$, which is not intersecting $\widetilde{\mathcal{C}}$ and is the resolution of the singular point of type $\frac{1}{3}(1,1)$ lying over the point pt. Furthermore, X' is a smooth minimal surface and if we contract those three (-3)-curves on X', then we obtain the surface X, by construction. In other words, $X' \longrightarrow X$ is the minimal resolution of the singularities of X, $\widetilde{X} = X'$. Thus, we have exhibited an elliptic fibration $\pi' \circ q \colon \widetilde{X} \longrightarrow \mathbb{P}^1$, which according to Lemma 6.3 is the canonical map of $\widetilde{X}$, since they have the same general fibres.

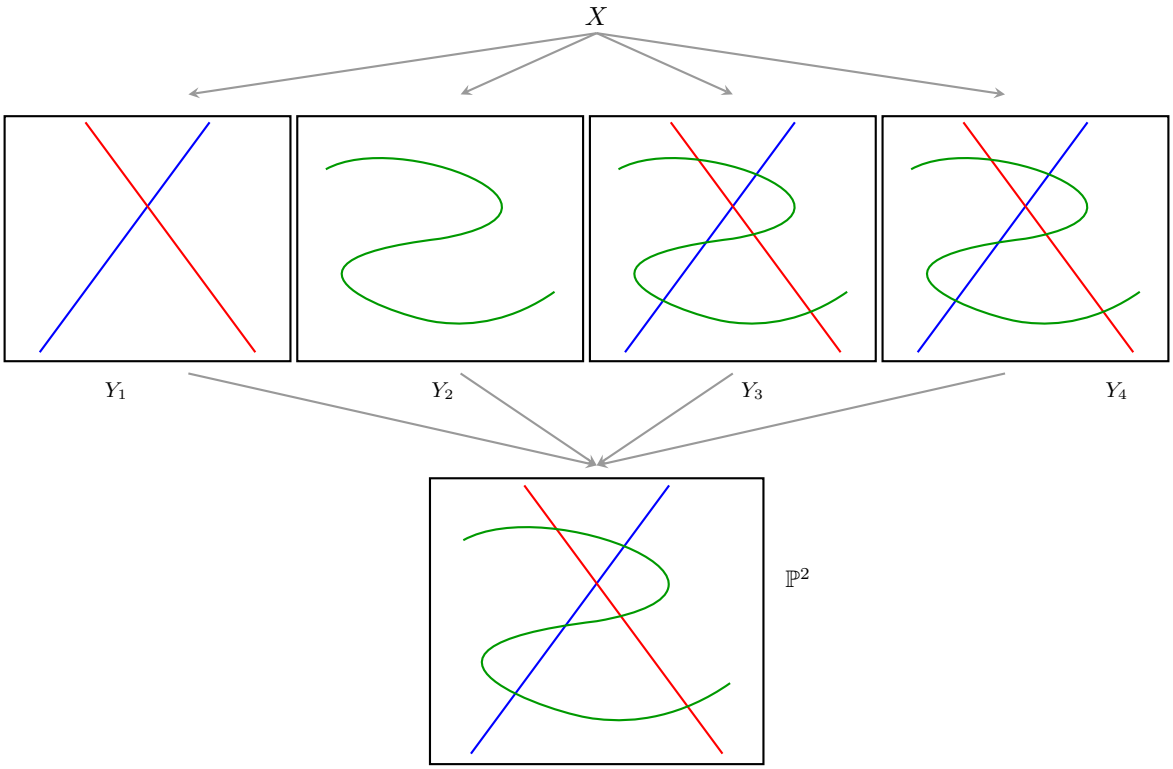


FIGURE 8. The branch locus of the case $B-(3,2)1)$ together with the branch loci of its intermediate $(3,1)$-planes $Y_1, \ldots, Y_4$.

We now examine more closely the weight–2 Hodge structure of $\widetilde{X}$. By Corollary 5.8, the transcendental lattice $T(\widetilde{X})_{\mathbb{Q}}$ decomposes as direct sum of the corresponding contributions coming from the intermediate $(3,1)$-planes that are K3 surface, hence Y_3 and Y_4. Looking at the case $A-(3,1)2)$ in Table 10, then we have $\big(h^{2,0}(Y_i),\, h^{1,1}(Y_i),\, h^{0,2}(Y_i)\big)_{tra} = (1,10,1)$ for $i = 3,4$. To complete the study of $H^2(\widetilde{X},\mathbb{Q})$ we need to look at Y_1 and Y_2. Indeed, for these two rational surfaces we have $h^{1,1}(Y_1) = 1$ and $h^{1,1}(Y_2) = 7$, see Table 3. Thus, $\big(h^{2,0}(\widetilde{X}),\, h^{1,1}(\widetilde{X}),\, h^{0,2}(\widetilde{X})\big)_{tra} = (2,20,2)$ and $\rho(\widetilde{X}) = 7+3 = 10$, where 7 is the contribution given by Y_2 and 3 are the three (-3)-curves of the resolution of X. We point out that the

computation holds for a very general member $\widetilde{X}$ of the family. The Hodge diamond of $\widetilde{X}$ is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 2 & & 30 & & 2 \\ & 0 & & 0 & \\ & & 1 & & \end{array}.$$

The same kind of result can be obtained with different methods using one among [**22**], [**42**], and [**31**] to the (singular) triple cover $X \longrightarrow Y_3$.

To conclude this example we present X in a suitable weighted projective space according to the method explained in Subsection 1.3. Let $(x_0 : x_1 : x_2)$ be homogeneous coordinates on $\mathbb{P}^2$ and let

$$f_{e_1} \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3)), \qquad f_{e_2}\, f_{2e_2}, \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1))$$

be defining equations of the branch components D_{e_1} (smooth cubic) and D_{e_2}, D_{2e_2} (lines), respectively. We recall the degrees of the $\mathcal{L}_\chi$, $\chi \in G^*$, listed in Equation 49.

Thus, we embed X into the weighted projective space $\mathbb{P}(1,1,1,1,1,2,2,2,3,3)$ with weighted homogeneous coordinates

$$(x_0 : x_1 : x_2 : \ y_{\epsilon_1} : y_{\epsilon_2} : \ y_{\epsilon_2^2} : \ y_{\epsilon_1^2} : \ y_{\epsilon_1\epsilon_2^2} : y_{\epsilon_1\epsilon_2} : y_{\epsilon_1^2\epsilon_2^2} : y_{\epsilon_1^2\epsilon_2}),$$

with $\deg x_i = 1$, $\deg y_{\epsilon_1} = \deg y_{\epsilon_2} = \deg y_{\epsilon_2^2} = 1$, $\deg y_{\epsilon_1^2\epsilon_2^2} = \deg y_{\epsilon_1^2\epsilon_2} = 3$, and the remaining y_χ have degree 2.

Note that from Theorem 1.3 (cf. Subsection 1.3) five of the equations defining X are:

$$y_{\epsilon_1}^2 = y_{\epsilon_1^2}, \qquad y_{\epsilon_1} y_{\epsilon_2^2} = y_{\epsilon_1\epsilon_2^2}, \qquad y_{\epsilon_1} y_{\epsilon_2} = y_{\epsilon_1\epsilon_2}, \qquad y_{\epsilon_1^2} y_{\epsilon_2^2} = y_{\epsilon_1^2\epsilon_2^2}, \qquad y_{\epsilon_1^2} y_{\epsilon_2} = y_{\epsilon_1^2\epsilon_2}.$$

This tells us that we can eliminate the last five variables and consider X defined in $\mathbb{P}^5$, by projection. Thus, X is defined by the following equations, see Subsection 1.3:

$$\operatorname{rk}\begin{pmatrix} f_{2e_2} & y_{\epsilon_2} & y_{\epsilon_2^2} \\ y_{\epsilon_2} & y_{\epsilon_2^2} & f_{e_2}, \end{pmatrix} \leq 1, \qquad y_{\epsilon_1}^3 = f_{e_1}. \tag{50}$$

Finally, the projection onto the first three coordinates gives the $(3,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$.

Furthermore, one can easily see that the elliptic fibration studied above is the (unique) fibration $\widetilde{X} \longrightarrow \mathbb{P}^1$ obtained by resolving the indeterminacy locus of the rational map

$$X \dashrightarrow \mathbb{P}^1, \quad (x_0 : x_1 : x_2 : \ y_{\epsilon_1} : y_{\epsilon_2} : \ y_{\epsilon_2^2} : \ y_{\epsilon_1^2}) \mapsto (f_{2e_2}(x_0, x_1, x_2) : -f_{e_2}(x_0, x_1, x_2)).$$

8. Appendix: Tables

In this appendix, we collect the complete list of SNC (p,k)-planes $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★ produced by the algorithm described in Section 2, together with the Picard number of the minimal model and the transcendental part of the Hodge structure of a very general member of the family of $\widetilde{X}_{min}$. The results are organized into several tables according to the corresponding values of (p,k).

For each admissible pair (p,k), we list all possible configurations of divisors $\{D_g\}_{g\in G}$. Cases arising from the same collection of degrees $\{d_g\}_{g\in G}$ are grouped together. Indeed, for fixed degrees d_g, the divisor D_g may decompose into several smooth irreducible components, provided that their total degree is d_g and that the overall branch divisor remains in simple normal crossing position.

For each case, we also record the basket of singularities of the corresponding (p,k)-plane, computed using Proposition 1.12. Any divisor D_g not explicitly appearing in the tables is understood to be zero.

Each case is identified by a letter, the pair (p,k), and a numerical index. The letter distinguishes cases corresponding to different collections of degrees $\{d_g\}_{g\in G}$, while the numerical index separates cases with identical degree data but distinct choices for the decomposition of the divisors $\{D_g\}_{g\in G}$.

case	No.	D_{e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,1)$	1)	sextic	$\emptyset$	1	(1,19,1)
	2)	quintic + line	$1/2^5$	6	(1,14,1)
	3)	quartic + conic	$1/2^8$	9	(1,11,1)
	4)	quartic + 2 lines	$1/2^9$	10	(1,10,1)
	5)	cubic + cubic	$1/2^9$	10	(1,10,1)
	6)	cubic + conic +line	$1/2^{11}$	12	(1,8,1)
	7)	cubic + 3 lines	$1/2^{12}$	13	(1,7,1)
	8)	3 conics	$1/2^{12}$	13	(1,7,1)
	9)	2 conics + 2 lines	$1/2^{13}$	14	(1,6,1)
	10)	conic + 4 lines	$1/2^{14}$	15	(1,5,1)
	11)	6 lines	$1/2^{15}$	16	(1,4,1)

Table 9. Building data of a SNC $(2,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X)=1$.

case	No.	D_{e_1}	D_{2e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(3,1)$	1)	quartic	line	$1/3^4$	4	(1,16,1)
	2)	cubic + line	line	$1/3^4, 2/3^3$	10	(1,10,1)
	3)	2 conics	line	$1/3^4, 2/3^4$	12	(1,8,1)
	4)	conic + 2 lines	line	$1/3^4, 2/3^5$	14	(1,6,1)
	5)	4 lines	line	$1/3^4, 2/3^6$	16	(1,4,1)

Table 10. Building data of a SNC $(3,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X)=1$.

case	No.	D_{3e_1}	D_{4e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(5,1)$	1)	line	cubic	$3/5^3$	6	(1,14,1)
	2)	line	conic + line	$3/5^3, 4/5^2$	14	(1,6,1)
	3)	line	3 lines	$3/5^3, 4/5^3$	18	(1,2,1)

TABLE 11. Building data of a SNC $(5,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{4e_1}	D_{5e_1}	D_{6e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(7,1)$	1)	line	line	conic	$2/7^2, 3/7^2, 4/7$	10	(1,10,1)
	2)	line	line	2 lines	$2/7^2, 3/7^2, 4/7, 6/7$	16	(1,4,1)

TABLE 12. Building data of a SNC $(7,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{6e_1}	D_{8e_1}	D_{9e_1}	D_{10e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(11,1)$	1)	line	line	line	line	$2/11, 3/11,$ $4/11, 5/11,$ $6/11, 7/11$	12	(1,8,1)

TABLE 13. Building data of a SNC $(11,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{8e_1}	D_{9e_1}	D_{10e_1}	D_{12e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(13,1)$	1)	line	line	line	line	$2/13, 3/13,$ $4/13, 5/13,$ $7/13, 9/13$	10	(1,10,1)

TABLE 14. Building data of a SNC $(13,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{10e_1}	D_{12e_1}	D_{14e_1}	D_{15e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(17,1)$	1)	line	line	line	line	$2/17, 3/17,$ $5/17, 7/17,$ $9/17, 13/17$	6	(1,14,1)

TABLE 15. Building data of a SNC $(17,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{12e_1}	D_{14e_1}	D_{15e_1}	D_{16e_1}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(19,1)$	1)	line	line	line	line	$2/19, 3/19,$ $4/19, 5/19,$ $7/19, 13/19$	4	(1,16,1)

TABLE 16. Building data of a SNC $(19,1)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ with $p_g(X) = 1$.

case	No.	D_{e_1}	D_{e_2}	$D_{e_{12}}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,2)$	1)	quartic	conic	0	$\emptyset$	9	(1,11,1)
	2)	cubic + line	conic	0	$1/2^6$	12	(1,8,1)
	3)	2 conics	conic	0	$1/2^8$	13	(1,7,1)
	4)	conic + 2 lines	conic	0	$1/2^{10}$	14	(1,6,1)
	5)	4 lines	conic	0	$1/2^{12}$	15	(1,5,1)
	6)	quartic	2 lines	0	$1/2^2$	10	(1,10,1)
	7)	cubic + line	2 lines	0	$1/2^8$	13	(1,7,1)
	8)	2 conics	2 lines	0	$1/2^{10}$	14	(1,6,1)
	9)	conic + 2 lines	2 lines	0	$1/2^{12}$	15	(1,5,1)
	10)	4 lines	2 lines	0	$1/2^{14}$	16	(1,4,1)
$B-(2,2)$	1)	cubic	cubic	line	$\emptyset$	9	(1,10,1)
	2)	cubic	conic + line	line	$1/2^4$	11	(1,8,1)
	3)	cubic	3 lines	line	$1/2^6$	12	(1,7,1)
	4)	conic + line	conic + line	line	$1/2^8$	13	(1,6,1)
	5)	conic + line	3 lines	line	$1/2^{10}$	14	(1,5,1)
	6)	3 lines	3 lines	line	$1/2^{12}$	15	(1,4,1)
$C-(2,2)$	1)	quintic	line	line	$\emptyset$	1	(2,28,2)
	2)	quartic + line	line	line	$1/2^8$	9	(2,20,2)
	3)	cubic + 2 lines	line	line	$1/2^{14}$	15	(2,14,2)
	4)	cubic + conic	line	line	$1/2^{12}$	13	(2,16,2)
	5)	conic + 3 lines	line	line	$1/2^{18}$	19	(2,10,2)
	6)	2 conics + line	line	line	$1/2^{16}$	17	(2,12,2)
	7)	5 lines	line	line	$1/2^{20}$	21	(2,8,2)
$D-(2,2)$	1)	quartic	conic	conic	$\emptyset$	4	(2,22,2)
	2)	cubic + line	conic	conic	$1/2^6$	10	(2,16,2)
	3)	2 conics	conic	conic	$1/2^8$	12	(2,14,2)
	4)	conic + 2 lines	conic	conic	$1/2^{10}$	14	(2,12,2)
	5)	4 lines	conic	conic	$1/2^{12}$	16	(2,10,2)
	6)	quartic	2 lines	conic	$1/2^2$	5	(2,21,2)
	7)	cubic + line	2 lines	conic	$1/2^8$	11	(2,15,2)
	8)	2 conics	2 lines	conic	$1/2^{10}$	13	(2,13,2)
	9)	conic + 2 lines	2 lines	conic	$1/2^{12}$	15	(2,11,2)
	10)	4 lines	2 lines	conic	$1/2^{14}$	17	(2,9,2)
	11)	quartic	2 lines	2 lines	$1/2^4$	6	(2,20,2)
	12)	cubic + line	2 lines	2 lines	$1/2^{10}$	12	(2,14,2)
	13)	2 conics	2 lines	2 lines	$1/2^{12}$	14	(2,12,2)
	14)	conic + 2 lines	2 lines	2 lines	$1/2^{14}$	16	(2,10,2)
	15)	4 lines	2 lines	2 lines	$1/2^{16}$	18	(2,8,2)
$E-(2,2)$	1)	cubic	cubic	cubic	$\emptyset$	1	(3,30,3)
	2)	conic+line	cubic	cubic	$1/2^4$	5	(3,26,3)
	3)	3 lines	cubic	cubic	$1/2^6$	7	(3,24,3)
	4)	conic+line	conic+line	cubic	$1/2^8$	9	(3,22,3)
	5)	conic+line	3 lines	cubic	$1/2^{10}$	11	(3,20,3)
	6)	3 lines	3 lines	cubic	$1/2^{12}$	13	(3,18,3)
	7)	3 lines	3 lines	conic+line	$1/2^{16}$	17	(3,14,3)
	8)	conic+line	conic+line	conic+line	$1/2^{12}$	13	(3,18,3)
	9)	conic+line	conic+line	3 lines	$1/2^{14}$	15	(3,16,3)
	10)	3 lines	3 lines	3 lines	$1/2^{18}$	19	(3,12,3)

TABLE 17. Building data of a SNC $(2,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{e_1+e_2}$	$D_{e_1+e_3}$	$D_{e_2+e_3}$	$D_{e_1+e_2+e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,3)$	1)	conic	conic	conic	0	0	0	0	$\emptyset$	13	(1,7,1)
	2)	2 lines	conic	conic	0	0	0	0	$1/2^4$	14	(1,6,1)
	3)	2 lines	2 lines	conic	0	0	0	0	$1/2^8$	15	(1,5,1)
	4)	2 lines	2 lines	2 lines	0	0	0	0	$1/2^{12}$	16	(1,4,1)
$B-(2,3)$	1)	line	cubic	line	0	0	0	line	$\emptyset$	13	(1,7,1)
	2)	line	conic + line	line	0	0	0	line	$1/2^8$	15	(1,5,1)
	3)	line	3 lines	line	0	0	0	line	$1/2^{12}$	16	(1,4,1)
$C-(2,3)$	1)	quartic	line	line	0	0	line	0	$\emptyset$	8	(3,30,3)
	2)	cubic+line	line	line	0	0	line	0	$1/2^{12}$	17	(3,21,3)
	3)	2 conics	line	line	0	0	line	0	$1/2^{16}$	20	(3,18,3)
	4)	conic+2 lines	line	line	0	0	line	0	$1/2^{20}$	23	(3,15,3)
	5)	4 lines	line	line	0	0	line	0	$1/2^{24}$	26	(3,12,3)
$D-(2,3)$	1)	line	line	conic	cubic	0	0	0	$\emptyset$	12	(2,16,2)
	2)	line	line	conic	conic+line	0	0	0	$1/2^8$	16	(2,12,2)
	3)	line	line	conic	3 lines	0	0	0	$1/2^{12}$	18	(2,10,2)
	4)	line	line	2 lines	cubic	0	0	0	$1/2^4$	14	(2,14,2)
	5)	line	line	2 lines	conic+line	0	0	0	$1/2^{12}$	18	(2,10,2)
	6)	line	line	2 lines	3 lines	0	0	0	$1/2^{16}$	20	(2,8,2)
$E-(2,3)$	1)	conic	conic	conic	conic	0	0	0	$\emptyset$	11	(3,21,3)
	2)	2 lines	conic	conic	conic	0	0	0	$1/2^4$	13	(3,19,3)
	3)	2 lines	2 lines	conic	conic	0	0	0	$1/2^8$	15	(3,17,3)
	4)	2 lines	2 lines	conic	2 lines	0	0	0	$1/2^{12}$	17	(3,15,3)
	5)	conic	conic	2 lines	conic	0	0	0	$1/2^4$	14	(3,18,3)
	6)	2 lines	conic	2 lines	conic	0	0	0	$1/2^8$	16	(3,16,3)
	7)	2 lines	2 lines	2 lines	conic	0	0	0	$1/2^{12}$	18	(3,14,3)
	8)	2 lines	2 lines	2 lines	2 lines	0	0	0	$1/2^{16}$	20	(3,12,3)
$F-(2,3)$	1)	conic	line	line	conic	0	line	0	$\emptyset$	12	(1,6,1)
	2)	2 lines	line	line	conic	0	line	0	$1/2^4$	13	(1,5,1)
	3)	2 lines	line	line	2 lines	0	line	0	$1/2^8$	14	(1,4,1)
$G-(2,3)$	1)	line	conic	cubic	line	0	line	0	$\emptyset$	9	(3,23,3)
	2)	line	conic	conic+line	line	0	line	0	$1/2^8$	15	(3,17,3)
	3)	line	conic	3 lines	line	0	line	0	$1/2^{12}$	18	(3,14,3)
	4)	line	2 lines	cubic	line	0	line	0	$1/2^4$	11	(3,21,3)
	5)	line	2 lines	conic+line	line	0	line	0	$1/2^{12}$	17	(3,15,3)
	6)	line	2 lines	3 lines	line	0	line	0	$1/2^{16}$	20	(3,12,3)
$H-(2,3)$	1)	line	line	line	0	0	quartic	line	$\emptyset$	2	(4,40,4)
	2)	line	line	line	0	0	cubic+line	line	$1/2^{12}$	14	(4,28,4)
	3)	line	line	line	0	0	2 conics	line	$1/2^{16}$	18	(4,24,4)
	4)	line	line	line	0	0	conic+2 lines	line	$1/2^{20}$	22	(4,20,4)
	5)	line	line	line	0	0	4 lines	line	$1/2^{24}$	26	(4,16,4)
$I-(2,3)$	1)	conic	conic	line	cubic	0	0	line	$\emptyset$	4	(5,38,5)
	2)	conic	conic	line	conic+line	0	0	line	$1/2^8$	12	(5,30,5)
	3)	conic	conic	line	3 lines	0	0	line	$1/2^{12}$	16	(5,26,5)
	4)	conic	2 lines	line	cubic	0	0	line	$1/2^4$	7	(5,35,5)
	5)	conic	2 lines	line	conic+line	0	0	line	$1/2^{12}$	15	(5,27,5)
	6)	conic	2 lines	line	3 lines	0	0	line	$1/2^{16}$	19	(5,23,5)
	7)	2 lines	2 lines	line	cubic	0	0	line	$1/2^8$	10	(5,32,5)
	8)	2 lines	2 lines	line	conic+line	0	0	line	$1/2^{16}$	18	(5,24,5)
	9)	2 lines	2 lines	line	3 lines	0	0	line	$1/2^{20}$	22	(5,20,5)

TABLE 18. Building data of a SNC $(2,3)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{e_1+e_2}$	$D_{e_1+e_3}$	$D_{e_2+e_3}$	$D_{e_1+e_2+e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$J-(2,3)$	1)	conic	line	line	line	line	conic	0	$\emptyset$	10	(2,12,2)
	2)	conic	line	line	line	line	2 lines	0	$1/2^4$	12	(2,10,2)
	3)	2 lines	line	line	line	line	2 lines	0	$1/2^8$	14	(2,8,2)
$K-(2,3)$	1)	line	line	conic	line	conic	conic	0	$\emptyset$	7	(4,25,4)
	2)	line	line	conic	line	conic	2 lines	0	$1/2^4$	10	(4,22,4)
	3)	line	line	conic	line	2 lines	2 lines	0	$1/2^8$	13	(4,19,4)
	4)	line	line	2 lines	line	2 lines	2 lines	0	$1/2^{12}$	16	(4,16,4)
$L-(2,3)$	1)	line	line	line	line	line	cubic	line	$\emptyset$	4	(4,28,4)
	2)	line	line	line	line	line	conic+line	line	$1/2^8$	12	(4,20,4)
	3)	line	line	line	line	line	3 lines	line	$1/2^{12}$	16	(4,16,4)
$M-(2,3)$	1)	line	line	conic	conic	line	line	conic	$\emptyset$	2	(6,36,6)
	2)	line	line	conic	conic	line	line	2 lines	$1/2^4$	6	(6,32,6)
	3)	line	line	conic	2 lines	line	line	2 lines	$1/2^8$	10	(6,28,6)
	4)	line	line	2 lines	2 lines	line	line	2 lines	$1/2^{12}$	14	(6,24,6)

TABLE 19. Building data of a SNC $(2,3)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,4)$	1)	line	line	conic	line	line	$\emptyset$	15	(1,5,1)
	2)	line	line	2 lines	line	line	$1/2^8$	16	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(2,4)$	1)	line	line	cubic	line	line	$\emptyset$	18	(4,28,4)
	2)	line	line	conic+line	line	line	$1/2^{16}$	26	(4,20,4)
	3)	line	line	3 lines	line	line	$1/2^{24}$	30	(4,16,4)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$C-(2,4)$	1)	line	line	conic	conic	line	$\emptyset$	18	(3,18,3)
	2)	line	line	conic	2 lines	line	$1/2^8$	21	(3,15,3)
	3)	line	line	2 lines	2 lines	line	$1/2^{16}$	24	(3,12,3)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$D-(2,4)$	1)	line	line	line	line	conic	line	$\emptyset$	16	(2,10,2)
	2)	line	line	line	line	2 lines	line	$1/2^8$	18	(2,8,2)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_3}$	$D_{e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$E-(2,4)$	1)	conic	line	line	line	conic	line	$\emptyset$	16	(5,28,5)
	2)	conic	line	line	line	2 lines	line	$1/2^8$	20	(5,24,5)
	3)	2 lines	line	line	line	2 lines	line	$1/2^{16}$	24	(5,20,5)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_2+e_3}$	$D_{e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$F-(2,4)$	1)	conic	conic	line	line	line	line	$\emptyset$	15	(5,29,5)
	2)	conic	2 lines	line	line	line	line	$1/2^8$	19	(5,25,5)
	3)	2 lines	conic	line	line	line	line	$1/2^8$	20	(5,24,5)
	4)	2 lines	2 lines	line	line	line	line	$1/2^{16}$	24	(5,20,5)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_3}$	$D_{e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$G-(2,4)$	1)	line	line	line	cubic	line	line	$\emptyset$	12	(6,42,6)
	2)	line	line	line	conic+line	line	line	$1/2^{16}$	24	(6,30,6)
	3)	line	line	line	3 lines	line	line	$1/2^{24}$	30	(6,24,6)

TABLE 20. Building data of a SNC $(2,4)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$H-(2,4)$	1)	conic	conic	line	line	conic	line	$\emptyset$	10	(9,54,9)
	2)	conic	conic	line	line	2 lines	line	$1/2^8$	16	(9,48,9)
	3)	conic	2 lines	line	line	2 lines	line	$1/2^{16}$	22	(9,42,9)
	4)	2 lines	2 lines	line	line	2 lines	line	$1/2^{24}$	28	(9,36,9)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_3+e_4}$	$D_{e_1+e_3+e_4}$	$D_{e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$I-(2,4)$	1)	line	line	line	line	line	line	line	$\emptyset$	12	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_4}$	$D_{e_2+e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$J-(2,4)$	1)	line	line	line	line	conic	line	line	$\emptyset$	14	(4,20,4)
	2)	line	line	line	line	2 lines	line	line	$1/2^8$	18	(4,16,4)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_3}$	$D_{e_2+e_3}$	$D_{e_1+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$K-(2,4)$	1)	line	line	conic	line	conic	line	line	$\emptyset$	10	(8,44,8)
	2)	line	line	conic	line	2 lines	line	line	$1/2^8$	16	(8,38,8)
	3)	line	line	2 lines	line	2 lines	line	line	$1/2^{16}$	22	(8,32,8)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_3}$	$D_{e_1+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$L-(2,4)$	1)	cubic	line	line	line	line	line	line	$\emptyset$	4	(9,60,9)
	2)	conic+line	line	line	line	line	line	line	$1/2^{16}$	20	(9,44,9)
	3)	3 lines	line	line	line	line	line	line	$1/2^{24}$	28	(9,36,9)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_4}$	$D_{e_1+e_2+e_3}$	$D_{e_2+e_3+e_4}$	$D_{e_1+e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$M-(2,4)$	1)	line	line	line	line	line	line	line	line	$\emptyset$	12	(3,12,3)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_2+e_3}$	$D_{e_3+e_4}$	$D_{e_1+e_2+e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$N-(2,4)$	1)	conic	line	line	line	line	line	line	line	$\emptyset$	10	(7,34,7)
	2)	2 lines	line	line	line	line	line	line	line	$1/2^8$	16	(7,28,7)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_4}$	$D_{e_2+e_4}$	$D_{e_1+e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$O-(2,4)$	1)	line	line	conic	line	line	line	line	line	$\emptyset$	9	(7,35,7)
	2)	line	line	2 lines	line	line	line	line	line	$1/2^8$	16	(7,28,7)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_3}$	$D_{e_3+e_4}$	$D_{e_1+e_2+e_4}$	$D_{e_2+e_3+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$P-(2,4)$	1)	line	line	line	line	line	line	line	line	line	$\emptyset$	10	(6,24,6)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_1+e_4}$	$D_{e_2+e_4}$	$D_{e_3+e_4}$	$D_{e_1+e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$Q-(2,4)$	1)	line	line	line	conic	line	line	line	line	line	$\emptyset$	4	(11,52,11)
	2)	line	line	line	2 lines	line	line	line	line	line	$1/2^8$	12	(11,44,11)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{e_1+e_2}$	$D_{e_2+e_3}$	$D_{e_3+e_4}$	$D_{\sum_{i=1}^3 e_i}$	$D_{\sum_{i=2}^4 e_i}$	$D_{\sum_{i=1}^4 e_i}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$R-(2,4)$	1)	line	line	line	line	line	line	line	line	line	line	$\emptyset$	6	(10,40,10)

TABLE 21. Building data of a SNC $(2,4)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_2+e_3+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,5)$	1)	line	line	line	line	line	line	$\emptyset$	16	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_2+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(2,5)$	1)	line	line	conic	line	line	line	$\emptyset$	27	(5,25,5)
	2)	line	line	2 lines	line	line	line	$1/2^{16}$	32	(5,20,5)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_4+e_5}$	$D_{e_1+e_2+e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$C-(2,5)$	1)	line	line	line	line	line	line	line	$\emptyset$	20	(3,12,3)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_2}$	$D_{e_1+e_3+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$D-(2,5)$	1)	conic	line	line	line	line	line	line	$\emptyset$	24	(9,44,9)
	2)	2 lines	line	line	line	line	line	line	$1/2^{16}$	32	(9,36,9)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_3}$	$D_{e_2+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$E-(2,5)$	1)	line	line	line	conic	line	line	line	$\emptyset$	23	(9,45,9)
	2)	line	line	line	2 lines	line	line	line	$1/2^{16}$	32	(9,36,9)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_2+e_5}$	$D_{e_1+e_2+e_5}$	$D_{e_2+e_3+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$F-(2,5)$	1)	line	line	line	line	line	line	line	line	$\emptyset$	20	(7,28,7)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_3}$	$D_{e_2+e_4}$	$D_{e_2+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$G-(2,5)$	1)	line	line	line	line	line	line	conic	line	$\emptyset$	16	(15,72,15)
	2)	line	line	line	line	line	line	2 lines	line	$1/2^{16}$	28	(15,60,15)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_3}$	$D_{e_2+e_5}$	$D_{e_3+e_5}$	$D_{e_3+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$H-(2,5)$	1)	line	line	line	line	line	line	line	line	line	$\emptyset$	16	(13,52,13)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_3}$	$D_{e_1+e_5}$	$D_{e_1+e_3+e_5}$	$D_{e_2+e_3+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$I-(2,5)$	1)	line	line	line	line	line	line	line	line	line	$\emptyset$	16	(13,52,13)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	$D_{e_1+e_2}$	$D_{e_1+e_4}$	$D_{e_2+e_4}$	$D_{e_3+e_5}$	$D_{e_1+e_2+e_4}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$J-(2,5)$	1)	line	line	line	line	line	line	line	line	line	line	$\emptyset$	8	(21,84,21)

TABLE 22. Building data of a SNC $(2,5)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	D_{e_6}	$D_{e_1+e_2+e_3+e_4+e_5+e_6}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(2,6)$	1)	line	line	line	line	line	line	line	$\emptyset$	36	(7,28,7)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	D_{e_6}	$D_{e_3+e_6}$	$D_{e_1+e_2+e_4+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(2,6)$	1)	line	line	line	line	line	line	line	line	$\emptyset$	36	(15,60,15)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	D_{e_5}	D_{e_6}	$D_{e_1+e_6}$	$D_{e_2+e_4}$	$D_{e_3+e_5}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$C-(2,6)$	1)	line	line	line	line	line	line	line	line	line	$\emptyset$	28	(27,108,27)

TABLE 23. Building data of a SNC $(2,6)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(3,2)$	1)	conic	conic	line	$\emptyset$	11	(1,8,1)
	2)	conic	2 lines	line	$2/3^3$	13	(1,6,1)
	3)	2 lines	2 lines	line	$2/3^6$	15	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	D_{2e_2}	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(3,2)$	1)	cubic	line	line	$1/3^3$	10	(2,20,2)
	2)	conic+line	line	line	$1/3^3$, $2/3^6$	18	(2,12,2)
	3)	3 lines	line	line	$1/3^3$, $2/3^9$	22	(2,8,2)

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+e_2}$	$D_{2(e_1+e_2)}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$C-(3,2)$	1)	line	line	line	conic	$1/3^6$	14	(1,6,1)
	2)	line	line	line	2 lines	$1/3^6$, $2/3^3$	16	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+2e_2}$	$D_{2(e_1+e_2)}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$D-(3,2)$	1)	line	line	cubic	line	$\emptyset$	1	(3,30,3)
	2)	line	line	conic+line	line	$2/3^6$	13	(3,18,3)
	3)	line	line	3 lines	line	$2/3^9$	19	(3,12,3)

case	No.	D_{e_1}	D_{e_2}	D_{2e_2}	$D_{2e_1+e_2}$	$D_{2(e_1+e_2)}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$E-(3,2)$	1)	conic	line	line	line	line	$1/3^3$	10	(2,12,2)
	2)	2 lines	line	line	line	line	$1/3^3$, $2/3^3$	14	(2,8,2)

TABLE 24. Building data of a SNC $(3,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{2e_1+e_2+2e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(3,3)$	1)	line	conic	line	line	$\emptyset$	19	(3,18,3)
	2)	line	2 lines	line	line	$2/3^9$	25	(3,12,3)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{e_1+2e_2}$	$D_{e_1+2e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(3,3)$	1)	line	line	line	line	line	$\emptyset$	13	(1,4,1)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{2e_2}	$D_{2e_1+2e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$C-(3,3)$	1)	line	line	line	line	line	$1/3^9$	22	(2,8,2)

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{2e_1+e_2}$	$D_{2e_1+2e_2}$	$D_{e_1+2e_2+2e_3}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$D-(3,3)$	1)	line	line	line	line	line	line	$\emptyset$	13	(5,20,5)

TABLE 25. Building data of a SNC $(3,3)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	D_{e_4}	$D_{2(e_1+e_2+e_3+e_4)}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(3,4)$	1)	line	line	line	line	line	$\emptyset$	31	(5,20,5)

TABLE 26. Building data of a SNC $(3,4)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+e_2}$	$D_{3e_1+3e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(5,2)$	1)	line	line	line	line	$2/5^5$	18	(1,2,1)

case	No.	D_{e_1}	D_{e_2}	$D_{3e_1+4e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(5,2)$	1)	conic	line	line	$\emptyset$	17	(2,12,2)
	2)	2 lines	line	line	$4/5^5$	25	(2,4,2)

TABLE 27. Building data of a SNC $(5,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	D_{e_3}	$D_{4(e_1+e_2+e_3)}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(5,3)$	1)	line	line	line	line	$\emptyset$	37	(4,8,4)

TABLE 28. Building data of a SNC $(5,3)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+2e_2}$	$D_{5e_1+4e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$A-(7,2)$	1)	line	line	line	line	$\emptyset$	13	(2,8,2)

case	No.	D_{e_1}	D_{e_2}	$D_{e_1+e_2}$	$D_{5e_1+5e_2}$	Sing(X)	$\rho(\widetilde{X}_{min})$	$h^2(X)_{tra}$
$B-(7,2)$	1)	line	line	line	line	$2/7^7$	26	(3,12,3)

TABLE 29. Building data of a SNC $(7,2)$-plane $\pi\colon X \longrightarrow \mathbb{P}^2$ satisfying Property ★.

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Federico Fallucca, INdAM, Università degli Studi di Trento, Dipartimento di Matematica, I-38123 Trento, Italy
e-mail federico.fallucca@unitn.it

Matteo Penegini, Università degli Studi di Pavia, Dipartimento di Matematica *"Felice Casorati"*, I-27100 Pavia, Italy
e-mail matteo.penegini@unipv.it

Anna Ulivi, Università degli Studi di Genova, DIMA Dipartimento di Matematica, I-16146 Genova, Italy
e-mail anna.ulivi@edu.unige.it